# A tailored exact method for static bicycle redistribution with inventory-ratio dispersion

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Round131 frozen S2 benchmark reading draft, 23 October 2026. Section 6.11 is the primary final-version SC benchmark; Sections 6.12-6.13 report separate tiny overhead and V35 scope. Historical observations retain their original versions. The maintained mathematical appendix is Appendix A; full published-result tables and three clocks are in Supplement S. The complete algorithm supplement is Appendix B. Citation keys resolve in references.bib.

## Abstract

Static bicycle redistribution couples attainable station inventories with capacity- and time-feasible vehicle routes. We minimize inventory-to-target ratio Gini dispersion plus weighted two-sided target deviation under empty departure and unique nonzero one-way service. The sole proposed method is GI-TBC in its uniformly fixed pair-support implementation, MS2 (S2). It integrates independently paid full H, inventory-state products with implicit service integrality, valid static strengthening and complete scoped Gini-interval coverage around native

LP/MIP calls. Triple support is disabled on every input. Certificates follow a declared floating-point evidence policy, distinct from ideal mathematical validity.

On the six known SC problems at two Seeds (12 matched observations), S2 and P certify 10 and 5 observations, respectively; the strict S2/P directions are 10 Win, 0 Lose, 0 Tie and 2 undetermined. The tiny layer records 9 strict Win, 3 strict Lose and 0 Tie across twelve matched observations, with 0 observed >=10-second delays and 0 P-only certification risks. At V35, S2 and P have 0 and 0 nonzero certificates across two Seed0 observations, with 0 P-only cases; both-open cases have unknown final certification-time ordering. All ten inputs are already-known constructed problems from two station tables; the three evidence layers are not pooled. The method and full workflow are fixed across sizes. R129 selection and R130 adoption remain settled, while prior same-H/outer-control evidence supports only scoped attribution. There is no final-version V50/V60 claim, universal dominance claim or new single-component causal claim.

## 1. Introduction

An overnight redistribution plan has two coupled tasks: choose final station inventories and realize them by feasible vehicle routes. A desired inventory vector may look balanced yet require more travel, capacity or handling time than the fleet can provide. Conversely, optimizing routes toward fixed targets can obscure the allocation decision when those targets cannot all be attained. We consider this interaction directly. A final inventory is assessed by its ratio to a station-specific target, and routes determine which combinations of ratio changes are physically attainable.

Two objective components are useful for different reasons. Dispersion of inventory-to-target ratios measures how unevenly proportional attainment is distributed across stations. Weighted relative target deviation measures the magnitude of departure from the individual targets. Dispersion alone cannot express target attainment: a uniform 50% shortage has zero dispersion. Target deviation alone permits concentrating the remaining discrepancy where its weights are small. Their weighted sum defines a precise optimization preference without claiming that it identifies customer service, revenue or socioeconomic equity. Those broader meanings require additional demand data and a justified mapping from station targets to user outcomes.

The objective also changes the computational structure. Its Gini denominator is the endogenous sum of ratios, rather than a constant conserved stock total. Station stock need not be conserved because vehicles can return loaded. The numerator couples every station pair, and the denominator couples all stations, while feasible inventory changes are supported by an ordered load-feasible fleet. This is a different interaction from minimizing travel subject to exact target restoration or demand-interval satisfaction. An inventory relaxation can explain an objective floor, but an inventory solution does not establish route feasibility.

We evaluate a complete tailored exact method, a tailored branch-and-cut framework with Gini-interval decomposition (GI-TBC), whose fixed paper implementation identity is uniform pair-support MS2 (S2), for this stated problem. “Exact” identifies the formulation and complete proof obligations of the method; reported certificates remain numerical certificates under a declared floating-point policy. The method combines an independently paid feasible-solution startup with exact inventory-state products, a service representation retaining binary direction, valid strengthening and a Gini-interval controller. Gurobi provides native branching, generic cuts and presolve. The project strengthening is mainly added statically within each generated model; the outer Gini-interval cover coordinates multiple native LP/MIP calls rather than one native search tree. The controller decides which complete domains to solve or subdivide and how their returned bounds cover the original problem. A good incumbent from the startup is valuable, but is not an optimality proof.

The computational question is deliberately narrower than a claim of general superiority. Does this single integrated method complete more optimality certifications than the cold original compact formulation under matched complete production budgets, and reduce full certification cost where both certify? The complete frozen-version

main panel is Section 6.11; tiny overhead and finite V35 scope are reported separately in Sections 6.12 and 6.13. Sections 6.9 and 6.10 retain the earlier selection-time and known-role attribution evidence. Earlier small-instance panels retain their full-M-B predecessor identity. Earlier endpoint experiments supply an extended-range comparison. A separate longer window on one known positive-objective instance tests a proof boundary, rather than enlarging the independent sample. Both successes and losses are retained. S2 is adopted from the passed R129 prospective selection; full M-B and ENS-C are historical predecessor/representation comparators, and no method is selected separately for each instance.

The paper has three contributions.

1. It gives a complete, source-faithful formulation and a problem-specific derivation of the inventory-state/service representation. The service quantities can be continuous without changing integer solutions under explicit uniqueness and direction premises; state/service equalities are proved valid. The physical and auxiliary Gini values are kept distinct.
2. It organizes the complete integrated algorithm and its proof contract. Objective-preserving embeddings, cutoff complementation and a maintained cover explain when local native information can support an original-problem bound. Numerical closure, finite termination and optimality certification have separate meanings.
3. It supplies a frozen E3 S2/P main certification table on all six known SC problems at both Seeds (24 arms), a separate two-input tiny overhead layer (24 arms, three Seeds by two repeats), and two Seed0 V35 scope observations (four arms). R129 selection, R130 attribution and older full-M-B/ENS measurements retain their original protocols and roles; no aggregate cross-layer success rate or average speedup is reported.

The constituent tools include standard product linearization, McCormick envelopes, multistart construction, local exchange, MIP Starts and bound pruning. We do not claim new general branch-and-cut theory for these tools. The scientific object is their particular integration for a nonseparable inventory objective and its measured behavior under one budget. Literature positioning in Section 2 limits the contribution relative to neighboring models; Appendix A makes the formulation and mathematical premises explicit. Sections 3-5 describe the problem, representations and algorithm; Section 6 presents direct certification evidence; Section 7 retains the extended historical evidence; Section 8 discusses its limits.

## 2. Related work and contribution position

Static bicycle rebalancing covers several materially different problems. Raviv, Tzur and Forma (Raviv et al. 2013) use station dissatisfaction and routing considerations with time resources. Their arc and sequence formulations have different service restrictions; the early full preprint inspected here must not be collapsed into one blanket single-visit model. Our objective is the specified ratio dispersion and relative target deviation, without a travel-cost objective term. Final-article metadata and the early preprint are distinguished in the literature ledger.

The publisher abstract of Chemla, Meunier and Wolfler Calvo (Chemla et al. 2013) describes a single capacitated vehicle restoring a target distribution with revisits and temporary storage. It distinguishes branch-and-cut lower bounding of a relaxation from tabu-search feasible upper bounds. We use only these abstract-supported features; its detailed initial-load and handling constraints were not obtained. Our unique one-way station service excludes both revisits and buffering, while permitting target deviations.

Erdoğan, Laporte and Wolfler Calvo (Erdoğan et al. 2014) study interval inventory requirements with one vehicle, route/handling cost and exact branch-and-cut formulations, including Benders decomposition. Their fixed-route network explicitly accounts for depot supply and handling. Erdoğan, Battarra and Wolfler Calvo (Erdoğan et al. 2015) instead address target restoration with revisits and temporary storage using an exact routing/feasibility

framework. Its depot imbalance and target-equality semantics differ from empty departure and optional target attainment here. Neither problem can be adopted unchanged as a matched benchmark.

Schuijbroek, Hampshire and van Hoeve (Schuijbroek et al. 2017) connect service-level inventory intervals with multi-vehicle routing. Their routing objective is makespan, with given initial vehicle inventories and an artificial ending node; their formulation permits more service flexibility than unique one-way visits, subject to its transshipment limits. The cluster-first framework is an approximate multi-vehicle approach, alongside MIP and CP models. This provides a close allocation/routing comparison, but not the same physical model, objective or certificate scope.

The openly available Ikonen, Heljanko and Harjunkoski article (Ikonen et al. 2025) reformulates routing/load disjunctions into a MILP for demand intervals. Duplicated stations permit split visits without temporary storage; prescribed initial load and open/loop options are explicit. Its single-vehicle exact model can be combined with approximate multi-vehicle clustering. Its disjunctions and our Gini proof cover address different structures, despite shared terminology. Reported larger station counts cannot establish relative speed across the different objectives, instances, hardware and thread limits.

Its certificates concern its finite duplication model; solutions requiring additional visits or temporary operations can lie outside that generated model.

Equity requires specifying the resource being balanced. Matl, Hartl and Vidal (Matl et al. 2018) analyze workload-equity functions and consequences in vehicle routing. We use their distinction as modeling context; driver-workload conclusions are not transferred as theorems about station inventory-to-target ratios. The bibliography consistently uses the 2018 volume year and records first online publication in 2017.

The search was updated through 18 October 2026. Qiao et al. (Qiao et al. 2026) describe two-echelon coordination with task-duration fairness, an exact first-echelon subproblem and a hybrid second-echelon method. Caggiani et al. (Caggiani et al. 2026) combine free-floating electric relocation, battery swapping and equity evaluation. These are abstract/excerpt-level checks; their complete constraints remain unverified here. They establish that fairness-aware rebalancing is an active prior topic, and preclude any broad claim that introducing fairness or Gini into rebalancing is itself our novelty.

**Table 1. Closest model distinctions.** "Unknown" means not established by the accessed source. The three middle detailed comparisons were checked against their full model sections. Table entries do not assert interchangeable feasible sets.

| Work | Objective / target requirement | Fleet and visits | Temporary or split service | Initial/return and time semantics | Exact scope |
|---|---|---|---|---|---|
| Chemla 2013 | Target distribution | Single; revisits | Buffering allowed | Detailed constraints unknown from abstract | Relaxation bound plus feasible search |
| Erdoğan 2014 | Travel/handling; hard inventory intervals | Single; at most once | No split visit | Depot stock bounds departure; depot handling charged | Full and Benders branch-and-cut |
| Erdoğan 2015 | Travel; exact targets | Single; revisits | Temporary storage allowed | Depot imbalance participates; not our empty-departure model | Routing plus time-extended feasibility |
| Schuijbroek 2017 | Makespan; service-level intervals | Multiple; sequence indexed | Limited transshipments; service not uniquely assigned as here | Given initial load; artificial open end; handling cost | Joint MIP/CP; clustering heuristic |
| Ikonen 2025 | Time/distance/handling; intervals | Single model; duplicated visits | Split permitted; temporary storage forbidden | Given initial load; open or loop option | MILP per vehicle; clustered fleet approximate |
| Present model | Ratio Gini plus weighted relative target deviation | Multiple; one nonzero service per station | Neither split nor temporary service | Empty departure; loaded closed return with unloading; hard route horizon | Complete joint original-model numerical cover |

The relevant standard mathematics is also explicit. Bilinear envelopes are attributed to the established global-relaxation literature (McCormick 1976); their four inequalities are derived in Appendix A. An exact product with a selected discrete inventory is an application of known disaggregation, not a generic new linearization theorem. Likewise, continuous declaration of a variable already determined integrally by the remaining integer decisions is a formulation choice whose premises must be proved. The empirical question is whether the resulting method helps this problem.

The project lineage is narrower than a literature genealogy. The current cold P is the repository's canonical compact writer and original objective evaluator (TailoredExact project 2026b). Inspection of the relevant writer, evaluator, startup bridge and representation headers found no verified mapping that identifies one of the public BRP formulations above as its exact original model. Shared routing ideas are insufficient evidence for that identification. H retains the project's inherited decoder and later constructive/exchange components; the paper does not claim to have invented a full genetic algorithm or to have proved H optimal. The code and evidence maps in the appendix and reproducibility notes separate these inherited components from the tailored representation and complete-method comparison.

## 3. Problem definition and objective interpretation

Let $N = \{1, \ldots, n\}$ contain service stations and 0 be the depot. Each station has capacity $C_i$, initial inventory $b_i$, target inventory $D_i > 0$, and weight $\omega_i$. Each vehicle k has capacity $Q_k$, begins empty and follows at most one depot-closed route. A station may be left unserved or visited once by one vehicle, with an integer nonzero pickup or drop. Simultaneous pickup and drop at the same station, split service and temporary storage operations are excluded. These are physical restrictions of the problem being evaluated, not conclusions about all rebalancing applications.

Write $p_{ki}$ and $d_{ki}$ for pickup and drop quantities. Final station inventory obeys

$$Y_i = b_i + \sum_k (d_{ki} - p_{ki}), \qquad 0 \le Y_i \le C_i. \tag{1}$$

Every route prefix and its return load must lie in $[0, Q_k]$. A vehicle can collect, deliver and collect again; cumulative pickup is therefore not bounded by $Q_k$. Loaded return is allowed. If $R_k = \sum_i (p_{ki} - d_{ki})$, empty departure and nonnegative return imply

$$\sum_i Y_i + \sum_k R_k = \sum_i b_i. \tag{2}$$

Station inventory can decrease through depot returns. The capacity/initial values serialized for the depot are excluded from the objective and do not supply a preloaded fleet. In particular, a large depot stock entry cannot repair a station shortage in this model.

Travel time is the Euclidean distance between the serialized coordinates divided by 1.5. With unit handling times $t_p, t_d$, total duration of route k is

$$\text{travel}_k + t_p \sum_i p_{ki} + t_d \left( \sum_i d_{ki} + R_k \right) = \text{travel}_k + (t_p + t_d) \sum_i p_{ki} \le T. \tag{3}$$

The equality follows from load conservation and includes depot unloading. Thus the form used in the writer is not an approximation that ignores return handling. All reported fleets are replayed with their assigned capacities,

ordered operations and complete durations. The parameter $T$ is physical route time; the computational cap is a separate resource.

Set $r_i = Y_i/D_i$, $S = \sum_i r_i$, and $A = \sum_{i<j} |r_i - r_j|$. The objective is

$$G_{\text{true}}(Y) = \begin{cases} A/(nS), & S > 0, \\ 0, & S = 0, \end{cases} \qquad F(Y) = G_{\text{true}}(Y) + \lambda \sum_i \omega_i |r_i - 1|. \tag{4}$$

The denominator depends on the final inventory vector. The convention at $S = 0$ sets dispersion to zero but retains all relative target penalties. For nonnegative ratios, $0 \le G_{\text{true}} \le (n-1)/n$. The target penalty is two-sided: inventories above as well as below target are penalized. We consistently call it relative target deviation, rather than observed unmet demand.

A zero Gini indicates identical ratios, including the stipulated all-zero case. It does not indicate that stations meet their targets. For example, a common ratio .5 has zero dispersion but positive target deviation when weights and lambda are positive. Conversely, under strictly positive lambda and weights, $F = 0$ requires every ratio to equal one. The study's positive-objective instances cannot be interpreted through a zero-dispersion shortcut.

The weights are max-normalized within the constructed inputs rather than normalized to sum one. Therefore $\lambda \sum_i \omega_i$ varies across the panel. A fixed lambda does not imply a fixed aggregate tradeoff between target deviation and Gini as station count or the weight distribution changes. This matters both for interpreting objective magnitudes and for the difficulty of proving optimality. Cross-instance values of F are not a common operational service score.

The static assumption freezes inventories during redistribution; customer rentals and returns do not alter them while trucks operate. Coordinates have a real station-source basis, but stock levels, targets, weights and the centroid depot follow a synthetic, declared construction. We therefore evaluate a physical optimization model on traceable constructed instances, not an end-to-end deployment validated against realized rental outcomes.

## 4. Formulations and problem-specific strengthening

### 4.1 Why the inventory representation matters

The cold baseline P-GRB is the canonical compact MILP for the physical problem in Section 3. Binary arcs and assignments, unique station service, direction indicators, MTZ ranks, prefix loads and the complete duration constraint couple routing to final inventory. Appendix A.1-A.3 gives the full numbered formulation in one place. It retains apparently redundant rows actually written by the implementation; it does not substitute a more familiar routing model. The difficulty is not just route selection: integer station stocks enter both the numerator and the variable denominator of ratio Gini.

Write $H_{ij} \ge \pm(r_i - r_j)$, $e_i \ge \pm(r_i - 1)$, $P = \sum_i \omega_i e_i$, and introduce a Gini epigraph g. Exact products $Z_i = gY_i$ give

$$n \sum_i Z_i/D_i \ge \sum_{i<j} H_{ij}, \qquad \min f = g + \lambda P. \tag{5}$$

Cold P uses binary inventory bits and binary-continuous products. S2 selects one integer inventory from the necessary domain $\mathcal{Y}_i$ in each interval $[\alpha, \beta]$:

$$\sum_{y \in \mathcal{Y}_i} s_{iy} = 1, \quad Y_i = \sum_y y s_{iy}, \quad \alpha s_{iy} \le q_{iy} \le \beta s_{iy}, \quad \sum_y q_{iy} = g, \quad Z_i = \sum_y y q_{iy}. \tag{6}$$

At integer selectors one state y=Y_i is active; all other q are zero and the active q equals g, proving $Z_i = gY_i$. This is an application of discrete product disaggregation, not a new general product theorem. Its LP relaxation differs from the bit model. The actual mode retains the original ratio and deviation constraints rather than replacing them with state-based deviation equalities.

S2 declares pickup/drop continuous while retaining integer Y, unique service, binary assignment and binary direction. At an unserved station both quantities are zero. At the sole serving vehicle either d=0 and p=b−Y, or p=0 and d=Y−b. The nonzero quantity is therefore integer. Integer load declarations are retained but are unnecessary for this inference. Multiple serving vehicles, simultaneous pickup/drop or fractional inventory would invalidate it. Independently, one-way nonzero service proves the state/service equalities

$$\sum_k (p_{ki} + d_{ki}) = \sum_y |b_i - y| s_{iy}, \qquad \sum_k z_{ki} + s_{i,b_i} = 1. \tag{7}$$

A missing initial-inventory state means $s_{i,b_i} = 0$. These equalities link movement and visit to the selected state; they are not extra premises of the integrality argument. ENS-C keeps integer p/d and omits these two equalities, while sharing H, the static package and the controller. Neither a representation comparison nor an ideal implication between rows identifies the independent computational contribution of a constituent without its ablation.

### 4.2 Physical strengthening and the role of integer service

Conservation gives $\sum_i b_i - \sum_k Q_k \le \sum_i Y_i \le \sum_i b_i$, because the fleet's return stock lies between zero and total vehicle capacity. Visit/inventory links fix Y=b when unserved. Summing route durations gives global handling $h \sum_{ki} p_{ki} \le mT$. Single-station reachability intersects the inventory domain with the largest operation any one vehicle can support after paying its necessary depot round trip. These physical bounds hold before any Gini interval or incumbent is chosen.

Two less immediate families encode route support and supply compatibility. For an active two-station support W, the smallest explicit depot-cycle travel $\tau(W)$ is the minimum of its two orders. The general three-station validity proof is retained in Appendix A for the full-M-B predecessor, but no three-station support row is emitted by S2. Metric shortcutting makes it a lower bound on any full tour containing W. If all W are served, their integer nonzero operations and empty departure yield

$$|W| \le \sum_{i \in W} (p_{ki} + d_{ki}) \le P_k + D_k^{\text{drop}} \le 2P_k, \quad P_k \ge \lceil |W|/2 \rceil. \tag{M1}$$

Thus $\tau(W) + h\lceil |W|/2 \rceil > T$ excludes their simultaneous service. Here P_k is **all** pickup on the route; the supplies may be outside W. The alternative conditional row is

$$h \sum_{i \in W} p_{ki} + M_{kW} \sum_{i \in W} z_{ki} \le T - \tau(W) + M_{kW}|W|, \quad M_{kW} = T + \tau(W) + h \max(1, \sum_{i \in W} \min(b_i, Q_k)). \tag{M2}$$

All served cancels the M terms and leaves a necessary travel/handling inequality. With one missing service, its available right side is at least $2T + h \max(1, \sum_W \min(b_i, Q_k))$, enough for every bounded pickup on W. Appendix A.5.4 proves both branches and gives the exact shared 50000-support enumeration. Capacity limits prefixes, never cumulative pickup.

For transfer, label collected units and match every delivered unit to an earlier pickup on its vehicle; nonnegative prefix load makes such matching possible. A source i for delivery j implies the ordered subsequence 0,i,j,0, whose metric travel plus at least one unit of handling is necessary. Therefore

$$d_{kj} \le \sum_{i \ne j:\ \tau_{0i}+\tau_{ij}+\tau_{j0}+h \le T} p_{ki}. \tag{M3}$$

An empty compatible source set forbids delivery; an unvisited source contributes zero; extra stations cannot manufacture stock. Loaded return consists of unmatched pickups and does not break the argument. This row supplies a relaxation-level connection between stock decisions and route time, while the original ordered loads remain indispensable. Appendix A states the emitted numerical guards, indices and all remaining physical rows.

### 4.3 Gini and cutoff strengthening through a physical embedding

For nonnegative ratios let $A = \sum_{u<v} |r_u - r_v|$. Fix i<j. For every other u, triangle inequality gives $|r_i - r_u| + |r_j - r_u| \ge |r_i - r_j|$. These n−2 contributions, the pair itself and nonnegative omitted pairs prove

$$A \ge (n-1)|r_i - r_j|, \qquad (n-1)|r_i - r_j| \le n\beta S \quad \text{when } G_{true} \le \beta. \tag{M4}$$

The static H-spread row uses $(n-1)H_{ij} \le n\beta S$. Combining it with the original H lower rows implies both signed variable-S centering rows; $S \le \bar{S}$ then implies fixed centering. The ideal containment is useful for understanding, not a reason to remove current rows or claim measured redundancy. The direct interval cap/floor $n\alpha S \le H_\Sigma \le n\beta S$ and these spread rows admit the **tight physical completion** H=absolute differences.

That qualification is essential. Every physical fleet in the stated true-G interval and cutoff has a completion with g=true G, exact e,H, Z=gY and W_SP=SP, preserving F. An arbitrary native completion may inflate g,e,H or W, and need not survive the added rows. Conversely native integer objective is at least decoded physical F. The proof contract is objective-preserving physical projection, not preservation of every original auxiliary assignment as required by an unrestricted user-cut claim. At S=0 all ratios/differences are zero; g=0 realizes the original convention and the penalty remains payable.

From true $F \le c$ and $G \ge \alpha$, nonnegative penalties give $\omega_i |Y_i/D_i - 1| \le (c-\alpha)/\lambda$. Outward-rounded integer intersections and reachability give the domain rectangle, its penalty floor $\underline{P}$ and S bounds. The active writer performs the finite sequence in Appendix A.5.1; a preset field requesting two propagation rounds does not implement an infinite fixed point or even two feedback rounds. Required movement follows immediately from inventory balance on this tightened rectangle.

For positive S, multiplying F≤c yields $A + n\lambda SP \le ncS$. Replacing S in the denominator by its upper bound gives $A + n\bar{S}\lambda P \le n\bar{S}c$. Four standard McCormick inequalities (McCormick 1976), obtained by expanding the four nonnegative rectangle products, relax W_SP=SP and support the lifted row $H_\Sigma + n\lambda W_{SP} \le ncS$. Appendix A.5.7 gives every coefficient, bound and derivation. The original objective cutoff is retained: at S=0 cross multiplication becomes tautological and would otherwise forget the positive penalty. These are cutoff-scoped rows, regardless of a “Global” implementation metadata label.

### 4.4 Native and project responsibilities

The project supplies the original model, inventory representation, static physical/interval rows, H, complete verified Starts and domain controller. Gurobi supplies native LP/MIP optimization, presolve, cuts and branching. Native monitoring callbacks can terminate on a mathematical bound target; S2 does not implement dynamic

separation of these static families or custom native branching. Benders, pricing, dynamic resource cuts, D1, AMF/RC, single-child contraction and timed-quanta research branches are off. Standard triangle, cover, product and McCormick reasoning is not claimed as new general theory. The method specification matrix maps the finite active list to numbered proofs and actual generation points.

## 5. Complete S2 algorithm and certificate logic

### 5.1 Startup and maintained information

H is a finite constructive and neighborhood procedure. It constructs a verified fleet by joint insertion; generates 24 shuffled station/separator orders under seed 20260626 and one completed constructive order; fully decodes each; performs first-improvement guided order descent; then alternates strict physical insertion/quantity closure with neutral balanced relocation/equal-net exchange. Appendix B.6-B.11 supplies the encoding, legal amounts, compensation, full compaction, candidate generation, cache policy and actual ties. This active path does not execute genetic generations. H supplies an own upper bound, never a lower certificate. Every compared integrated arm pays for its own H; saved or cross-arm routes are not free incumbents.

The controller stores own verified fleet, U, its true G, incumbent epoch and a coverage registry. Each unit carries interval, qualified bound and source scope, cutoff, parent/children, depth and status. Runtime caches separately bind canonical matrix identity and complete LP evidence to their generation/epoch. Maintaining this distinction prevents a good LP point from becoming a physical incumbent or a child-domain bound from being promoted to its parent.

### 5.2 Two bound targets and complete decision procedure

After the selected parent's LP return, let its updated bound be b and the other relevant unit bounds be b_j. Requeue first if any $b_j + \epsilon < b$. Otherwise, until the unit's NEXT milestone has been reached, request

$$t_{next} = \min\{b_j : b_j > b + \epsilon\}, \qquad \epsilon = 10^{-7}, \tag{M5}$$

if this set is nonempty. Once the milestone is reached, or no strict higher competitor exists, allow child lookahead. NEXT therefore raises a controlling unit toward the next frontier level before paying for subdivisions. It is not the child target and not a time slice.

For two qualified finite raw child LP bounds $\ell$_j, put

$$\Delta = \max(U - b, \max(\epsilon, 10^{-12})), \quad v_j = \mathrm{clip}((\ell_j - b)/\Delta, 0, 1), \quad \eta = \min(v_0, v_1), \quad \mu = (v_0 + v_1)/2. \tag{8}$$

If $b_+ = \min(\ell_0, \ell_1) \le b + \epsilon$, run terminal MIP. If it strictly improves b and ημ plus the explicitly defined floating score tolerance (B2) reaches .08, split. If the gain is positive but score insufficient, retain the parent and run its MIP with target b+. Recompute decisions from the actual return. With an infeasible child, the frozen policy installs both children atomically and marks the proved empty ones; it does not invoke the dormant contraction variant.

The following algorithms specify the complete active flow. `Native` means the request contract in B.4 (actual model identity, type restoration, remaining global deadline, optional validated complete Start, native return and request cleanup). `Qualify` tests its stated identity/finalization/settings/consistency gates; it is not a new oracle. Geometry, tie order, cache tests and all return cases are fully defined in B.1-B.4.

```
Algorithm 1  FrozenMB
Input: instance and physical parameters; frozen method settings; one process deadline
Output: own fleet/U, qualified complete-cover L, unresolved units and evidence status
1  Run H (B.6), admitting only complete physically verified fleets.
   If own verified zero and the inherited zero-evidence policy passes,
       finalize its independent F>=0 certificate and return.
   If no admissible fleet exists, retain U=+infinity and report startup failure;
       RETURN failure without initializing a finite-cutoff proof.
2  Create root L0=[0,min(U,(n-1)/n)], bound 0, Open, epoch 0;
   retain outside-range analytic floor U. For n=1 use root [0,0].
3  While time remains and unresolved relevant coverage remains:
       select a controlling unit with B.2's epsilon-plateau cyclic tie rule;
       if no unit selectable, stop and retain all remaining obligations;
       outcome := ProcessLeaf(unit);
       FinalizeAndUpdateCover(outcome);
       if deadline, engineering failure or terminal interruption: stop.
4  Finish native cleanup and lifecycle outputs; qualify sources and own witness.
   Return the raw complete-cover minimum L without clipping to U;
       distinguish its numerical policy from exact complement accounting (B.4).
   Return own U/L and all unresolved coverage; certify only if the inherited
       numerical gap, source, physical and lifecycle gates all pass.
```

```
Algorithm 2  ProcessLeaf / ChooseAction
Precondition: selected unit is still represented; no pending native call.
1  Validate/rebuild artifact for current epoch; reuse a qualified same-epoch
   parent LP or request its complete LP relaxation.
   On deadline/invalid LP return OPEN_STOP/ERROR, with parent unchanged.
   On qualified infeasible LP return CLOSE(parent,cutoff evidence).
   Merge a finite optimal parent-LP bound on this same domain.
   If b >= current U-epsilon return FATHOM with its bound source.
2  Apply NEXT rule (M5) in its stated priority order:
       noncontrolling -> REQUEUE;
       strict target -> Native(parent MIP,target NEXT), then RETURN_NATIVE;
       otherwise proceed to 3.
3  If depth>=8 or width<=1e-4+1e-12:
       Native(complete parent MIP,no target), then RETURN_NATIVE.
4  Form midpoint children [alpha,mid],[mid,beta], ids parent.0,parent.1.
   Obtain both complete current LP results or reuse B.2's valid cached pair.
   A child starts with max(parent bound,its qualified finite LP bound),
       but retain raw child LP fields for AM. Children are not active leaves.
   On either incomplete/invalid result return OPEN_STOP/ERROR; keep parent.
5  If either child is qualified infeasible: return SPLIT(children,empty flags).
   Otherwise compute (8), (B2), b+=min(raw child bounds):
       b+<=b+epsilon -> Native(complete parent MIP,no target);
       score+score_tolerance>=.08 -> return SPLIT(both children);
       otherwise -> Native(parent MIP,target b+,kind CHILD).
   For a native action return its actually finalized outcome and source identity.
Postcondition: no unresolved domain removed; no target value admitted as proof
   merely because it was requested; no speculative child counted as active.
```

```
Algorithm 3  FinalizeAndUpdateCover
Input: ProcessLeaf outcome, current cover, own witness/epoch/runtime caches
1  For native outcomes require actual return and request cleanup first.
   On failed engineering qualification: stop ERROR; leave unresolved region.
   Admit any finite qualified native bound only on its recorded domain/cutoff.
   Retain the source cutoff for proof accounting in A.6; do not clip the
       raw stored numerical bound (B.4).
2  Replay a candidate fleet and original objective. If own U improves by >1e-9,
       replace own witness, increment epoch and tighten current cutoffs.
       Keep qualified old scoped floors; invalidate retained models/LP/children
       on next access using B.2; never transfer a child-only floor upward.
3  SPLIT: validate parent identity, both child ids/lineage/endpoints, shared
       midpoint, inherited bounds and full union. Install both in one transaction,
       mark parent Replaced, then mark proved infeasible children Empty.
       Discard replaced/empty backend models; failed transaction retains parent.
4  CLOSE/FATHOM or qualified native OPTIMAL/INFEASIBLE: close only this
       cutoff obligation and keep its complement floor and evidence.
   Target reached + termination requested + bound>=target-epsilon:
       set NEXT milestone only for NEXT, then requeue parent (no forced split).
   REQUEUE: retain all state/coverage, return to selection.
   TimeLimit/interrupted before proof/target: retain eligible partial bound,
       keep unit Open, stop common lifecycle. Unknown/error is never Empty.
5  Recompute full-cover minimum; output unresolved units if any remain.
Postcondition: every potential physical improver is still covered by an open
   obligation or a qualified closed/complement floor.
```

Own-U updates and native-bound admission share a source ledger but not a meaning: a feasible fleet can remain valid even if a dependent proof source is later rejected. Start ineligibility is neither infeasibility nor evidence that a low-G region lacks feasible fleets. The next strict target has priority before geometry eligibility; a narrow/deep leaf can still do a NEXT phase, but eventually uses complete MIP rather than deleting its region.

### 5.3 Why the returned bounds cover the original problem

For a source bound $\ell$ on an interval model with objective-preserving embeddings of all physical points in $F \le c$, points inside that cutoff have $F \ge \ell$, while points outside have $F > c$. Its unconditional regional floor is therefore min($\ell$,c). Infeasible cutoff models yield floor c, not an infinite original-region bound. This is why old cutoffs remain in the evidence scope after U improves.

For a complete covering family $\mathcal{C}$ with such floors β_I, plus the analytic outside-range floor U,

$$L = \min\{U, \min_{I \in \mathcal{C}} \beta_I\} \tag{9}$$

is a lower bound. Every physical point belongs to at least one covering region or the complement. Both closed midpoint children preserve the parent's union, including their shared endpoint. Max may combine independent valid floors on the same domain; min combines siblings. An unsplit parent or incomplete lookahead remains an obligation. These observations prove coverage inductively, without treating depth limits, milestones, target termination or repeated U as certificates. Appendix B.5 gives both synthetic teaching states and one labelled R123 atomic-cover event.

### 5.4 Exact mathematics and numerical evidence

The embedding, triangle, support and cover arguments above concern real/exact arithmetic under their explicit physical premises. Actual matrices use binary64 coefficients and Gurobi tolerances; the retained evidence policy

is $10^{-7}$. The implemented penalty closure test $\alpha + \lambda \underline{P} \geq c - 10^{-9}$ can close a non-strict cutoff containing a mathematical point at or just below c. It is a **numerical closure**, not proof of strict rational infeasibility. A local $10^{-9} < 10^{-7}$ comparison does not bound global accumulated error. H's internal floating descent likewise does not prove exact rational monotonicity, although its admitted physical neutral moves preserve the exact recomputed objective and improve the entire duration tuple.

Reported original-problem certificates additionally need own verified U, qualified complete coverage, source/return identities and normal lifecycle completion. The sealed nonzero policy requires −1e−7≤U−L≤1e−7 without clipping; this is stronger than the native certificate function's one-sided test. B.4.1 states every admission gate, the no-native clean own-zero branch $U \leq 1e-12$, and the exact-zero source-reduced fallback. Invalid proof dependencies cannot be repaired by a good incumbent or independent zero floor; source-reduced outcomes remain separate. The single deadline guarantees an operational end, not certification within every window. No solver tolerance or historical certificate field is changed in this revision. The finite sample and nonfactorial design in Section 6 limit empirical attribution; these mathematical details add no new performance evidence.

## 6. Versioned certification and attribution evidence

The current sole method is S2/MS2. The principal final-version benchmark is Section 6.11 (A); Sections 6.12 (B) and 6.13 (C) separately report absolute tiny overhead and finite V35 scope. Section 6.9 preserves R129 selection-time evidence and Section 6.10 preserves R130 known-role attribution. Sections 6.1-6.8 retain predecessor/development identities. Old full-M-B/ENS times are never relabelled S2 or mixed into a common final-version clock cohort.

### 6.1 Historical R126 predecessor design and one-method question

At execution of the historical R126 panel, full three-station M-B was the sole proposed method; the current paper method is S2/MS2. Before generating any Round126 input or seeing its difficulty, we fixed the six problems, two method entries, order, Seed and budgets in Table SC1. Cold P-GRB is the complete original compact model. It receives no H, incumbent, route, bound, Start, certificate, cache or prior solve from M-B. M-B pays for its own unchanged H, VD-P, implicit integer service, A/B, static strengthening, NEXT/AM, full cover and outputs on every problem. ENS-C is a predecessor/representation comparator retained in the historical section; it is not selected per instance or run in this panel. D1 selection is closed, D0 is not a fallback, and no engine switch or fusion is used.

**Table SC1. The entire fixed Seed0 panel.** Pickup/drop are 60/60 seconds and lambda is .15 in every row. P denotes cold P-GRB. Each method occupies the first position three times.

| Problem | Source | n/m | Geometry / stock | Q | Physical T (s) | Production cap (s) | Order |
|---|---|---|---|---|---|---|---|
| SC-C15-S | CITI | 15/2 | compact / shortage | [30, 30] | 3600 | 600 | P, M-B |
| SC-B15-S | CABI | 15/2 | regional / shortage | [20, 40] | 5400 | 600 | M-B, P |
| SC-C15-B | CITI | 15/2 | regional / balanced | [30, 30] | 5400 | 600 | P, M-B |
| SC-B20-S | CABI | 20/2 | compact / shortage | [30, 30] | 5400 | 1200 | M-B, P |
| SC-C20-S | CITI | 20/2 | regional / shortage | [20, 40] | 7200 | 1200 | P, M-B |
| SC-B20-U | CABI | 20/2 | compact / surplus | [30, 30] | 7200 | 1200 | M-B, P |

**Table SC1b. All six station-stock and weight totals.** Depot placeholders are excluded. Weights are maximum-normalized, not sum-normalized.

| Problem | sum initial | sum target | sum weights | lambda sum weights |
|---|---|---|---|---|
| SC-C15-S | 157 | 178 | 9.599999 | 1.43999985 |
| SC-B15-S | 99 | 114 | 8.98 | 1.347 |
| SC-C15-B | 276 | 276 | 6.844233 | 1.02663495 |
| SC-B20-S | 191 | 217 | 6.98892 | 1.048338 |
| SC-C20-S | 231 | 263 | 7.721076 | 1.1581614 |
| SC-B20-U | 149 | 129 | 6.270832 | 0.9406248 |

All six inputs were generated once from the two already safe frozen station tables. CITI source identity is `9f9aad24e61d661971c24c6f5ec78a69081edf01ffc433563e52a70b2eaca2d0`. CABI source identity is `d4e664c3ecb8c580e95d39541555be9cf592666a515cb854a857fb1d117ff0b8`, with safe-table identity `d982051e5871c575ecdf9df538605401bd98341ae9b3b48112482eff80d42ffa`. The family is `round126-small-certification-v1|source=<source>|source_sha256=<identity>`, and each selection ID is `r126_<problem>_<geometry>_r<replicate>`. The family enters the original anchor hash and the real selection ID enters target/inventory hash material. The adapter is `round126-thin-offline-adapter-v1`.

The original four anchors, compact/regional geometry, tie rules, three-decimal coordinates, target fractions [.44,.56], total correction/sign repair, maximum-normalized six-decimal weights, centroid depot and distances are retained. Only the declared family/ID and table parameters change. Source capacities and coordinates are geographical data; inventories, targets, weights, depot and fleet settings are constructed. No H, LP, MIP or difficulty filter chose or redrew a problem. Station overlaps are recorded. The six problems are not a complete factorial design, IID city samples or six cities.

For the four shortage problems, service stations alone satisfy s=sum D-sum b>0. Empty departure and nonnegative loaded return imply sum Y<=sum b. Thus, with delta=min(lambda*omega_i/D_i)>0,

$$F(Y) \geq \sum_i (\lambda \omega_i / D_i)|Y_i - D_i| \geq \delta \sum_i |Y_i - D_i| \geq \delta s > 0.$$

**Table SC2. Independent positive-objective arithmetic.** The coefficients are the exact decimal interpretation of the serialized weights and lambda=.15; parsed binary64 values are also recorded. These floors are not injected into the solver or substituted for its reported L. Empty fleets remain legal, so shortage does not mean infeasibility.

| Shortage problem | sum b | sum D | s | Exact-decimal positive floor |
|---|---|---|---|---|
| SC-C15-S | 157 | 178 | 21 | 0.111999825 |
| SC-B15-S | 99 | 114 | 15 | 0.09 |
| SC-B20-S | 191 | 217 | 26 | 0.0864264375 |
| SC-C20-S | 231 | 263 | 32 | 0.089256 |

Balanced and surplus objectives were not presumed zero or positive. A qualified H zero termination is labelled startup-zero and supports termination of the complete method, without credit to a nonzero native proof or interval backend.

### 6.2 Complete production clocks and certificate admission

The production executable and DLL are the unchanged frozen identities specified in Section 5 and the reproducibility record. Both entries use Gurobi Seed0, Threads1, PresolveAuto and the original remaining parameters, requesting relative/absolute MIP gaps zero. Evidence tolerance remains 1e-7. H retains seed 20260626. The 30-second native reserve, cap-minus6 CLI request, cap-minus2 external hard stop and common absolute deadline

are inherited; preparation is not moved outside the clock. A larger cap can change a path and is not treated as a freely reusable prefix.

T_produce runs from full method start through normal output sealing, including own H, all models/LP/MIP work, static rows, Start, online verification, callbacks, cleanup and required output. T_verify is independent post-seal validation without new optimization or route search. T_total is their sum; cross-arm reconciliation is separately billed group work. Define T_cert=T_produce only when the final certificate passes every own physical, numerical, identity, scope, cover, return and lifecycle gate; otherwise T_cert is null. It is not first incumbent discovery, Gurobi Runtime or the final MIP call's runtime.

Clean nonzero certification uses the inherited signed-gap admission -1e-7<=U-L<=1e-7 alongside all other gates. Raw L is not clipped to U. Clean own-zero and damaged-source exact-zero retain their distinct inherited policies. Same-input cross-arm witnesses are reconciled only after both own productions, using the unchanged 1e-7 scope/consistency rule; they never supply the other arm's starting information. A damaged bound cannot be repaired by a good fleet or another certificate.

The five main categories are BOTH_CERTIFIED, MB_ONLY_CERTIFIED, P_ONLY_CERTIFIED, BOTH_OPEN and UNEVALUABLE. Both-certification speedup is T_cert(P)/T_cert(M-B), so values above one favor M-B. We display every exact observed time difference and ratio; the descriptive materiality threshold is max(30 seconds,.10*T_cert(P)), producing MB_FASTER/P_FASTER/TIME_CLOSE. This threshold is neither statistical significance nor an engineering gate. Open U/L/gap comparisons are supplementary and generate no main WIN. A normal open native return before its nominal cap is reported at its actual duration: neither the cap nor actual open duration is substituted as a hypothetical certification time, exact speedup or unproved speedup lower bound.

### 6.3 All twelve arms, six pairs and evidence limits

**Table SC3. Complete own endpoints.** All signed gaps are retained. Full-precision CSV and source/call/model/fleet indices are part of the package.

| Problem | Method | Own U | Qualified L | Raw signed U-L | Certificate / evidence |
|---|---|---|---|---|---|
| SC-C15-S | P-GRB | 0.462806408538 | 0.462806408538 | 1.11022302463e-16 | yes / CLEAN_NONZERO_CERTIFICATE |
| SC-C15-S | M-B | 0.462806408538 | 0.462806408538 | 5.55111512313e-17 | yes / CLEAN_NONZERO_CERTIFICATE |
| SC-B15-S | M-B | 0.94700467634 | 0.94700467634 | 1.55431223448e-15 | yes / CLEAN_NONZERO_CERTIFICATE |
| SC-B15-S | P-GRB | 0.94700467634 | 0.94700467634 | 8.881784197e-16 | yes / CLEAN_NONZERO_CERTIFICATE |
| SC-C15-B | P-GRB | 0.0067607771599 | 0.0067607771599 | -1.73472347598e-17 | yes / CLEAN_NONZERO_CERTIFICATE |
| SC-C15-B | M-B | 0.0067607771599 | 0.0067607771599 | 1.56125112838e-17 | yes / CLEAN_NONZERO_CERTIFICATE |
| SC-B20-S | M-B | 0.315389121053 | 0.315389106185 | 1.48683525492e-08 | yes / CLEAN_NONZERO_CERTIFICATE |
| SC-B20-S | P-GRB | 0.340249880289 | 0.209049610255 | 0.131200270034 | no / RELIABLE_POSITIVE_LB_OPEN |
| SC-C20-S | P-GRB | 0.280579842119 | 0.161051808386 | 0.119528033733 | no / RELIABLE_POSITIVE_LB_OPEN |
| SC-C20-S | M-B | 0.216430184034 | 0.207366884943 | 0.00906329909051 | no / RELIABLE_POSITIVE_LB_OPEN |
| SC-B20-U | M-B | 0.173761789119 | 0.173761789119 | 4.16333634234e-16 | yes / CLEAN_NONZERO_CERTIFICATE |
| SC-B20-U | P-GRB | 0.221769088877 | 0.0133968443922 | 0.208372244485 | no / RELIABLE_POSITIVE_LB_OPEN |

**Table SC4. All three clocks and certification costs (seconds).** Unknown certification time is shown as unknown, never replaced by the cap.

| Problem | Method | Cap | T_produce | T_verify | T_total | T_cert |
|---|---|---|---|---|---|---|
| SC-C15-S | P-GRB | 600 | 464.293373 | 6.0499092 | 470.343282 | 464.293373 |
| SC-C15-S | M-B | 600 | 41.5624639 | 1.1568301 | 42.719294 | 41.5624639 |
| SC-B15-S | M-B | 600 | 6.85265371 | 0.9079541 | 7.76060781 | 6.85265371 |
| SC-B15-S | P-GRB | 600 | 12.9786823 | 0.3325261 | 13.3112084 | 12.9786823 |
| SC-C15-B | P-GRB | 600 | 243.923894 | 0.3062603 | 244.230154 | 243.923894 |
| SC-C15-B | M-B | 600 | 44.0815251 | 0.9998445 | 45.0813696 | 44.0815251 |
| SC-B20-S | M-B | 1200 | 1069.00171 | 5.1936419 | 1074.19536 | 1069.00171 |
| SC-B20-S | P-GRB | 1200 | 1170.67074 | 23.7145084 | 1194.38525 | unknown |
| SC-C20-S | P-GRB | 1200 | 1173.28631 | 1.3285943 | 1174.6149 | unknown |
| SC-C20-S | M-B | 1200 | 1170.76201 | 4.9794916 | 1175.7415 | unknown |
| SC-B20-U | M-B | 1200 | 18.8735701 | 1.7018773 | 20.5754474 | 18.8735701 |
| SC-B20-U | P-GRB | 1200 | 1170.65504 | 1.7020734 | 1172.35712 | unknown |

**Table SC5. All six fixed paired outcomes.** Positive P-minus-M-B means less M-B production time. Single-sided categories are qualitative censored observations; failures are UNEVALUABLE rather than ordinary censoring.

| Problem | Main category | P minus M-B (s) | T_cert P/M-B | Time description | T_total P/M-B |
|---|---|---|---|---|---|
| SC-C15-S | BOTH_CERTIFIED | 422.730909 | 11.1709781 | MB_FASTER | 11.0100902 |
| SC-B15-S | BOTH_CERTIFIED | 6.12602859 | 1.89396442 | TIME_CLOSE | 1.71522756 |
| SC-C15-B | BOTH_CERTIFIED | 199.842369 | 5.5334722 | MB_FASTER | 5.41754069 |
| SC-B20-S | MB_ONLY_CERTIFIED | unknown | unknown | censored / no time ratio | unknown |
| SC-C20-S | BOTH_OPEN | unknown | unknown | censored / no time ratio | unknown |
| SC-B20-U | MB_ONLY_CERTIFIED | unknown | unknown | censored / no time ratio | unknown |

The fixed denominator is six problems and twelve arms. The pairs comprise 3 both-certified, 2 M-B-only, 0 P-only, 1 both-open and 0 unevaluable observations. M-B certifies 5/6 problems, including 5 nonzero and 0 zero objectives (0 startup-zero). Cold P certifies 3/6, including 3 nonzero and 0 zero objectives. A problem certified by both remains one problem. All certified physical values appear in Table SC3 and the summary, rather than being pooled across instances.

The BOTH_CERTIFIED intersection is {SC-C15-S, SC-B15-S, SC-C15-B}, size 3. Its observed speedup median is 5.5334722 and geometric mean 4.89200527. This conditional intersection does not estimate a whole-panel mean. The neighboring single-sided and open counts above remain part of the result; no cap filling, PAR-2 time, best-of-Seed or selected-problem average is used. T_total reverses the observed production-cost ordering on SC-C20-S. Both cost views are retained. On an open pair this is an output-cost observation, with no certification-time ordering or speedup inference; T_produce remains the main certification clock where qualified.

The evidence provides limited direct certification support on SC-C15-S, SC-C15-B, SC-B20-S, SC-B20-U. No problem meets the predeclared opposite material-time or P-only category. This does not turn the remaining problems into wins or establish a population runtime claim.

For the historical Round126 Seed0 cohort, the preregistered evidence classification is `LIMITED_DIRECT_CERTIFICATION_SUPPORT`, with w=4 and l=0, and the execution state is `COMPLETE`. These are bounded evidence labels, not acceptance standards or universal stability tests. The mathematical propositions establish representation, valid embedding and complete proof obligations under their premises; the measured speed/certification counts are empirical statements. This cold-baseline comparison measures the complete method and does not identify the separate causal contribution of cuts, AM or H. All four shortage cases remain visible, including the both-open SC-C20-S.

### 6.4 Seed confirmation with the original executable

Round127 v3 retains the six problems and original complete caps, changes Gurobi Seed from 0 to 1, and independently pays both original executable E0 arms. These are six problems under two Seeds, with no best-of-Seed selection. The new symmetric direction uses strict unrounded complete certification cost on both-certified pairs, or a qualified certificate against a valid open comparator. Both-open is undetermined. The historical materiality labels in Section 6.3 remain historical; Table R127-S reports the additional symmetric direction.

**Table R127-S. E0 Seed1 certification and transition from the same six Seed0 problems.** Times are complete production seconds, present only for qualified certificates.

| Problem | Seed0 status / direction | Seed1 status / direction | MB cert s | P cert s |
|---|---|---|---|---|
| SC-C15-S | Both certified / Win | Both certified / Win | 35.556 | 474.432 |
| SC-B15-S | Both certified / Win | Both certified / Win | 2.917 | 16.098 |
| SC-C15-B | Both certified / Win | Left only certified / Win | 65.243 | open |
| SC-B20-S | Left only certified / Win | Left only certified / Win | 975.413 | open |
| SC-C20-S | Both open / Undetermined | Both open / Undetermined | open | open |
| SC-B20-U | Left only certified / Win | Left only certified / Win | 15.455 | open |

Seed1 gives BOTH_CERTIFIED: 2, LEFT_ONLY_CERTIFIED: 3, BOTH_OPEN: 1. MB has 5 observed wins, 1 undetermined comparison. The conditional geometric mean P/MB ratio is 8.582 over 2 both-certified problems; this intersection does not represent the open runs.

The symmetric directions remain five wins and one undetermined comparison across the two Seeds. MB retains five certifications; P changes from three to two because SC-C15-B is valid open at Seed1. This preserves the limited confirmation while retaining the observed certification-status change.

### 6.5 Historical full-M-B tiny cases, repeated Seeds and absolute delay risk

Two once-generated geographic cases, SM-C8-S (V8/M1, shortage, Q=20, T=2400) and SM-B10-B (V10/M2, balanced, Q=15/25, T=3600), use E0 and a 90-second complete cap. Each uses Seeds 0,1,2 and two independent repetitions with reversed method order. No warmup or cross-run fleet, bound, Start or basis is shared. Twelve matched observations still concern two problems. The 10-second axis is an absolute observed-delay warning, not an internal method switch or statistical significance threshold. A smaller positive delay is still a direction Lose.

**Table R127-T. All tiny observations.** Delta is MB minus P, only when both certify. Displayed values are rounded; directions and the >=10-second test use original unrounded times.

| Problem | Seed / repeat | Status | Direction | MB cert s | P cert s | Delta s | Delta >=10 s |
|---|---|---|---|---|---|---|---|
| SM-C8-S | 0/1 | Both certified | Win | 3.407 | 5.661 | -2.254 | False |
| SM-C8-S | 0/2 | Both certified | Win | 3.723 | 5.230 | -1.507 | False |
| SM-C8-S | 1/1 | Both certified | Win | 2.832 | 3.893 | -1.061 | False |
| SM-C8-S | 1/2 | Both certified | Win | 2.469 | 4.214 | -1.745 | False |
| SM-C8-S | 2/1 | Both certified | Win | 2.650 | 4.693 | -2.043 | False |
| SM-C8-S | 2/2 | Both certified | Win | 2.975 | 4.335 | -1.360 | False |
| SM-B10-B | 0/1 | Both certified | Win | 0.814 | 0.869 | -0.055 | False |
| SM-B10-B | 0/2 | Both certified | Win | 0.481 | 1.170 | -0.689 | False |
| SM-B10-B | 1/1 | Both certified | Win | 0.534 | 1.069 | -0.536 | False |
| SM-B10-B | 1/2 | Both certified | Lose | 0.830 | 0.746 | 0.083 | False |
| SM-B10-B | 2/1 | Both certified | Lose | 0.900 | 0.824 | 0.077 | False |
| SM-B10-B | 2/2 | Both certified | Win | 0.471 | 1.134 | -0.663 | False |

SM-C8-S: 6/6 both-certified, 0 P-only, 0 observed delays at least 10 seconds. Median delta -1.626 seconds (all six pairs); repeated warning=no, all-six median warning=no. Per-Seed paired differences in seconds (repeat1, repeat2) are Seed0: (-2.253792, -1.507034); Seed1: (-1.060989, -1.744591); Seed2: (-2.043333, -1.360336). No cap is substituted for a missing certification time.

SM-B10-B: 6/6 both-certified, 0 P-only, 0 observed delays at least 10 seconds. Median delta -0.295 seconds (all six pairs); repeated warning=no, all-six median warning=no. Per-Seed paired differences in seconds (repeat1, repeat2) are Seed0: (-0.054771, -0.689162); Seed1: (-0.535587, +0.083291); Seed2: (+0.076776, -0.662628). No cap is substituted for a missing certification time.

The twelve observations give ten wins and two losses for MB. The losses are SM-B10-B Seed1/repeat2 (+0.083291 seconds) and Seed2/repeat1 (+0.076776 seconds). Thus no 10-second warning or P-only event was observed in this finite sample, while the small timing regressions remain explicit; the data do not establish their absence on other runs.

The SM-B10-B Seed0/repeat1 MB producer was sealed once at 0.814122304 seconds. A legacy comparison treated H's omitted empty vehicles and the committed explicit [0,0] unused routes as different. The correction normalizes only this handoff comparison after validating unique in-range vehicle identities; all physical/model/type/return/coverage checks remain. The original failed audit and each read-only recovery attempt are preserved. No native solve was repeated. Its actual verification span remains 650.179386500 seconds and total 650.993508804 seconds, including the recovery delay; these values are not substituted for its original certification production cost. Machine fees separately measure actual verification processes and retain both failed and successful parent segments.

### 6.6 Grouped mechanism diagnostics on three known roles

All core methods use the same frozen E1 executable and DLL and fresh Seed1 runs. Each MB-type arm independently pays the unchanged full H pipeline; its observed completion is reported below. P and PT remain cold compact runs without H. K1 is SC-C15-S (600 seconds), K2 SC-B20-S (1200), and K3 PB-B35-B (1200). E1 pays its own P and MB comparators; the same-named E0 observations are not reused. MR deletes only support cover/conditional handling and ordered compatible-supply rows C_R. MG deletes only explicit H-spread, variable-S/fixed centering, objective estimators and the dedicated W_SP block C_G. Existing derived inventory domains, exact Gini/state products, cap/floor, penalty-LB closure and required movement remain. MS2 deletes only three-station support, retaining pairs and ordered compatibility. Native Gurobi Cuts retain their original settings in all MB-type arms. These are deletion diagnostics around the same H and representation, not alternative proposed algorithms.

The required MD specification did not pass mathematical qualification. The retained numerical closure can emit an impossible row when $\gamma_L + \lambda P_{LB} \geq U_0 - 10^{-9}$. As already explained in Appendix A.5.6/A.7, that closure establishes no strict improver under the declared numerical policy; it need not retain an equal-U0 physical solution or its Start. A legal V2, zero-stock, T=60, handling-unit=120 boundary has forced Y=0, P_LB=2 and U0=0.3. Its equal-U0 empty fleet violates only the impossible row. Thus the requested inclusive root embedding and retained own-U0 Start cannot both be guaranteed while preserving that closure. Three MD arms were blocked before formal evaluation. This is a specification incompatibility, not evidence that interval control is beneficial or harmful, and does not by itself revoke the existing no-strict-improver proof policy. No replacement experiment was added.

K3 is a fresh 1200-second evaluation on a known role. It does not continue the historical 2400-second run or reuse that run as a comparator or presumed prefix.

**Table R127-C. All predeclared core comparisons.** Left is the evaluated method; ratios use the both-certified intersection only. PT aliases, when applicable, are marked and not counted independently.

| Role | Left / right | Status | Direction | Left cert s | Right cert s | Right / left |
|---|---|---|---|---|---|---|
| SC-C15-S | MB/P | Both certified | Win | 35.558 | 473.821 | 13.325 |
| SC-C15-S | MB/PT | Left only certified | Win | 35.558 | open | N/A |
| SC-C15-S | MB/MD | Unevaluable | Unevaluable | 35.558 | not run | N/A |
| SC-C15-S | MB/MR | Both certified | Lose | 35.558 | 35.251 | 0.991 |
| SC-C15-S | MB/MG | Both certified | Win | 35.558 | 40.673 | 1.144 |
| SC-C15-S | MB/MS2 | Both certified | Lose | 35.558 | 35.242 | 0.991 |
| SC-C15-S | PT/P | Right only certified | Lose | open | 473.821 | N/A |
| SC-B20-S | MB/P | Left only certified | Win | 966.283 | open | N/A |
| SC-B20-S | MB/PT | Left only certified | Win | 966.283 | open | N/A |
| SC-B20-S | MB/MD | Unevaluable | Unevaluable | 966.283 | not run | N/A |
| SC-B20-S | MB/MR | Both certified | Lose | 966.283 | 743.130 | 0.769 |
| SC-B20-S | MB/MG | Left only certified | Win | 966.283 | open | N/A |
| SC-B20-S | MB/MS2 | Both certified | Lose | 966.283 | 679.866 | 0.704 |
| SC-B20-S | PT/P | Both open | Undetermined | open | open | N/A |
| PB-B35-B | MB/P | Both open | Undetermined | open | open | N/A |
| PB-B35-B | MB/PT | Both open | Undetermined | open | open | N/A |
| PB-B35-B | MB/MD | Unevaluable | Unevaluable | open | not run | N/A |
| PB-B35-B | MB/MR | Both open | Undetermined | open | open | N/A |
| PB-B35-B | MB/MG | Both open | Undetermined | open | open | N/A |
| PB-B35-B | MB/MS2 | Both open | Undetermined | open | open | N/A |
| PB-B35-B | PT/P | Both open | Undetermined | open | open | N/A |

**Table R127-M. Original root exposure and cost.** Native seconds are diagnostics within the complete production clock. Unknown means unavailable from the original run; no extra Optimize was used.

| Role / mode | H complete | Root rows / cols | Root LP L | LP native s | MIP native s | Splits / target hits |
|---|---|---|---|---|---|---|
| SC-C15-S/P | N/A | 2199/976 | unknown | 0.000 | 472.444 | 0/0 |
| SC-C15-S/MR | True | 3904/2083 | 0.18563301 | 0.181 | 33.584 | 0/1 |
| SC-C15-S/MB | True | 5054/2083 | 0.18563301 | 0.205 | 33.683 | 0/1 |
| SC-C15-S/PT | N/A | 2199/976 | unknown | 0.000 | 569.926 | 0/0 |
| SC-C15-S/MG | True | 4523/2080 | 0.18563301 | 0.245 | 38.908 | 0/1 |
| SC-C15-S/MS2 | True | 4144/2083 | 0.18563301 | 0.169 | 33.498 | 0/1 |
| SC-B20-S/MG | True | 7967/3099 | 0.11721380 | 0.366 | 1167.708 | 0/0 |
| SC-B20-S/PT | N/A | 3629/1551 | unknown | 0.000 | 1169.890 | 0/0 |
| SC-B20-S/MS2 | True | 6643/3102 | 0.11721380 | 0.368 | 676.869 | 0/0 |
| SC-B20-S/MB | True | 8923/3102 | 0.11721380 | 0.401 | 962.891 | 0/0 |
| SC-B20-S/MR | True | 6223/3102 | 0.11721380 | 0.444 | 740.106 | 0/0 |
| SC-B20-S/P | N/A | 3629/1551 | unknown | 0.000 | 1169.891 | 0/0 |
| PB-B35-B/MS2 | True | 23500/9893 | 0.00000000 | 0.962 | 1163.970 | 0/0 |
| PB-B35-B/P | N/A | 13955/5434 | unknown | 0.000 | 1169.650 | 0/0 |
| PB-B35-B/MG | True | 40154/9890 | 0.00000000 | 1.802 | 1163.219 | 0/0 |
| PB-B35-B/PT | N/A | 13955/5434 | unknown | 0.000 | 1169.614 | 0/0 |
| PB-B35-B/MR | True | 21610/9893 | 0.00000000 | 1.273 | 1164.145 | 0/0 |
| PB-B35-B/MB | True | 43135/9893 | 0.00000000 | 1.718 | 1161.300 | 0/0 |

Actual family counts, nonzeros, H/fleet identities, native calls/nodes and first logged own-witness availability are in mechanism_costs.csv. Saved original root-LP points are tested against actual deleted rows without a new solve. For MG, W is eliminated using its serialized bounds and every W-containing inequality; an empty interval identifies only non-extendibility of that saved point. A nonempty interval or equal LP objective does not establish global redundancy. These root diagnostics do not establish certification-time causality. Full H/scope matching and actual row/column differences are recorded per role. Pair/triple totals 1120,2660,21420 are below the shared 50000 cap, so MS2 does not redistribute a binding truncation budget.

**Table R127-D. Component interpretation after checking actual exposure.** Eligible roles require complete identical H actions/fleet, matched root scope and actual removal of the declared rows. Timing directions in unexposed roles cannot establish a component effect.

| Comparison | Eligible known roles | Eligible directions | Disposition |
|---|---|---|---|
| MB/MD | none | none | Blocked before formal |
| MB/MR | SC-C15-S, SC-B20-S, PB-B35-B | 2 observed losses, 1 undetermined comparison | Reverse certification observations |
| MB/MG | SC-C15-S, SC-B20-S, PB-B35-B | 2 observed wins, 1 undetermined comparison | Direct certification increment in this panel |
| MB/MS2 | SC-C15-S, SC-B20-S, PB-B35-B | 2 observed losses, 1 undetermined comparison | Reverse certification observations |

All three observed roles have complete identical H actions/fleet, matched root scope and the prescribed actual row deletions. C_R and the three-station package show reverse certification-time observations: on K1 the deleted variants are faster by 0.307 and 0.316 seconds; on K2 MR certifies at 743.130 seconds and MS2 at 679.866, versus MB at 966.283. These losses remain losses under the symmetric rule. C_G has direct certification support within this panel: MB is faster than MG on K1 (35.558 versus 40.673 seconds), and only MB certifies on K2. All six executed K3 methods remain valid open, so that role supplies no certification-time direction. The mixed component evidence does not establish that every row package is necessary.

**Table R127-E. Checks at the saved original root-LP points.** Delta L is deleted-model minus full-model root LP bound. Missing points remain unknown. The W result fixes every surviving original variable and checks the actual serialized W block.

| Role / mode | Matched H / scope | Delta L | Deleted rows | Ordinary-point check | W extension |
|---|---|---|---|---|---|
| SC-C15-S/MR | True | -1.110e-16 | 1150 | No violation at saved point | Not applicable / unavailable |
| SC-C15-S/MG | True | -1.110e-16 | 531 | No violation at saved point | Extension interval nonempty at saved point |
| SC-C15-S/MS2 | True | -2.776e-17 | 910 | No violation at saved point | Not applicable / unavailable |
| SC-B20-S/MR | True | 1.388e-17 | 2700 | No violation at saved point | Not applicable / unavailable |
| SC-B20-S/MG | True | 2.776e-17 | 956 | No violation at saved point | Extension interval nonempty at saved point |
| SC-B20-S/MS2 | True | 8.327e-17 | 2280 | No violation at saved point | Not applicable / unavailable |
| PB-B35-B/MR | True | 0.000e+00 | 21525 | No violation at saved point | Not applicable / unavailable |
| PB-B35-B/MG | True | 0.000e+00 | 2981 | No violation at saved point | Extension interval nonempty at saved point |
| PB-B35-B/MS2 | True | 0.000e+00 | 19635 | No violation at saved point | Not applicable / unavailable |

At these nine saved points, the deleted ordinary rows have no detected violation, all three MG W-extension intervals are nonempty, and the root LP values agree within the recorded numerical tolerance. These observations explain no additional exclusion at the retained root optima; they do not show global redundancy or explain away the opposing certification-time results.

### 6.7 A bounded, separately trained compact baseline

The six fixed configurations vary only MIPFocus (0/1/2/3) and native Cuts (-1/2). S12, PB-B20-S and R98-C2 each use Seeds 2 and 3, totaling 36 E1 cold-P runs in a precommitted cyclic order. None is a core, main or tiny evaluation input. Actual model environments read back the settings after presets, with zero Starts. Selection maximizes qualified certifications N, then the minimum per-Seed count, then the declared censored penalty score and tie-breakers. A nondefault requires no loss in N or its per-Seed minimum, and either more certifications or at least a 10% score improvement when N ties above zero. The score's open-run value 1 is an artificial training penalty; it is never a certification time or speedup. Complete training clocks use the uniformly recorded parent admission marker; original sealed receipts and the initial import-prefix correction are preserved separately.

**Table R127-Q. All six predeclared configurations.** Each denominator is six training positions; the two per-Seed counts each have denominator three.

| Config | Focus / Cuts | N Seed2 / Seed3 | N / N_min | Score |
|---|---|---|---|---|
| q0 | 0/-1 | 2 / 1 | 3/1 | 0.742 |
| q1 | 1/-1 | 1 / 1 | 2/1 | 0.894 |
| q2 | 2/-1 | 2 / 2 | 4/2 | 0.657 |
| q3 | 3/-1 | 0 / 0 | 0/0 | 1.000 |
| q4 | 2/2 | 2 / 2 | 4/2 | 0.658 |
| q5 | 3/2 | 0 / 0 | 0/0 | 1.000 |

The committed selection is q2 (MIPFocus=2, Cuts=-1). This one parameter file is used unchanged in all three held-out known roles, including adverse outcomes. This is parameter holdout confirmation on known instances, not an independently sampled benchmark or an optimally tuned general solver. The main comparison MB/P changes the whole integrated method and cannot identify one component. Shared MB arms across seven core comparisons are not independent replications.

PT/P gives RIGHT_ONLY_CERTIFIED: 1, BOTH_OPEN: 2 among 3 nonalias typed entries: 1 observed loss, 2 undetermined comparisons, plus 0 aliases. The conditional right/left geometric mean is N/A over 0 both-certified roles. The per-role outcomes, including any reverse direction or valid open endpoint, remain in Table R127-C.

MB/PT gives LEFT_ONLY_CERTIFIED: 2, BOTH_OPEN: 1 among 3 nonalias typed entries: 2 observed wins, 1 undetermined comparison, plus 0 aliases. The conditional right/left geometric mean is N/A over 0 both-certified roles. The per-role outcomes, including any reverse direction or valid open endpoint, remain in Table R127-C.

Although q2 has four training certifications against q0's three, it does not strengthen the baseline on the three held-out roles: PT has no certification, while default P certifies K1; both remain open on K2/K3. MB certifies K1/K2 against valid-open PT and both are open on K3. This supports MB's bounded competitiveness against the unchanged selected file, with no both-certified MB/PT ratio and no claim about optimal Gurobi tuning.

### 6.8 Inclusive-root control and a bounded support Seed check (Round128)

Round128 follows the separately preserved PARTIAL Round127 study. It evaluates MD-R (inclusive-root control v1), a newly corrected research control, rather than relabelling the three unstarted historical MD positions. At the Round128 stage, frozen full M-B / GI-TBC was the sole proposed method; P-GRB remains the primary

benchmark. One separately sealed E2 build supplies all 16 formal arms: twelve Seed0 arms on the same three known research roles, followed by only MB/MS2 on K1 and K2 at Seed1. These yield 15 core typed comparisons and two additional support comparisons, not 17 independent trials or five distinct problems. Each Seed has one matched observation per role. E1 Seed1 times are historical context, never same-build comparators.

The old named penalty-closure shortcut can write $0 \leq -1$ at an equal-cutoff lower bound. A legal empty-stock two-station fleet at $T = 60$ has $F = U_0 = .3$ and violates that shortcut. On an actual interval $I = [a, b]$, suppose the cutoff domain supports $F(x) \geq L_a := a + \lambda \underline{P}$. The complement has $F(x) > c$, so coverage of the two domains gives the regional bound

$$B_I = \min\{L_a, c\}.$$

This minimum follows from coverage; it is not numerical clipping. If $L_a > c$, the inclusive cutoff domain is empty, subject to investigating any contradicting known witness first. At $L_a = c$, strict improvement is excluded but an equal-$c$ physical solution may remain. When $c - 10^{-9} \leq L_a < c$, the directly supported bound remains $L_a$; it is not exact $c$ and does not rule out every arbitrarily small improvement. The shortcut is therefore an encoding of proof/control closure rather than an ordinary valid inequality for every non-strict cutoff solution. This finite counterexample does not itself revoke every frozen M-B numerical result. In the three saved R127 core MB call histories, none of the nine actual matrices contains the shortcut.

MD-R suppresses only that semantically marked pseudo-infeasibility row, in both shared and fallback emission paths. It retains the physical model, all valid domains and strengthening, $P \geq \underline{P}$, the same complete independently paid H, and fixed own-$U_0$ root $I_0 = [0, \min\{U_0, (n-1)/n\}]$ with $F_{model} \leq U_0$ and the original $n = 1$ convention. Before LP, its own entire physical fleet is mapped and checked against actual columns, original types, bounds, every row and objective. It then pays one complete root LP, restores integer VTypes and submits the complete legal own Start to one complete root MIP. It has no child LP, target handoff, split or root rebuild. The original startup-zero policy is retained. A numerical relevance threshold cannot waive the nonzero two-call obligation; TIME_LIMIT/INTERRUPTED cannot become native optimality. Raw signed-gap and full physical/source/return/coverage checks remain, without a new analytic certificate or rational error claim. Finite native boundary, nonempty-fleet, open-return and full E1/E2 model/H/parameter bridges precede the E2 seal.

**Table R128.1. All core Seed0 certification times (seconds).** Open entries have null T_cert in the authority, not the cap or a native Runtime substituted as certification time.

| Known role, Seed0 | P T_cert | MB T_cert | MD-R T_cert | MS2 T_cert | MB splits |
|---|---|---|---|---|---|
| SC-C15-S | 464.967 | 41.554 | 40.670 | 33.427 | 0 |
| SC-B20-S | open | 1066.781 | 1067.337 | 991.422 | 0 |
| PB-B35-B | open | open | open | open | 0 |

Every formal arm independently pays its entire method. T_produce includes admission, H where applicable, modeling, LP/MIP, Start checks and required output; T_verify and T_total remain separate. Only final qualified certificates receive T_cert=T_produce. Actual same-H action/fleet, initial U0, root scope and complete matrix differences are retained in the current evidence. All prescribed component comparisons pass that finite eligibility check. The per-arm call counts distinguish actual splits, child probes, target calls and terminal calls. An observed absence of splits restricts the outer-control interpretation to the exposed probing, target handoff and scheduling work; it cannot demonstrate multi-leaf decomposition speedup.

**Table R128.2. The complete same-E2 two-Seed support observations (seconds).** A positive MS2 saving means MB takes longer. Directions use unrounded complete certification time and certification priority.

| Known role | Seed | MB T_cert | MS2 T_cert | MS2 saving (s) | MB/MS2 direction |
|---|---|---|---|---|---|
| SC-C15-S | 0 | 41.554 | 33.427 | 8.127 | Lose |
| SC-C15-S | 1 | 35.497 | 35.737 | -0.241 | Win |
| SC-B20-S | 0 | 1066.781 | 991.422 | 75.359 | Lose |
| SC-B20-S | 1 | 965.849 | 680.009 | 285.841 | Lose |

MS2 deletes only three-station support cover/conditional-handling rows. Pair rows, ordered supply, enumeration order and the shared 50000 cap remain; the three roles' 1120, 2660 and 21420 candidates do not bind that cap. K3's endpoint and certification risks remain visible in Table R128.1 and the complete endpoint authority. Deleting rows need not improve MIP work monotonically, and the same native Seed does not force identical search paths. Fixed modeling/Start cost and actual MIP work are reported separately rather than attributing every elapsed-time difference to strengthening. Two Seed observations on known roles do not establish general statistical stability.

All three Seed0 MB arms actually pay one root LP, two child LPs and one terminal MIP; only K1 adds one child-bound-target MIP. No NEXT call or split occurs. Each MD-R pays exactly its root LP and root MIP. At K2 Seed0, MB/MS2 native MIP seconds are 1063.046/988.337, work is 2204.035/1952.267, and nodes are 65227/70275. Model generation plus native read costs are 1.100/0.807 seconds; Start checks/submission cost 0.0057/0.0055 seconds. At K2 Seed1, MB/MS2 native MIP seconds are 962.110/676.970, work is 1974.645/1320.987, and nodes are 55838/51800. Model generation plus native read costs are 1.106/0.809 seconds; Start checks/submission cost 0.0065/0.0056 seconds. Thus the K2 difference mainly accompanies changed MIP work rather than fixed row-writing cost; even the node-count direction differs between Seeds. The observed MB/MS2 root LP bounds agree within 1e-12 on every role/Seed. This finite root observation does not prove global redundancy of the deleted rows. Native Runtime is nested solver work, not certification time. Missing compact modeling subclocks remain explicitly unavailable rather than inferred from the clock remainder.

The complete core comparisons are:

| Comparison, core Seed0 | Typed statuses | Directions |
|---|---|---|
| MB/P | both certified: 1; left only certified: 1; both open: 1 | Win: 2; UNDETERMINED: 1 |
| MB/MD-R | both certified: 2; both open: 1 | Lose: 1; Win: 1; UNDETERMINED: 1 |
| MB/MS2 | both certified: 2; both open: 1 | Lose: 2; UNDETERMINED: 1 |
| MD-R/P | both certified: 1; left only certified: 1; both open: 1 | Win: 2; UNDETERMINED: 1 |
| MS2/P | both certified: 1; left only certified: 1; both open: 1 | Win: 2; UNDETERMINED: 1 |

The default compact benchmark is newly paid in E2 at Seed0. No old E1 P time is borrowed for the additional support observations. The earlier six-configuration training choice q2 and its adverse held-out PT result remain in Section 6.7: PT certifies 0/3 versus default P's 1/3 in that separate E1 study. Round128 adds no tuning, tiny rerun, evaluation input or longer window, and does not establish superiority over the strongest possible tuned compact baseline. Artificial boundary fixtures are separately identified qualification work.

The bounded stage decision is `BOUNDED_SIMPLIFICATION_SIGNAL`. Certification risk remains unresolved on K3: all four core methods are qualified open there, so neither a certification loss nor uniform simplification safety is established. K1 is mixed across Seeds: MS2 saves 8.127 seconds at Seed0 but is 0.241 seconds slower at Seed1. Against that retained mixed evidence, K2 supplies the bounded repeated signal: both methods certify at each Seed, and MS2 saves 75.359 seconds at Seed0 (7.1%) and 285.841 seconds at Seed1 (29.6%). Those are two matched observations on one known role, not population evidence. MD-R establishes a legal control, not a performance victory: it is 0.883 seconds faster than MB on K1, 0.556 seconds slower on K2, and both are open on K3; no MB split occurs. That stage retains two MB/P Wins and one BOTH_OPEN comparison, with no claimed advantage over the strongest possible tuned compact baseline. At that stage, the K2 support signal warranted one future uniform confirmation design and full M-B remained the paper method. Section 6.9 records that completed R129 selection; Section 6.10 adopts S2 and gives the distinct current attribution study.

The historical Round128 next-priority recommendation, subsequently executed in Section 6.9, was substantive review of one then-unexecuted uniform simplification confirmation design: fix a single candidate that removes triple support for every evaluated case while retaining the frozen H, representation, pair support, ordered supply, C_G, controller, numerical policy and default parameters. Any later authorization should freeze its matched

certification-first panel before execution, retain every open or adverse result, independently pay complete methods, and forbid role-specific switching or retuning. No such confirmation, adoption, additional Seed/input or longer window is executed in Round128. Until that decision, retain frozen M-B / GI-TBC, P-GRB as the primary benchmark and the historical ENS-C default; restrict the D claim to the actual probing/target/scheduling exposure and do not claim multi-leaf speedup.

### 6.9 Prospective confirmation of a uniform pair-support candidate (Round129)

This section preserves the R129 selection-time record: the four inputs were new at that stage, MS2 was a candidate, and the five conditions passed. Section 6.10 adopts that version and treats the reused roles as known attribution diagnostics. The E2 numbers, raw-field meanings and original selection criteria below retain their historical identities.

Round128 supplied a bounded development signal on known K2/SC-B20-S, while retaining the K1 timing reversal and open K3. Round129 is a separately preregistered finite confirmation of a single already implemented candidate. MS2 omits three-station support on every input and retains the full frozen H, R, pair support, ordered supply, C_G, controller and numerical policy. No MD-R variant is run and no MD-R pseudo-row suppression is applied to MB/MS2. P remains the original independently paid cold compact Gurobi baseline. All thirty arms use the original same E2 PE and Gurobi 13.0.2 DLL; the only varied native parameter is the prespecified Seed0/1.

The method remains a tailored branch-and-cut framework with Gini-interval decomposition. Its project strengthening is generated statically for each actual scoped model; Gurobi performs the native branching, presolve and native cut generation. The framework name or abbreviation supplies no separate novelty claim.

The four new problems are UC-C15-S (CITI, compact/r1, shortage, V15/M2, Q=[30,30], T=3600, cap600), UC-B20-B (CABI, regional/r2, balanced, V20/M2, Q=[20,40], T=7200, cap1200), UC-C20-U (CITI, regional/r1, surplus, V20/M2, Q=[30,30], T=7200, cap1200), and UC-B30-S (CABI, compact/r2, shortage, V30/M3, Q=[25,25,25], T=7200, cap1200). The known PB-C20-B protection input (V20/M2, Q=[30,30], T=3600, cap900) is reported separately and cannot supply a prospective gain. These are draws from two existing source tables, not new cities, IID samples or a full factorial design. Seeds add observations of the same problems.

The recipe, dependencies, all ten method orders, caps and five candidate rules were committed before once-only offline generation. Original R126/R122/R109/R57 anchor, tie, target, inventory, weight, depot and travel-time rules are retained. No H/LP/MIP or difficulty/reachability filter selected the data. All lawful station overlap is retained. Shortage arithmetic supplies positive necessary objective floors of 12/275 and 464877/2500000 for UC-C15-S and UC-B30-S respectively, excluding the depot; these decimal-data checks are not injected into any model, Start or reported native lower bound. Balanced/surplus outcomes remain regardless of zero objectives or backend exposure.

Each method independently pays its complete H/admission/model/native/online-output workflow under the original complete deadline, shutdown reserve and numerical settings. No solution, U, Start, basis, tree or cache is shared. The opposite Seed reverses each role's method order. Thirty arms form ten matched groups and thirty typed shared-data comparisons (MS2/P, MS2/MB, MB/P), not thirty independent trials. The new layer contains 24 arms/eight groups/24 comparisons; protection contains six/two/six.

**Table R129.1. Every fixed complete-method observation (seconds).** C denotes a qualified certificate and its complete T_cert=T_produce. O denotes a qualified open result and observed T_produce in parentheses, with T_cert=null. U would identify unqualified evidence. The last two rows are the known protection role.

| Role | Seed | P | MB | MS2 |
|---|---|---|---|---|
| UC-C15-S | 0 | O (571.153) | O (570.623) | O (570.589) |
| UC-C15-S | 1 | O (570.546) | O (570.698) | O (570.797) |

| Role | Seed | P | MB | MS2 |
|---|---|---|---|---|
| UC-B20-B | 0 | C 316.976 | C 4.018 | C 3.631 |
| UC-B20-B | 1 | C 317.545 | C 3.778 | C 3.594 |
| UC-C20-U | 0 | O (1170.625) | C 701.525 | C 554.378 |
| UC-C20-U | 1 | O (1170.568) | C 632.520 | C 481.105 |
| UC-B30-S | 0 | O (1171.634) | O (1170.425) | O (1170.476) |
| UC-B30-S | 1 | O (1170.482) | O (1170.431) | O (1170.723) |
| PB-C20-B | 0 | O (870.586) | C 393.721 | C 369.589 |
| PB-C20-B | 1 | O (870.721) | C 413.767 | C 294.156 |

**Table R129.2. New-input certification counts, keeping Seeds separate (denominator four per Seed).**

| Seed | P certified | MB certified | MS2 certified |
|---|---|---|---|
| 0 | 1 | 2 | 2 |
| 1 | 1 | 2 | 2 |

The numbers of new problems certified at both Seeds are P: 1, MB: 2, MS2: 2. At Seed0 the P-certified/method-open counts are MB: 0, MS2: 0; at Seed1 they are MB: 0, MS2: 0. Full per-observation U, qualified L, raw signed gap, all three clocks and every reverse comparison remain in the accompanying arms/pairs authorities and Supplement S. Open endpoints are not certification Wins, and open times are not replaced by caps.

**Table R129.3. All matched MS2/MB directions, including the separate protection role.** Savings are T_MB-T_MS2 only when both certify; N/A carries no unobserved speed ratio.

| Role | Seed | MS2/MB status | Direction | MS2 saving (s) |
|---|---|---|---|---|
| UC-C15-S | 0 | BOTH_OPEN | UNDETERMINED | N/A |
| UC-C15-S | 1 | BOTH_OPEN | UNDETERMINED | N/A |
| UC-B20-B | 0 | BOTH_CERTIFIED | Win | 0.387 |
| UC-B20-B | 1 | BOTH_CERTIFIED | Win | 0.185 |
| UC-C20-U | 0 | BOTH_CERTIFIED | Win | 147.147 |
| UC-C20-U | 1 | BOTH_CERTIFIED | Win | 151.415 |
| UC-B30-S | 0 | BOTH_OPEN | UNDETERMINED | N/A |
| UC-B30-S | 1 | BOTH_OPEN | UNDETERMINED | N/A |
| PB-C20-B | 0 | BOTH_CERTIFIED | Win | 24.133 |
| PB-C20-B | 1 | BOTH_CERTIFIED | Win | 119.611 |

Component attribution is conditional on actual complete H/actions/fleet, initial U and root scope, native settings, and duplicate-preserving full matrix/column/type/bound/objective equality after the triple deletion. Every actual root is checked after its own H. Subsequent legitimate own-UB trajectories can differ. Missing roots, incomplete H or unmatched scopes restrict the observation to a complete-method comparison. The all-candidate counts (1120, 2660, 2660, 13485 and 2660) are below the original shared cap50000, so this panel does not reassign a binding enumeration budget. Actual cover/conditional counts and split/target, LP/MIP work/nodes, model/Start and witness-availability diagnostics are retained in the mechanism evidence. An absent split neither invalidates completeness nor demonstrates multi-leaf speedup. No new D control effect was measured in R129; the final pair-only outer-control comparison is reported separately in Section 6.10.

All ten matched MB/MS2 observations have the same complete H actions and fleets, initial U, root scope and settings. After triple deletion, the remaining rows and all columns, types, bounds and objective match exactly. The actual triple cover/conditional counts removed at each Seed are 0/910 (UC-C15-S), 2204/76 (UC-B20-B), 0/2280 (UC-C20-U), 3969/8211 (UC-B30-S), and 0/2280 (PB-C20-B). UC-B20-B naturally has three splits in each MB and MS2 arm at both Seeds; every other MB/MS2 arm has zero splits. This new exposure coexists with the historical R128 no-split observations. Without its own D control, R129 did not identify a net decomposition speed benefit; the distinct final-version control is in Section 6.10. At UC-C20-U Seed0, MB and MS2

respectively use 698.099 and 551.181 native MIP seconds, 1412.023 and 1058.789 work units, and 49499 and 54557 nodes. Model-generation plus native-read seconds are 1.145 and 0.850. At UC-C20-U Seed1, MB and MS2 respectively use 629.024 and 478.057 native MIP seconds, 1272.487 and 911.262 work units, and 41225 and 45006 nodes. Model-generation plus native-read seconds are 1.150 and 0.849. The C20 saving accompanies lower MIP time/work despite more MS2 nodes at both Seeds; fixed row-writing/read savings are about 0.3 seconds. All observed matched root LP bounds agree within 1e-12. This finite agreement does not prove global redundancy of triple rows. Native costs remain nested explanatory measurements rather than T_cert.

UC-C15-S and UC-B30-S remain open for all three methods at both Seeds. UC-B30-S has an open-endpoint reversal: MS2 and MB U are respectively 0.381251 and 0.429057 at Seed0 but 0.429057 and 0.365930 at Seed1. UC-C15-S Seed1 also has slightly worse MS2 U than MB. Both-open comparisons remain UNDETERMINED with no certification ratio or Win. The three tiny negative signed gaps and all fifteen Seed1 raw top-level LB=0/certificate=false records are preserved. Qualified final endpoints use the original source/return/coverage and numerical policy; no raw field, LB or gap is clipped or rewritten.

In the new layer, MS2/P and MB/P each have two both-certified observations, two method-only certificates and four both-open observations. Their certification-time ratio geometric means (right/left) are 87.827903 and 81.425709 respectively, solely on UC-B20-B at the two Seeds. MS2/MB has four both-certified observations (UC-B20-B and UC-C20-U at both Seeds) and four both-open observations; its right/left geometric mean is 1.179515 on those four observations only. In protection, MS2/P and MB/P are method-only certified at both Seeds and have no ratio; MS2/MB is both-certified at both Seeds with a 1.224121 right/left geometric mean. These intersections do not represent the full panel.

For a fixed valid scope with identical domains and objective, deleting valid triple rows preserves the physical integer optimum and can only weaken or preserve the mathematical optimal LP lower bound. No theorem makes deletion faster. Native Runtime, work, nodes, root LP time, fewer rows or a smaller open gap do not replace complete certification time. Ratios and their geometric means are restricted to explicitly labelled both-certified intersections.

The fixed research decision magnitude requires a certified MS2 saving of at least max(10 seconds,10% of MB time), or MS2-only certification. A material time cost uses the symmetric faster-method reference; these labels do not alter strict unrounded Win/Lose or claim statistical significance. Advancement also requires no additional per-Seed certification loss or P-only risk, protection of every current MB certificate, no two-Seed large protection slowdown, one newly exposed role supporting MS2/MB materially at both Seeds, and one new role giving direct MS2/P certification Wins at both Seeds. Development K2, protection, subsecond directions and open gaps cannot substitute for the new signal.

| Fixed condition | Requirement | Satisfied |
|---|---|---|
| 1 | Retain per-Seed and both-Seed new-problem certification counts relative to MB | True |
| 2 | Add no per-Seed P-certified/method-open risk relative to MB | True |
| 3 | Retain every protected MB certificate and avoid a large slowdown at both Seeds | True |
| 4 | One new input has eligible triple deletion and material MS2/MB support at both Seeds | True |
| 5 | One new input gives direct MS2/P certification Wins at both Seeds | True |

The finite scientific recommendation is `RECOMMEND_UNIFORM_MS2_AS_PAPER_CANDIDATE`. The five fixed conditions support recommending uniform MS2 as the single method candidate for the next paper revision. This was the recommendation at the R129 stage. R130 now adopts uniform MS2 as the sole method; historical full-M-B

numbers retain their original labels. The exact mode mapping is original E2 `--round127-mode mb` to original E2 `--round127-mode ms2`; only three-station cover/conditional support is omitted uniformly. No old MB number is relabelled MS2. R130 resolves this version map by labelling earlier full-M-B tables as predecessor evidence and restricting current main-method performance to qualified actual MS2 observations. It does not relabel old results or rerun the larger historical panels. The exact evidence layers and input/Seed/program/clock identities are listed in paper_version_manifest.json and evidence_migration.csv. This small prospective panel does not establish V50/V60 safety. The frozen paper method, P benchmark, historical ENS-C default, R127 PT negative findings and all R128 reverse/open/no-split facts remain identifiable. No extra confirmation is automatically appended.

### 6.10 Fixed-version same-start and outer-control attribution (Round130)

Round129's five conditions passed before this study. The user now adopts **GI-TBC (uniform pair-support implementation, MS2)** as the sole proposed method, short ID **S2**. Its algorithm identity is the original E2 `--round127-mode ms2`: pair support is enabled uniformly and triple support is never emitted. H, representation, ordered supply, C_G, A/B, Gini control, numerical tolerances and default native parameters are unchanged. Historical full M-B measurements retain their three-station identity; ENS-C remains the repository default and is not the paper reproduction entry. This is a version adoption, not a third algorithm or another candidate-selection round.

The fixed panel contains UC-B20-B, UC-C20-U and UC-B30-S from R129, with their original input bytes, Q vectors and physical T=7200 seconds. These are deliberately known attribution roles: a previously splitting nonzero-certified role, a further nonzero-certified risk role, and an open shortage role. They are not unseen prospective validation inputs, independent replications of R129 or a difficulty-blind sample. Balanced, surplus and shortage refer to total inventory scenarios, not identical local station regimes. No input, Seed, parameter, tuning file or cap was added after outcomes.

Each method independently pays its complete workflow under the same 1200-second cap, original 30-second shutdown reserve, six-second native offset and two-second hard-stop offset. Threads=MIPThreads=1, affinity mask4, H_seed=20260626, per-group Gurobi Seed0/1, PresolveAuto, MIPFocus=0, Cuts=-1 and requested zero relative/absolute MIP gaps are retained. Each native call receives only the remaining complete deadline. Three problems, two Seeds and four methods give 24 serial arms in six matched groups. The five prespecified comparisons in each group yield 30 typed shared-data entries, not 30 independent trials. No arm uses another arm's fleet, upper bound, Start, basis, tree or cache.

For each role, the Seed1 method order reverses Seed0. This limited counterbalancing does not remove all host-load noise or turn two paths into a statistical robustness study.

PS uses the existing independently paid H-to-original-compact entry, not P-tuned. Its complete Start mapper checks every actual column/type/bound/row and the original objective, submits and reads back the full vector, and leaves the mathematical model unchanged. P stays cold. A Start that does not create a new native incumbent is not automatically invalid: native heuristics can already have an equally good or better solution. Actual submission/readback and native adoption are reported separately. H and mapping costs are included in PS production time.

The inherited verified-zero startup scope is retained for PS/S2/S2R. Such a qualified startup-zero outcome has no backend exposure and is not same-H backend attribution; no new zero shortcut is introduced. The normal nonzero PS path has one original-compact MIP. Every formal role here has nonzero own H.

S2R uses the off-by-default unique `s2-root` mode, retaining pair support and all valid root strengthening. Its own H defines the inclusive root

$$I_0 = [0, \min\{U_0, (n-1)/n\}], \qquad F_{\text{model}} \le U_0.$$

It suppresses only the semantically identified interval-no-improver proof-closure row, validates its own full Start against actual original integer types before LP relaxation, pays one complete root LP, restores the integer types, and pays one root MIP with one submitted own Start. No child LP, NEXT/AM target, split, epoch or restart follows. Nonzero chains cannot skip a call merely through a relevance tolerance. Native TIME_ LIMIT/INTERRUPTED remains open; a near-equal numerical bound does not replace the required complete native optimal-root return. An infeasible root despite its legal own witness is a consistency issue, not a shortcut certificate. All costs share the deadline.

For a physical solution, nonnegative penalties give G<=F and nonnegative ratios give G<=(n-1)/n. Thus every physical F<=U0 solution lies in the root; the cutoff complement cannot improve the own verified U0. A qualified root bound L combines with that complement as min(L,U0) by domain coverage. This identity does not excuse a bound contradicting its own legal witness. Equality of an analytic regional floor and cutoff excludes strict improvement only; a floor slightly below the cutoff retains its own value under the original numerical policy. The ordinary S2 proof shortcut is unchanged.

**Table R130.1. Every complete-method observation (seconds).** C is qualified T_cert=T_produce; O gives observed T_produce with T_cert=null. Open times are never replaced by caps.

| Known role / Seed | P | PS | S2 | S2R |
|---|---|---|---|---|
| UC-B20-B / 0 | C 318.453 | C 237.274 | C 3.559 | C 1.398 |
| UC-B20-B / 1 | C 316.817 | C 127.117 | C 3.504 | C 1.586 |
| UC-C20-U / 0 | O (1170.579) | O (1170.713) | C 554.709 | C 556.836 |
| UC-C20-U / 1 | O (1170.511) | O (1170.560) | C 481.820 | C 480.939 |
| UC-B30-S / 0 | O (1171.180) | O (1170.428) | O (1170.759) | O (1170.433) |
| UC-B30-S / 1 | O (1170.437) | O (1170.680) | O (1170.448) | O (1170.396) |

**Table R130.2. All primary, same-start and outer-control directions.** Win/Lose uses unrounded certificate time or a one-sided qualified certificate. Both open is UNDETERMINED. The remaining two fixed comparison types and all exact differences are retained in Supplement S and pairs.csv.

| Known role / Seed | S2/P | S2/PS | S2/S2R |
|---|---|---|---|
| UC-B20-B / 0 | Win | Win | Lose |
| UC-B20-B / 1 | Win | Win | Lose |
| UC-C20-U / 0 | Win | Win | Win |
| UC-C20-U / 1 | Win | Win | Lose |
| UC-B30-S / 0 | UNDETERMINED | UNDETERMINED | UNDETERMINED |
| UC-B30-S / 1 | UNDETERMINED | UNDETERMINED | UNDETERMINED |

Actual complete-H/action/capacity-assigned fleet equality across all same-H comparisons is True. Actual effective-root identity for S2/S2R is True. The matrix checks retain duplicate rows and compare complete objectives, column order/types/bounds, settings and all remaining rows. If a normal S2 root contains the omitted semantic shortcut or its H/scope differs, that result is retained as a complete-method comparison with restricted attribution. Complete per-role facts are in actual_method_differences.json and same_H.csv.

At B20 Seed0, S2/PS/S2R complete certification costs are 3.559/237.274/1.398 seconds; S2 produces 3 natural splits and S2R none. At B20 Seed1, S2/PS/S2R complete certification costs are 3.504/127.117/1.586 seconds; S2 produces 3 natural splits and S2R none. The H witness already attains the eventual B20 objective, 0.5773330699702928, before proof. It is not itself an optimality certificate. The original R129 coverage example is kept separately and is never injected into a formal run. The new actual observations therefore separate a

large backend/representation increment from any net contribution of the outer package; split counts alone cannot establish speedup.

UC-B20-B (same H backend): REPEATED_DIRECT_INCREMENT. UC-C20-U (same H backend): REPEATED_DIRECT_INCREMENT. UC-B30-S (same H backend): UNDETERMINED. UC-B20-B (outer control): CLOSE_AT_BOTH_SEEDS. UC-C20-U (outer control): CLOSE_AT_BOTH_SEEDS. UC-B30-S (outer control): UNDETERMINED.

The symmetric descriptive materiality rule is slower-faster >= max(10 seconds,0.10*faster), separate from strict Win/Lose and from one-sided certificates. It is not a statistical test or algorithm switch. Native LP/MIP seconds, Work, nodes, model/Start costs, target/split exposure, closure sources and first retained own final-U witnesses are explanatory measurements. Missing subclocks remain unknown; Work is not used as a stopping budget. Ratios and geometric means use only explicit both-certified intersections.

Both C20 Seed1 S2 and S2R retain the qualified signed gap U-L=-1.4988010832439613e-15, with L unchanged under the original signed 1e-7 numerical window. Original raw top-level claims remain distinct from qualified source/return/coverage endpoints. No bound clipping, new tolerance or rational-exactness claim is introduced. Cross-arm physical witnesses are checked only for contradictions and never transferred as free bounds or certificates.

**Table R130.3. Contribution interpretation matrix.**

| Comparison | Actual question | Scope boundary |
|---|---|---|
| S2/P | complete adopted method versus cold compact | all integrated differences; primary benchmark |
| PS/P | net independently paid H and full Start integration | neither a free solution nor guaranteed improvement |
| S2/PS | same-H complete backend/representation/control increment | does not isolate an individual cut or threshold |
| S2/S2R | same effective root, exposed outer package | probing, targets, domain-dependent rows, scheduling and splits together |
| S2R/PS | same-H single-root strengthened representation versus original compact | representation, LP preprocessing and cutoff domain remain coupled |

Execution is **COMPLETE**. The finite claim decision is **`SELECTED_MS2_WITH_SCOPED_OR_MIXED_ATTRIBUTION`**. The adopted S2 has repeated direct same-H backend support: on B20 it saves 233.71554430003744 and 123.61303300003055 seconds against independently paid PS, and on C20 it certifies at both Seeds while PS remains open. The outer package has no repeated certificate or material time increment in this panel. B20 exposes three natural splits and one reached partial target at each Seed, but S2R certifies in 1.3975899000652134 and 1.5858365014428273 seconds versus S2 in 3.5585215999744833 and 3.5044260000577196. C20 has no split/target exposure; its S2/S2R differences are also time-close. All four both-certified S2/S2R observations are TIME_CLOSE; their strict three Lose and one Win directions remain. All four methods stay open on B30 at both Seeds. No repeated material reverse reference role or proof/identity problem is observed. Retain S2 as the sole paper method and narrow decomposition-speed claims to this limited evidence; no additional experiment is required or authorized. This does not reconsider R129's passed selection or combine per-instance winners. PS remains a useful reference without needing to dominate P. This panel supplies no new single-cut C_G or C_R ablation for final S2, no V50/V60 validation, and no population robustness claim. R127's adverse selected P-tuned result, R128's timing reversal/no-split facts, R129's open C15/B30 results and its Seed-dependent endpoint reversals remain visible. The fixed attribution study ends without additional experiments.

### 6.11 Primary frozen-S2 certification benchmark (Round131 A)

The algorithm adopted in R130 is fixed. This measures the entire original six-problem SC set at Seeds 0 and 1 with the same frozen E3 S2 and cold P: 24 fresh independently paid arms and twelve matched observations. These are exposed known inputs, not new or unseen validation. No old M-B time is used as an S2 measurement. Each method pays admission, its own H if applicable, modeling, every LP/MIP/Start check and required outputs under one complete deadline. Full argv, native readbacks, physical/source-return-cover checks and three clocks remain in the reproducibility package. Caps are 600 seconds for V15 and 1200 for V20, with the original 30-second reserve and 6/2-second native/hard-stop offsets. Both-open results are UNDETERMINED; unqualified evidence is UNEVALUABLE; unknown certification times stay null.

**Table R131.A. All twelve main S2/P observations.** Time is complete qualified certification time in seconds. Delta and ratio exist only for both-certified pairs; exact authorities and all U/L/three-clock fields are in Supplement S.

| Known problem / Seed | P T_cert (s) | S2 T_cert (s) | Class | S2/P direction | S2 - P (s) | P / S2 |
|---|---|---|---|---|---|---|
| SC-C15-S / S0 | 460.809 | 33.167 | BOTH_CERTIFIED | Win | -427.642 | 13.894 |
| SC-C15-S / S1 | 469.536 | 35.538 | BOTH_CERTIFIED | Win | -433.998 | 13.212 |
| SC-B15-S / S0 | 12.875 | 2.851 | BOTH_CERTIFIED | Win | -10.024 | 4.515 |
| SC-B15-S / S1 | 16.181 | 2.245 | BOTH_CERTIFIED | Win | -13.936 | 7.208 |
| SC-C15-B / S0 | 242.256 | 57.928 | BOTH_CERTIFIED | Win | -184.328 | 4.182 |
| SC-C15-B / S1 | open | 33.444 | S2_ONLY_ CERTIFIED | Win | null | null |
| SC-B20-S / S0 | open | 982.279 | S2_ONLY_ CERTIFIED | Win | null | null |
| SC-B20-S / S1 | open | 676.371 | S2_ONLY_ CERTIFIED | Win | null | null |
| SC-C20-S / S0 | open | open | BOTH_OPEN | UNDETERMINED | null | null |
| SC-C20-S / S1 | open | open | BOTH_OPEN | UNDETERMINED | null | null |
| SC-B20-U / S0 | open | 26.893 | S2_ONLY_ CERTIFIED | Win | null | null |
| SC-B20-U / S1 | open | 15.998 | S2_ONLY_ CERTIFIED | Win | null | null |

On the six known SC problems at two Seeds (12 matched observations), S2 and P certify 10 and 5 observations, respectively; the strict S2/P directions are 10 Win, 0 Lose, 0 Tie and 2 undetermined. The tiny layer records 9 strict Win, 3 strict Lose and 0 Tie across twelve matched observations, with 0 observed >=10-second delays and 0 P-only certification risks. At V35, S2 and P have 0 and 0 nonzero certificates across two Seed0 observations, with 0 P-only cases; both-open cases have unknown final certification-time ordering. The main conditional P/S2 geometric mean is 7.578 on the explicitly listed 5 both-certified observations only. Materiality describes magnitude, not direction or a method-selection gate. All losses remain in the table.

### 6.12 Absolute final-S2 tiny overhead (Round131 B)

The two original V8/V10 inputs each use three Seeds and two opposite-order repeats, for 24 arms and twelve comparisons under complete 90-second contracts. No free warm-up, shared H/Start/tree or extra Seed is used. The six repeated observations per problem measure absolute overhead and are not weighted into A. A strict subsecond loss is still Lose; the 10-second warning is practical, not a theorem about noise. The legal empty-fleet/structurally easy SM-B10-B remains in the table and does not stand for all small-instance difficulty.

**Table R131.B. All twelve tiny matched observations.** R denotes a full independent repeat at the same Seed. Missing certification times are null in the authority.

| Known problem / Seed / repeat | P T_cert (s) | S2 T_cert (s) | Class | S2/P direction | S2 - P (s) | P / S2 |
|---|---|---|---|---|---|---|
| SM-C8-S / S0 / R1 | 5.568 | 2.645 | BOTH_CERTIFIED | Win | -2.923 | 2.105 |
| SM-C8-S / S0 / R2 | 5.143 | 2.865 | BOTH_CERTIFIED | Win | -2.278 | 1.795 |
| SM-C8-S / S1 / R1 | 3.904 | 3.227 | BOTH_CERTIFIED | Win | -0.677 | 1.210 |
| SM-C8-S / S1 / R2 | 4.197 | 2.845 | BOTH_CERTIFIED | Win | -1.352 | 1.475 |
| SM-C8-S / S2 / R1 | 4.611 | 3.781 | BOTH_CERTIFIED | Win | -0.830 | 1.220 |
| SM-C8-S / S2 / R2 | 4.307 | 4.126 | BOTH_CERTIFIED | Win | -0.180 | 1.044 |
| SM-B10-B / S0 / R1 | 0.778 | 0.792 | BOTH_CERTIFIED | Lose | 0.014 | 0.983 |
| SM-B10-B / S0 / R2 | 1.122 | 0.461 | BOTH_CERTIFIED | Win | -0.661 | 2.433 |
| SM-B10-B / S1 / R1 | 1.067 | 0.511 | BOTH_CERTIFIED | Win | -0.556 | 2.088 |
| SM-B10-B / S1 / R2 | 0.708 | 0.801 | BOTH_CERTIFIED | Lose | 0.093 | 0.883 |
| SM-B10-B / S2 / R1 | 0.794 | 0.800 | BOTH_CERTIFIED | Lose | 0.006 | 0.992 |
| SM-B10-B / S2 / R2 | 1.129 | 0.461 | BOTH_CERTIFIED | Win | -0.667 | 2.447 |

There are 0 observed >=10-second delays and 0 P-only certification risks. Two-repeat warning roles: none.

| Problem | Both-cert n / 6 | Median S2-P (s) | S2-P range (s) | P median [range] (s) | S2 median [range] (s) |
|---|---|---|---|---|---|
| SM-C8-S | 6 | -1.091 | -2.923 .. -0.180 | 4.459 [3.904, 5.568] | 3.046 [2.645, 4.126] |
| SM-B10-B | 6 | -0.275 | -0.667 .. 0.093 | 0.930 [0.708, 1.129] | 0.652 [0.461, 0.801] |

The exact per-Seed two-repeat direction/range records and six-observation medians/ranges are in tiny_summary.json. A range is not estimated when fewer than two valid repeats exist.

### 6.13 Finite V35 scope (Round131 C)

Both original R122 V35 roles use their original Q, physical T and complete 2400-second contract, at Seed0 only, with opposite order across the two roles. These are fresh complete final-version runs, not continuations or pooled short-window clocks.

**Table R131.C. Both V35 Seed0 observations.** Both-open means final certification-time ordering is unknown. Endpoint U/L/gap is supplemental and supplies no certification-speed Win.

| Known problem / Seed | P T_cert (s) | S2 T_cert (s) | Class | S2/P direction | S2 - P (s) | P / S2 |
|---|---|---|---|---|---|---|
| PB-C35-U / S0 | open | open | BOTH_OPEN | UNDETERMINED | null | null |
| PB-B35-B / S0 | open | open | BOTH_OPEN | UNDETERMINED | null | null |

Final S2 has 0/2 nonzero V35 certificates. This does not extend observed nonzero certification to V35. C has one Seed, and there is no final-version V50/V60 validation. The source and scenario coverage remains finite and coupled: two station tables, constructed inventories/targets/weights, service-station total scenario labels, fixed lambda=.15, compact/regional Euclidean geography, and overlapping station selections. A=6 inputs/12 observations, B=2/12 and C=2/2 remain separate denominators. No new H, C_G or outer-controller causal claim is inferred. The R130 outer-package result and all predecessor negative/open findings are retained.

## 7. Extended endpoint evidence and the known long-window proof boundary

### 7.1 Earlier R122 prospective panel and measurement

The earlier R122 panel (TailoredExact project 2026c) was fixed before its six new inputs were generated once. It uses two frozen station sources: CITI with 442 eligible physical stations from 443 original rows, and CABI with 865 stations. Source identities, station selections and overlap records remain in the input manifest. The sources provide geographic station information; inventory landscapes and targets are constructed by the declared recipe.

The recipe retains four-anchor/hash selection and tie rules, target fractions .44-.56, coordinates serialized to three decimal places, weights to six decimals with maximum normalization, a centroid depot, stock-total correction and the declared heterogeneous or homogeneous capacity vectors. No H, LP or MIP screening selected problems by difficulty. There was no outcome-driven redraw, salt change or best-of-replicate selection.

**Table 2. Six new Seed0 problems.** Stock labels describe the construction. Lambda is .15 and pickup/drop are 60/60 throughout.

| Role | Source | n/m | Geometry / inventory | Q | Physical T $s$ | Cap $s$ | sum weights | lambda sum weights |
|---|---|---|---|---|---|---|---|---|
| PB-C20-B | CITI | 20/2 | compact / balanced | [30, 30] | 3600 | 900 | 7.288265 | 1.09323975 |
| PB-B20-S | CABI | 20/2 | regional / shortage | [20, 40] | 7200 | 900 | 7.284024 | 1.0926036 |
| PB-C35-U | CITI | 35/3 | regional / surplus | [20, 25, 30] | 10800 | 2400 | 11.590113 | 1.73851695 |
| PB-B35-B | CABI | 35/3 | compact / balanced | [25, 25, 25] | 7200 | 2400 | 14.72 | 2.208 |
| PB-C50-S | CITI | 50/4 | compact / shortage | [30, 30, 30, 30] | 10800 | 3600 | 20.216045 | 3.03240675 |
| PB-B50-U | CABI | 50/4 | regional / surplus | [20, 25, 35, 40] | 18000 | 3600 | 22.597221 | 3.38958315 |

Each new problem has four complete arms: cold P-GRB, ENS-C, M-B and MB-D1. Two additional fixed blocks repeat PB-B35-B and PB-C50-S with solver Seed1. Thus there are six new problems, eight method/Seed blocks, 32 arms and 48 directed pairs. The repeats are sensitivity observations on known inputs, not two new problems. Within a block all methods share the same physical input and production cap. Serial method order rotates across the planned blocks; the complete schedule is retained with the published design.

All methods use the same frozen executable and Gurobi 13.0.2 DLL, one native thread, one MIP thread, PresolveAuto, the stated Seed, FeasibilityTol/OptimalityTol $10^{-6}$, IntFeasTol $10^{-5}$, and requested MIPGap/MIPGapAbs zero. The evidence policy remains $10^{-7}$. The common 30-second native reserve is part of the one process deadline. Larger caps can change a method's search path, so a prefix of a longer run is not assumed to reproduce a shorter run.

Production time $T_{\mathrm{produce}}$ includes all own online startup, models, LP/MIP work, Starts, callbacks, cleanup and sealing. Post-seal verification is $T_{\mathrm{verify}}$; $T_{\mathrm{total}}$ is their sum. Cross-arm reconciliation is group cost, and nested native timers are not added again. A verification tail beyond the native reserve is not a producer timeout. We preserve original final output fleets as the primary U endpoints, even where a separately legal historical witness is slightly better.

The machine/software identity and full clocks are documented in Supplement S and the reproducibility notes. The comparison is within matched complete-method windows, rather than a mixture of historical executables, hardware states and caps. The finite nonfactorial design varies source, scale, geometry, stock, Q, T and aggregate weight together. It does not isolate their individual causal effects or estimate a population win probability.

### 7.2 Comparison and evidence tiers

The historical endpoint rule, retained only for these older tables, is certificate first. A candidate-only qualified certificate wins; a control-only certificate is a severe loss. When both certify, the time-materiality threshold is $\max(30\text{ s}, .10T_{\mathrm{control}})$. Severe certified regression additionally requires twice the control time and at least 120 seconds excess. Actual time ratios are meaningful only for two certified results.

For two open results let $\mathrm{gap} = U - L$, $a_U = \max(.001, .01|U_{\mathrm{control}}|)$, and $a_{\mathrm{gap}} = \max(.001, .10|\mathrm{gap}_{\mathrm{control}}|)$. Material improvement in U or gap without opposing material deterioration is WIN; the reverse is LOSS; opposing changes are MIXED; otherwise TIE. Severe open regression requires both specified U and gap deteriorations, with thresholds $\max(.001, .05|U_{\mathrm{control}}|)$ and $\max(.005, .25|\mathrm{gap}_{\mathrm{control}}|)$. Missing necessary evidence remains UNEVALUABLE. These are experiment-selection/materiality labels, not statistical significance tests.

We distinguish nonzero numerical certification, an independent own-zero certificate, reliable open bounds, source-reduced evidence and missing evidence. The R122 panel has seven nonzero certified arms and 25 open arms; it has no zero certificate, source-reduced arm or missing arm. All seven certificates are at 20 stations. The Seed0 layer contains 24 arms; the Seed1 layer contains eight open arms. PB-B35-B Seed0 cold P uses the explicitly independent lower floor zero, rather than a positive native proof. The tiny positive Seed1 bound is retained.

### 7.3 All historical outcomes and adverse cases

**Table 3. All 32 published R122 arms.** N denotes a nonzero numerical certificate and O a reliable open result, including the stated independent floor. U/L and signed gap are role-specific; caps are production windows. Supplement S1 retains full precision and $T_{\mathrm{verify}}, T_{\mathrm{total}}$ for every row. The displayed tiny signed gaps are not clipped.

| Role | Seed | Method | Cap $s$ | U | L | Signed gap | Tier | T_produce $s$ |
|---|---|---|---|---|---|---|---|---|
| PB-C20-B | 0 | P-GRB | 900 | 0.2246803118 | 0.1765783752 | 0.04810193667 | O | 872.58902 |
| PB-C20-B | 0 | ENS-C | 900 | 0.2246803118 | 0.2246803118 | -1.110223025e-16 | N | 430.55759 |
| PB-C20-B | 0 | M-B | 900 | 0.2246803118 | 0.2246803118 | 1.054711873e-15 | N | 395.85954 |
| PB-C20-B | 0 | MB-D1 | 900 | 0.2246803118 | 0.2246803118 | 2.498001805e-16 | N | 665.06878 |
| PB-B20-S | 0 | ENS-C | 900 | 0.8631523206 | 0.8631523206 | -2.220446049e-16 | N | 4.6224331 |
| PB-B20-S | 0 | M-B | 900 | 0.8631523206 | 0.8631523206 | -2.220446049e-16 | N | 2.6319677 |
| PB-B20-S | 0 | MB-D1 | 900 | 0.8631523206 | 0.8631523206 | -2.220446049e-16 | N | 2.6983887 |
| PB-B20-S | 0 | P-GRB | 900 | 0.8631523206 | 0.8631523206 | 1.110223025e-16 | N | 116.03278 |
| PB-C35-U | 0 | M-B | 2400 | 0.1696364022 | 0.08395185429 | 0.0856845479 | O | 2372.5264 |
| PB-C35-U | 0 | MB-D1 | 2400 | 0.1793004366 | 0.08353399353 | 0.09576644305 | O | 2370.6204 |
| PB-C35-U | 0 | P-GRB | 2400 | 0.1747718876 | 0.05970290317 | 0.1150689844 | O | 2370.6162 |
| PB-C35-U | 0 | ENS-C | 2400 | 0.1642374878 | 0.08157539377 | 0.08266209402 | O | 2370.5989 |
| PB-B35-B | 0 | MB-D1 | 2400 | 0.2345814099 | 0.09200798362 | 0.1425734262 | O | 2372.4404 |
| PB-B35-B | 0 | P-GRB | 2400 | 0.3055557782 | 0 | 0.3055557782 | O | 2370.5561 |
| PB-B35-B | 0 | ENS-C | 2400 | 0.2465044704 | 0.132287968 | 0.1142165024 | O | 2370.5527 |
| PB-B35-B | 0 | M-B | 2400 | 0.2465044704 | 0.1276343018 | 0.1188701686 | O | 2371.7862 |
| PB-C50-S | 0 | MB-D1 | 3600 | 0.3723417755 | 0.3568747399 | 0.01546703556 | O | 3572.8812 |
| PB-C50-S | 0 | M-B | 3600 | 0.3723417755 | 0.3568747399 | 0.01546703556 | O | 3570.5953 |
| PB-C50-S | 0 | ENS-C | 3600 | 0.3723417755 | 0.3579807999 | 0.01436097562 | O | 3571.8954 |
| PB-C50-S | 0 | P-GRB | 3600 | 0.4259500753 | 0.3492206958 | 0.07672937949 | O | 3583.0202 |
| PB-B50-U | 0 | P-GRB | 3600 | 1.176955759 | 0.06837938241 | 1.108576376 | O | 3572.991 |
| PB-B50-U | 0 | MB-D1 | 3600 | 0.677298419 | 0.4712348696 | 0.2060635494 | O | 3570.563 |
| PB-B50-U | 0 | M-B | 3600 | 0.6974735096 | 0.4553454 | 0.2421281096 | O | 3571.8624 |
| PB-B50-U | 0 | ENS-C | 3600 | 0.8325480152 | 0.4523603674 | 0.3801876478 | O | 3570.5731 |
| PB-B35-B | 1 | ENS-C | 2400 | 0.2345814099 | 0.1146410114 | 0.1199403984 | O | 2372.9734 |
| PB-B35-B | 1 | P-GRB | 2400 | 0.2797984344 | 2.027189664e-18 | 0.2797984344 | O | 2370.5117 |
| PB-B35-B | 1 | MB-D1 | 2400 | 0.2345814099 | 0.138322042 | 0.09625936783 | O | 2370.6168 |
| PB-B35-B | 1 | M-B | 2400 | 0.2465044704 | 0.0982245518 | 0.1482799186 | O | 2370.4896 |
| PB-C50-S | 1 | M-B | 3600 | 0.3723417755 | 0.357008334 | 0.01533344153 | O | 3572.6028 |
| PB-C50-S | 1 | ENS-C | 3600 | 0.3723417755 | 0.358161762 | 0.01418001348 | O | 3570.6597 |
| PB-C50-S | 1 | P-GRB | 3600 | 0.3917805586 | 0.3494508596 | 0.04232969893 | O | 3570.6723 |
| PB-C50-S | 1 | MB-D1 | 3600 | 0.3723417755 | 0.357008334 | 0.01533344153 | O | 3572.5642 |

**Table 4. Directed-pair classifications by denominator.** All 48 individual records, including the nine LOSS/MIXED observations, are in Supplement S2.

| Seed | Denominator | Pair | WIN | LOSS | MIXED | TIE | UNEVALUABLE |
|---|---|---|---|---|---|---|---|
| 0 | 6 new problems | MB-D1/P-GRB | 5 | 0 | 1 | 0 | 0 |
| 0 | 6 new problems | M-B/P-GRB | 6 | 0 | 0 | 0 | 0 |
| 0 | 6 new problems | ENS-C/P-GRB | 6 | 0 | 0 | 0 | 0 |
| 0 | 6 new problems | MB-D1/M-B | 1 | 2 | 1 | 2 | 0 |
| 0 | 6 new problems | MB-D1/ENS-C | 1 | 2 | 1 | 2 | 0 |
| 0 | 6 new problems | M-B/ENS-C | 1 | 1 | 0 | 4 | 0 |
| 1 | 2 fixed repeats | MB-D1/P-GRB | 2 | 0 | 0 | 0 | 0 |
| 1 | 2 fixed repeats | M-B/P-GRB | 2 | 0 | 0 | 0 | 0 |
| 1 | 2 fixed repeats | ENS-C/P-GRB | 2 | 0 | 0 | 0 | 0 |
| 1 | 2 fixed repeats | MB-D1/M-B | 1 | 0 | 0 | 1 | 0 |
| 1 | 2 fixed repeats | MB-D1/ENS-C | 1 | 0 | 0 | 1 | 0 |
| 1 | 2 fixed repeats | M-B/ENS-C | 0 | 1 | 0 | 1 | 0 |

M-B/P and ENS-C/P each produce six WIN on the six new roles. These are the historical complete-endpoint rule results, which combine certification and open-endpoint comparisons. They do not mean that all six problems certified faster, and they are not the direct certification outcome of Section 6. The two fixed Seed1 repeats retain WIN against cold P for each integrated method, but add no independent problems. M-B versus ENS-C has one WIN, one LOSS and four TIE on new roles; over all eight blocks it has one WIN, two LOSS and five TIE. Thus the chosen paper reference does not dominate the unchanged default instance by instance.

On PB-C20-B, all methods have the same final physical value, approximately .2246803118. M-B, ENS-C and D1 certify, while cold P remains open. M-B takes 395.8595 production seconds and ENS-C 430.5576; their difference is below the declared materiality threshold. D1 takes 665.0688 seconds and loses both incremental certified-time comparisons. Cold P's open run supplies no ratio of times to optimality.

PB-B20-S gives the other certification case. All four methods certify approximately .8631523206. Cold P takes 116.0328 production seconds, compared with 2.6320 for M-B, 4.6224 for ENS-C and 2.6984 for D1. This case permits actual certified-time comparisons. The small integrated-method differences fall below the 30-second minimum materiality floor; the panel's TIE labels should not be mistaken for identical measured times.

The V35 cases expose tradeoffs. PB-C35-U is open for every method. M-B has better U and gap than cold P, yet loses to ENS-C because its final U and gap are worse at the material scale. D1/P is MIXED: its weaker physical endpoint accompanies a stronger lower endpoint and smaller gap. D1 loses to both M-B and ENS-C. These outcomes remain even though the complete integrated family improves over cold P.

On PB-B35-B Seed0, D1 has a better U than M-B and ENS-C but a sufficiently weaker lower bound and larger gap that both incremental pairs are MIXED. With Seed1, those two D1 comparisons become WIN, and M-B/ENS-C changes from TIE to LOSS. All remain open. These fixed repeats show meaningful sensitivity without authorizing selection of the better Seed.

PB-B50-U supplies the sole new-role material D1/M-B WIN: D1 improves both physical U and the lower endpoint relative to M-B, and also improves on ENS-C. It nevertheless remains open. PB-C50-S supplies common integrated gains without natural D1 installation at either Seed. Such an unexposed candidate result is still a valid complete-method observation; no extra probe is forced to manufacture mechanism exposure.

D1's five new-role exposures and its two V20 certificates do not satisfy the declared promotion requirement of at least two exposed new roles with an incremental M-B win or added certificate. Only PB-B50-U qualifies. The completed panel therefore retains the MIXED selection conclusion, with M-B as the existing paper reference and ENS-C as project default. D1 keeps research/ablation value. D0 is not a reserve selection, and the discontinued MB-PC development is not revived.

Historical evidence is used only to constrain attribution. Earlier paid-start/backend comparisons distinguish the value of an own feasible startup from backend representation changes; they do not justify assigning an integrated win to an unexposed row. Earlier D1 C2 Seed0 severe loss and later Seed1 gain, and the F2 D0 Seed reversal, remain separately scoped adverse observations. We do not combine their executables or windows with this panel to construct a formal acceleration factor.

The completed five-role historical paid-start layer is MIXED for both integrated methods against P-S, which pays the same complete H before the original compact backend. Some tailored results improve proof completion after matched startup, while other roles favor P-S. Rejected native positive bounds and surviving independent own-zero witnesses keep their separate evidence grades. That layer also spans different measurement policies and cannot yield a pooled exact speedup. The finite attribution note identifies the concrete source reports and their limitations.

### 7.4 One known positive-objective long-window problem

The separate long-window study (TailoredExact project 2026a) concerns PB-C50-S only, with 50 stations, four capacity-30 vehicles, physical $T = 10800$, solver Seed0, and three freshly paid 18000-second caps in the fixed order P-GRB, M-B, ENS-C. No previous route, bound, tree, basis or cutoff is loaded. The study adds zero independent problem samples and does not include D1.

Its station totals are $\sum b_i = 495$ and $\sum D_i = 562$. Empty departure and nonnegative return imply $\sum Y_i \leq 495$; therefore $\sum_i |Y_i - D_i| \geq 67$. Positive lambda and weights give

$$F \geq 67 \min_i(\lambda\omega_i/D_i) > 0. \tag{10}$$

The prior exact-decimal explanation gives .24814831875, with a tighter inventory-only penalty floor .26064825 already documented. These are weaker than the published production bounds and are not injected into this method. Their purpose is to rule out an independent zero certificate for this role. No new inventory relaxation or route optimization is performed in preparing the paper.

**Table 5. One known-problem long-window study.** All three endpoints are reliable and open. No row is a time to optimality.

| Method | Cap $s$ | U | L | Signed gap | Certified | T_produce | T_verify | T_total |
|---|---|---|---|---|---|---|---|---|
| P-GRB | 18000 | 0.37764170855 | 0.350377933041 | 0.0272637755094 | False | 17971.7112989 | 1.96611939999 | 17973.6774183 |
| M-B | 18000 | 0.372341775499 | 0.358011485552 | 0.0143302899472 | False | 17970.5725766 | 40.964555 | 18011.5371316 |
| ENS-C | 18000 | 0.372341775499 | 0.358943666168 | 0.0133981093313 | False | 17970.8751446 | 58.2420274 | 18029.117172 |

Under the same fixed rule, M-B/P and ENS-C/P are WIN, and M-B/ENS-C is TIE. M-B improves U over cold P by .005299933051160555 and raises L by .00763355251102632; the signed gap decreases by .012933485562186875. ENS-C has the same final U as M-B and a .000932180615873357 higher L. The latter difference is below the pair's gap-materiality threshold. Arithmetic endpoint gains are not ratios of solver speed, and F* remains unknown.

M-B's final-quality fleet appears in the raw process record at 34.328 seconds; its qualified arm-availability upper time is 35.7173270817 seconds. No better own U is observed afterward. This does not establish that U is optimal, nor that all remaining work is pure proof. ENS-C has an early raw startup record at 34.312 seconds, but its clock-origin spread of 1.5845544376 seconds fails the predeclared one-second conversion check. Its four pre-end arm-clock checkpoints remain unknown. The sealed endpoint and separately measured QPC production/verification clocks remain qualified. We neither widen the guard nor compare those raw times as precise cross-method arm times.

The final coverage explains the outstanding proof rather than extrapolating from a plateau.

**Table 6. Actual final integrated-method covers.** “Empty” excludes improvement under the applicable cutoff; it does not assert unrestricted physical infeasibility.

| Method | Unit | True-G lower | True-G upper | Regional floor | State |
|---|---|---|---|---|---|
| M-B | L0.1 | 0.18617088775 | 0.372341775499 | 0.372341775499 | empty |
| M-B | L0.0.1 | 0.0930854438748 | 0.18617088775 | 0.372341775499 | empty |
| M-B | L0.0.0.0 | 0 | 0.0465427219374 | 0.358011485552 | open |
| M-B | L0.0.0.1 | 0.0465427219374 | 0.0930854438748 | 0.358827842293 | open |
| ENS-C | L0.1 | 0.18617088775 | 0.372341775499 | 0.372341775499 | empty |
| ENS-C | L0.0.1 | 0.0930854438748 | 0.18617088775 | 0.372341775499 | empty |
| ENS-C | L0.0.0.0 | 0 | 0.0465427219374 | 0.358943666168 | open |
| ENS-C | L0.0.0.1 | 0.0465427219374 | 0.0930854438748 | 0.366384964768 | open |

The low interval and its adjacent interval are

$$I_0 = [0, .046542721937380925],$$
$$I_1 = [.046542721937380925, .09308544387476185].$$

Both remain unresolved. The final own $G_{\text{true}} = .050119584194518604$ lies in $I_1$, so that fleet cannot provide an exact-g Start for $I_0$. M-B makes seven LP and one MIP request; its $I_0$ NEXT request uses about 17808.127 seconds. ENS-C makes nine LP and three MIP requests, including both interval targets and a subsequent low-interval terminal request. Child LP count and committed split count are different; counts do not establish retained native search trees or an internal cause of difficulty.

A fixed-endpoint counterfactual ledger illustrates why more than one obligation matters. Hold U and all other regional floors fixed. Raising only M-B's $I_0$ floor to U leaves $U - L_1 = .013513933206048268$; closing only $I_1$ leaves the global minimum at the old $I_0$ floor. This follows from the union minimum in (9). It is not a prediction that a different leaf schedule would perform better, and does not justify launching another experiment.

All three normal long-window observations are complete, qualified open results. The finite study records a proof limit, not an execution failure. The longer cap ends here. There is no final certification-time ratio, linear remaining-time prediction or automatic extension until a certificate appears.

## 8. Discussion, limitations and reproducibility

The principal final-version statement concerns the complete fixed Round131 A panel; B and C separately delimit tiny overhead and V35 scope. The earlier MS2 R129 selection observations and fixed R130 attribution panel retain their own roles. Time ratios condition on each explicitly both-certified intersection. The six-pair R126 certification distribution remains evidence for the full-M-B predecessor. Its scope is this original model, fixed small problem set and matched complete production budgets. That comparison is meaningful because the physical problem, input and full production costs are common. It does not establish state-of-the-art superiority against all algorithms solving this same physical model. Neighboring published methods differ in target feasibility, visits, depot supply, time treatment or objective. A matched adaptation would require its own justified design and validation.

The method's exact representation also has a precise scope. Implicit quantity integrality follows from the remaining integer decisions and unique one-way service. Its proof would fail under split service even if the same variable names were retained. Inventory-product exactness is an integer embedding property, while the SP estimator is a continuous relaxation. Local native success is usable globally only through cutoff complementation and full coverage. These distinctions identify what must be rederived before extending the model.

Certification is more limited than feasible-solution improvement. On the earlier R122 panel, all nonzero certificates occur at V20 and every V35/V50 outcome is open. Round126 adds the separately reported V15/V20 direct evidence. The known V50 long study keeps two integrated-method improvement obligations. This does not prove that V50 is intrinsically intractable or that no larger instance can certify; other historical cases remain valid in their own scope. It does rule out describing the present evidence as stable certification of all instances up to 50 stations.

H and the exact backend have complementary roles. H rapidly finds an effective physical fleet on the long role; its unchanged U leaves uncertainty about both primal suboptimality and lower-bound shortfall. Native MIP work dominates the recorded large-instance cost, but its internal search/proof allocation is not fully observable. The incompatible low-interval Start is a geometric fact about true Gini, not proof that all difficulty is primal. Inventory/ratio attainability and full fleet-route attainability must be distinguished before proposing an additional mechanism.

The numerical guarantee should be read literally. Qualified bounds and certificates use the frozen floating-point evidence policy, including the explicitly numerical closure guard. Exact coefficient arithmetic and physical replay support specific algebraic and witness checks; they are not an a posteriori rational certificate for every native optimization operation. The available evidence reveals no new same-domain counterexample, but the policy supplies no global accumulation theorem merely from the relative sizes of two tolerances.

Generalization is constrained by the experimental design. Each six-problem panel uses non-IID constructed problems from two station tables; neither panel represents six independent cities, and the two panels are not pooled as one certification sample. Station overlaps are recorded, and source, size, inventory regime, geometry, fleet and time vary together. The two R122 fixed-input Seed repeats expose sensitivity but cannot estimate a population probability. There is no best-of-Seed endpoint or instance-wise winner dispatcher. The tested lambda value and varying total weights do not establish robustness across tradeoff preferences. Objectives and time scales should therefore be reported per role.

Real station coordinates do not make the constructed inventories operational observations. Targets are not inferred service-level guarantees, the depot is synthetic, and Euclidean travel ignores the road network and traffic. The static model excludes concurrent customer demand and permits only the stated service semantics. Demographic fairness, station access, realized availability, staff workload and revenue would each need their own data and objective interpretation. The title's ratio dispersion should not be read as an empirical equity outcome.

The manuscript preserves unfavorable scientific evidence while moving engineering detail to its supporting files. Supplement S retains the historical R122 32 arms and 48 pairs, all three clocks, the three long endpoints and checkpoint unknowns, and adds the separately labelled 52 Round131 arms. The 26 current comparisons are shown in Tables R131.A-C. The original final-fleet endpoint is retained in the Seed1 cold-P case where an earlier legal witness was slightly better; its verification recovery and waiting costs remain in $T_{\mathrm{verify}}$. This distinction prevents changing an endpoint after observing its trajectory.

Reproduction has two levels. The frozen published campaigns retain detailed native source/call/model/return and fleet evidence (TailoredExact project 2026c, 2026a). R124's separate small dependency set supported three complete long-study endpoint replays and all 51 directed-pair classifications. R125 preserved those tables byte for byte and added a compact method-check package: rational inequalities, sampled saved matrix rows, finite decision states and declared H-move replay. Round126 retains those definitions and checks; its new work is the fixed certification panel and the complete updated reading draft. It does not repeat the full historical reconstruction. It does not claim a fresh native replication or revalidation of every historical matrix row. Exact input/software identities and the separate review scope are supplied in the reproducibility notes and review record.

The separately completed Round130 attribution study retains its scoped conclusion. Round131 closes the fixed final-version benchmark gap with the finite A/B/C evidence in Sections 6.11-6.13; this round ends without further solving, tuning, input, Seed or component selection. The paper is organized so a reader can judge whether the physical modeling choice, specific integration and finite evidence are scientifically useful, and whether one additional matched comparison or sensitivity question is essential. The present open proof alone supplies no mandate to resume solver time or invent another numbered component.

The Round127 diagnostic cohort comprises nine distinct evaluation inputs and three separate training inputs. E0/E1 layers and shared-arm comparisons remain distinct. Its execution status is PARTIAL because MD failed its mandatory inclusive-root qualification. The four evidence axes are reported separately; that historical specification issue is addressed by the new separately identified MD-R contract in Section 6.8, while its original blocked results remain unchanged. The maintained mathematical appendices retain the complete R125 derivations and declared numerical-closure distinction; the Round128 clarification and separate research-control note are labelled explicitly.

## 9. Conclusion

We have described a complete tailored method for static multi-vehicle redistribution with inventory-to-target ratio dispersion and relative target deviation. The formulation preserves the original routing, service, load and time semantics. Integer inventory states support exact products, and unique one-way service explains why the adopted S2 representation can relax quantity declarations. A complete scoped proof cover explains how returned native information can support an original-problem numerical bound.

GI-TBC in its uniform pair-support implementation, MS2 (S2), remains the sole proposed method. On the six known SC problems at two Seeds (12 matched observations), S2 and P certify 10 and 5 observations, respectively; the strict S2/P directions are 10 Win, 0 Lose, 0 Tie and 2 undetermined. The tiny layer records 9 strict Win, 3 strict Lose and 0 Tie across twelve matched observations, with 0 observed >=10-second delays and 0 P-only certification risks. At V35, S2 and P have 0 and 0 nonzero certificates across two Seed0 observations, with 0 P-only cases; both-open cases have unknown final certification-time ordering. The main SC measurements close a version-to-evidence gap while all unfavorable directions and finite scope limits remain explicit. R130 provides scoped same-H backend support without repeated material outer-package runtime gains. Empirical observations are separate from the ideal representation/coverage propositions and the unchanged floating-point evidence policy. The resulting certification statement is finite and case-dependent on these known inputs; it does not establish population-wide dominance.

The earlier six R122 endpoint WIN labels are preserved under their historical definition and cannot be read as six faster certifications. The known V50 long window remains open for every method. These capability boundaries do not erase accurately limited small-instance evidence, and smaller gaps do not substitute for optimality proof. ENS-C remains a historical comparison and the unchanged project default; no per-instance dispatch, D1 reopening or automatic follow-up experiment is introduced.

Round127 adds Seed1 confirmation, two tiny problems with twelve both-certified observations and two small timing losses, and mixed grouped-row evidence: C_R and three-station deletion variants are faster on the two certified roles, while the C_G comparison supports MB there. The separately selected PT file does not strengthen default P on the three held-out roles, although MB retains two certifications against PT's zero. Execution is PARTIAL and the Round127 scientific judgment is FOUNDATIONAL_ISSUE_REQUIRES_DECISION, confined to MD's inclusive-root/retained equality-closure specification. This does not itself revoke frozen MB's numerical proof policy or the unaffected observations. It does not establish universal component necessity, an absence of tiny regressions or superiority to the best possible Gurobi tuning. That remains the historical Round127 disposition. Section 6.8 supplies the separately qualified Round128 revision and all its fixed observations; it does not rewrite those blocked entries. That is the historical Round128 recommendation. Section 6.9 now reports the separately authorized finite confirmation and its candidate decision. The five fixed conditions support recommending uniform MS2 as the single method candidate for the next paper revision. This was the recommendation at the R129 stage. R130 now adopts uniform MS2 as the sole method; historical full-M-B numbers retain their original labels.

## References and supporting material

The thirteen citation entries are in references.bib. Access level, DOI, source URL, checked sections and limits are recorded in literature_evidence.csv and literature_scope.md. The complete mathematical specification is Appendix A. Supplement S retains the historical published arithmetic tables and adds the separately labelled Round129, Round130 and Round131 clocks; the present finite check script validates the new method examples. Reproducibility notes, current benchmark assessment, and review actions keep source identity, readiness and actual review scope separate from the scientific results.

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# Appendix A. Complete model, scoped strengthening and proof obligations

**Current-version map (Round130).** The sole method is S2/MS2. All R125 equations, physical embeddings and valid-cut derivations below are retained. Only two-station support is active in S2; the general three-station proof describes the historical full-M-B option and is not an active-row claim. Numerical proof-shortcut and inclusive-root qualifications retain their stated domains.

This appendix is the single maintained model specification for the manuscript. It transcribes the frozen writer, including constraints sometimes redundant for integer solutions. It does not replace the implemented model by a different vehicle-routing formulation. Equations are mathematical real-arithmetic expressions; A.7 states the separate numerical implementation boundary. Source locations refer to production snapshot 355eefd2be3fe5651195681faf7b3a63ee4032f8 (TailoredExact project 2026b).

## A.1 Sets, data and variables

Let $N = \{1, \dots, n\}$ be service stations, $N_0 = N \cup \{0\}$, $K = \{1, \dots, m\}$, and $A = \{(i, j) \in N_0^2 : i \neq j\}$. Depot 0 is absent from every inventory, ratio, weight and Gini sum. Vehicle indexing is shifted by one relative to source indices $0, \dots, M-1$. All vehicle-indexed formulas hold for each $k \in K$; all station formulas hold for each $i \in N$, unless otherwise stated.

Station capacity $C_i$ and initial inventory $b_i$ are integers with $0 \leq b_i \leq C_i$; target inventory $D_i$ is a positive integer no larger than $C_i$. The parser's depot capacity/stock fields do not provide departure supply. Vehicle capacity $Q_k > 0$ is integer. A vehicle starts empty and returns to the same depot, possibly loaded. Travel time $\tau_{ij} = \|v_i - v_j\|_2/1.5$ uses the serialized planar coordinates. The route horizon is $T$; unit pickup/drop times are $t_p, t_d$, with $h = t_p + t_d$. Penalty multiplier $\lambda \geq 0$ and station weights $\omega_i \geq 0$ are data; the study uses $\lambda = .15$, positive weights and $t_p = t_d = 60$. No inventory floor is imposed by the compatibility field min_ratio.

The binary variables are arc decisions $x_{kij}$, station service assignments $z_{ki}$, and pickup-direction indicators $a_{ki}$. Source calls the last variable mName(k,i); we rename it $a$ to avoid confusing it with the fleet size. Pickup $p_{ki}$, drop $d_{ki}$, final inventory $Y_i$, and post-service load $\ell_{ki}$ are nonnegative. The rank $u_{ki}$ is continuous. Define $\bar{p}_{ki} = \min(b_i, Q_k)$ and $\bar{d}_{ki} = \min(C_i - b_i, Q_k)$.

| Variable | Domain and bound in the active formulation |
|---|---|
| $x_{kij}, z_{ki}, a_{ki}$ | Binary, for all stated indices; no self arcs |
| $p_{ki}, d_{ki}$ | Integer for cold P and ENS-C; continuous for M-B; upper bounds $\bar{p}_{ki}, \bar{d}_{ki}$ |
| $\ell_{ki}$ | Integer in $[0, Q_k]$, retained by all three methods |
| $u_{ki}$ | Continuous in $[0, n]$ |
| $Y_i$ | Integer in $[\underline{y}_i, \bar{y}_i]$; plain model has $[0, C_i]$ |
| $r_i$ | Continuous in $[0, C_i/D_i]$ |
| $e_i$ | Continuous in $[0, \max(1, C_i/D_i - 1)]$ |
| $H_{ij}, i < j$ | Continuous in $[0, C_i/D_i + C_j/D_j]$ |
| $g, Z_i$ | Continuous; $g \in [0, 1]$ in the unsliced writer, $g \in [\alpha, \beta] \subseteq [0, 1]$ for an interval; $Z_i \in [0, C_i]$ |
| $f_{kij}$ | Continuous in $[0, n]$, F0 connectivity flow only |
| $s_{iy}, q_{iy}$ | Binary inventory selectors and continuous selected-G products, for $y \in \mathcal{Y}_i = \{\underline{y}_i, \dots, \bar{y}_i\}$, VD-P only |
| $b^{\text{bit}}_{it}, w_{it}$ | Binary inventory bits and continuous products, plain P only |
| $W_{SP}$ | Continuous in $[\max(0, \underline{SP}), \max\{\max(0, \underline{SP}), \bar{S}\bar{P}\}]$, when the aggregate SP estimator is active |

The model objective is $f = g + \lambda P$, where $P = \sum_i \omega_i e_i$; $f, S = \sum_i r_i$, and $H_\Sigma = \sum_{i<j} H_{ij}$ are expressions, not new columns. Constants $\underline{y}, \bar{y}, \underline{S}, \bar{S}, \underline{P}, \bar{P}$ are defined in A.5. Bound-only extrema columns r_min,r_max may

still be allocated by the writer; with the shared factory their centering inequalities use the projected pair form below and those columns have only their original bounds $[0, \max_i C_i/D_i]$. No additional vehicle-used binary exists: $v_k = \sum_j x_{k0j}$ is an expression.

### A.2 Routing, service and physical resources

The complete routing core is

$$\sum_{j\in N} x_{k0j} = \sum_{i\in N} x_{ki0}, \qquad \sum_{j\in N} x_{k0j} \le 1, \tag{A1}$$

$$\sum_{j\in N_0\setminus\{i\}} x_{kji} = \sum_{j\in N_0\setminus\{i\}} x_{kij} = z_{ki}, \qquad \sum_{k} z_{ki} \le 1, \tag{A2}$$

$$z_{ki} \le u_{ki} \le n z_{ki}, \qquad u_{ki} - u_{kj} + n x_{kij} \le n-1 \quad (i,j\in N,\ i\ne j). \tag{A3}$$

These are the writer's original depot degree, station degree and MTZ rows. Any selected station cycle contradicts the sum of (A3) around the cycle. Thus an integer nonempty selection has one depot-connected elementary tour; an unused vehicle has all arcs and assignments zero. A depot-to-depot empty route is represented without a self arc. The MTZ rows remain when F0 also adds the following connectivity flow:

$$0 \le f_{kij} \le n x_{kij} \quad ((i,j)\in A), \qquad \sum_{j\ne i} f_{kji} - \sum_{j\ne i} f_{kij} = z_{ki}, \tag{A4}$$

$$\sum_{j\in N} (f_{k0j} - f_{kj0}) = \sum_{i\in N} z_{ki}.$$

F0 here means the round20-current flow variant, including return-arc columns, with no $f \ge x$ normalization or start-coupling additions. A route visiting $q$ stations admits downstream-count flows $q, q-1, \dots, 1, 0$. These flows establish validity for every original route, independently of MTZ. Cold plain P does not receive this strengthening. The row package also called F0 has additional families in A.5; these two uses of the historical label must not be confused.

All protected service rows are retained:

$$p_{ki} \le \bar{p}_{ki} z_{ki}, \quad d_{ki} \le \bar{d}_{ki} z_{ki}, \quad a_{ki} \le z_{ki}, \quad p_{ki} \le \bar{p}_{ki} a_{ki}, \tag{A5}$$

$$d_{ki} \le \bar{d}_{ki}(z_{ki} - a_{ki}), \quad p_{ki} + d_{ki} \ge z_{ki}, \quad 0 \le \ell_{ki} \le Q_k z_{ki}.$$

The first pair is not silently dropped even where implied by direction rows. Zero directional bounds cause no division. Unique service, a binary direction and the last service inequality forbid split service, simultaneous pickup/drop and zero-operation visits.

Loads are the writer's original two-sided Big-M recurrences:

$$-Q_k(1 - x_{k0i}) \le \ell_{ki} - p_{ki} + d_{ki} \le Q_k(1 - x_{k0i}), \tag{A6}$$

$$-Q_k(1 - x_{kij}) \le \ell_{kj} - \ell_{ki} - p_{kj} + d_{kj} \le Q_k(1 - x_{kij}) \quad (i,j\in N,\ i\ne j), \tag{A7}$$

$$R_k := \sum_i (p_{ki} - d_{ki}) \ge 0, \qquad \sum_{(i,j)\in A} \tau_{ij} x_{kij} + h \sum_i p_{ki} \le T. \tag{A8}$$

The first selected arc fixes departure load to zero. Selected internal arcs propagate each post-service load. A return has no service operation: its load equals the last $\ell$, hence is at most $Q_k$. Summing the recurrence gives $R_k$ as that load, so its upper bound follows without a separate $R_k$ column or explicit upper row. Inactive recurrence rows are valid because their difference is between a feasible before-service load and another feasible post-service load, each in $[0, Q_k]$. Empty vehicles have $p = d = \ell = R = 0$.

In (A8), let $P_k = \sum_i p_{ki}$ and $D_k^{\text{drop}} = \sum_i d_{ki}$. Since $R_k = P_k - D_k^{\text{drop}}$, original handling time is

$$t_p P_k + t_d(D_k^{\text{drop}} + R_k) = (t_p + t_d)P_k. \tag{A9}$$

Return unloading is paid. Capacity $Q_k$ limits every load, not cumulative pickup $P_k$, which may exceed $Q_k$ after intermediate drops. There is one physical time limit per vehicle and a separate computational deadline.

Inventories and objective auxiliaries satisfy

$$Y_i + \sum_k (p_{ki} - d_{ki}) = b_i, \quad r_i = Y_i/D_i, \quad e_i \geq r_i - 1, \quad e_i \geq 1 - r_i, \tag{A10}$$

$$H_{ij} \geq r_i - r_j, \quad H_{ij} \geq r_j - r_i \quad (i < j). \tag{A11}$$

Consequently $\sum_i Y_i + \sum_k R_k = \sum_i b_i$. Loaded returns permit a decrease of station inventory; no external depot inventory can increase it.

### A.3 Original and disaggregated Gini objective

The physical objective is

$$F(Y) = G_{\text{true}}(Y) + \lambda \sum_i \omega_i |Y_i/D_i - 1|, \quad G_{\text{true}} = \begin{cases} \dfrac{\sum_{i<j} |r_i - r_j|}{n \sum_i r_i}, & S > 0, \\ 0, & S = 0. \end{cases} \tag{A12}$$

The native epigraph is

$$n \sum_i Z_i/D_i \geq H_\Sigma, \qquad \min f = g + \lambda \sum_i \omega_i e_i. \tag{A13}$$

Cold P uses inventory bits. With $B_i = \max(1, \lceil \log_2(C_i + 1) \rceil)$, for $t = 0, \ldots, B_i - 1$,

$$Y_i = \sum_t 2^t b_{it}^{\text{bit}}, \quad Z_i = \sum_t 2^t w_{it}, \quad 0 \leq w_{it}, \quad w_{it} \leq g, \quad w_{it} \leq b_{it}^{\text{bit}}, \quad w_{it} \geq g + b_{it}^{\text{bit}} - 1. \tag{A14}$$

The capacity bound removes unused bit encodings. These products are exact at binary bits. ENS-C and M-B instead use the finite value-disaggregated perspective block VD-P:

$$\sum_{y \in \mathcal{Y}_i} s_{iy} = 1, \quad Y_i = \sum_y y s_{iy}, \quad \alpha s_{iy} \leq q_{iy} \leq \beta s_{iy}, \quad \sum_y q_{iy} = g, \quad Z_i = \sum_y y q_{iy}. \tag{A15}$$

The active mode is VD-P, not VD-J: equalities reconstructing $r_i$ or $e_i$ from selectors are not substituted for (A10). The shared factory's bit-product McCormick rows are skipped for this block because those bit columns are absent. Products (A15) already use interval bounds. M-B additionally writes the original A/B rows

$$\sum_k (p_{ki} + d_{ki}) = \sum_{y \in \mathcal{Y}_i} |b_i - y| s_{iy}, \qquad \sum_k z_{ki} + s_{i,b_i} = 1. \tag{A16}$$

A missing initial state means $s_{i,b_i} = 0$, not a new selector. The independent ENS-Q option is false. The mode-linked direction equality of another variant is absent; M-B retains its original binary $a_{ki}$. ENS-C uses integer $p/d$ and omits (A16). These are the material representation differences; the complete H and controller are shared.

## A.4 Integrality, embedding and fairness propositions

**Lemma A1 (implicit service integrality).** With integer $b_i, Y_i$, (A2), (A5), and binary assignments/directions, M-B's continuous quantities are integer.

*Proof.* If $z = 0$, its quantities and direction are zero. At the only assigned vehicle, direction 1 forces $d = 0$, so $p = b_i - Y_i$ is integer. Direction 0 forces $p = 0$, so $d = Y_i - b_i$ is integer. No other vehicle contributes. Retained integer loads are not needed for this argument. Removing unique service, one-way direction or integer inventory invalidates the inference. The statement concerns integer feasible solutions, not equality of LP convex hulls. □

**Validity of A/B.** For an integer original service, total movement at a station equals $|b_i - Y_i|$, which its selected state reproduces. An assigned service is nonzero and one-way, so $Y_i \neq b_i$ exactly when the station is served. This proves both (A16); neither is an extra premise of Lemma A1.

**Lemma A2 (physical objective embedding).** Assume inventory domains and rows preserve all physical points assigned to a true-G interval $[\alpha, \beta]$ and cutoff $F \leq c$. Every such fleet has an extension satisfying the applicable original or VD-P groups, with $g = G_{\text{true}}$ and $f = F$, up to permutation of identical vehicle labels if symmetry rows are present.

*Proof.* Use the physical arcs, service, load prefixes, tour ranks and downstream flows. Select its integer inventory state, put $q_{iy} = g$ at the selected state and zero at others; (A15) yields $Z_i = gY_i$. Set $e_i = |r_i - 1|$ and $H_{ij} = |r_i - r_j|$. Then (A13) holds, with equality for $S > 0$. At $S = 0$ all inventories, ratios and differences are zero and $g = 0$ realizes the stipulated convention. Bounds and valid strengthening are satisfied by assumption. Conversely an integer product completion has $Z_i = gY_i$, hence $g \geq G_{\text{true}}$ for $S > 0$ and $f \geq F$; at zero sum, $g \geq 0$ suffices. □

Arbitrary native $g$ may exceed physical Gini. A fleet with $G_{\text{true}} < \alpha$ can have an inflated-g auxiliary completion. The proof assigns physical points by their true Gini and requires their objective-preserving extension; it does not classify every completion by its auxiliary g. An inventory vector without compatible routes is not an upper-bound witness.

For $n > 1$, triangle inequalities give $A := \sum_{i<j} |r_i - r_j| \leq (n-1)S$, so $G_{\text{true}} \leq (n-1)/n$. For $S > 0$, $G_{\text{true}} = 0$ means every ratio is equal; at $S = 0$ all are zero by nonnegativity. A common ratio below one is a common proportional shortage, not target attainment. If $\lambda > 0$ and all $\omega_i > 0$, $F = 0$ if and only if every $Y_i = D_i$. These are dispersion and target-deviation statements, with no implied demographic/service/revenue interpretation.

## A.5 Enabled strengthening: construction, proofs and scope

The finite list below describes the actual M-B path, not every option in the repository. All rows are assembled **statically when a canonical interval model is written**, before its LP or MIP request. A new cutoff epoch rewrites the relevant model; it does not separate these rows at fractional native nodes. The writer supplies physical rows, and the shared interval factory supplies the interval rows. Families registered by the factory are not all rows emitted by it. In particular, the factory's bit-product rows are filtered out for VD-P. The original objective and physical constraints remain present.

### A.5.1 Construction of the inventory rectangle (D1)

Let the current true-G interval be $I = [\alpha, \beta]$, the independently verified own incumbent be $U_e$, and the cutoff be $c = U_e$. **The active controller sets incumbent_epsilon to zero**, so this is a non-strict objective cutoff; a generic writer can subtract an epsilon, but that is not this frozen request. Start with $l_i = 0, u_i = C_i$. With $\lambda > 10^{-12}$, finite positive $U_e$, and $B = (c - \alpha)/\lambda \geq -10^{-10}$, for each $\omega_i > 10^{-12}$ intersect with

$$l_i \leftarrow \max\{l_i, \lceil D_i(1 - \max(0, B/\omega_i)) - 10^{-9} \rceil\}, \quad u_i \leftarrow \min\{u_i, \lfloor D_i(1 + \max(0, B/\omega_i)) + 10^{-9} \rfloor\}. \tag{A19}$$

This is followed by one intersection with physical movement reachability. Write $t_i = \tau_{0i} + \tau_{i0}$ and

$$\begin{aligned} M_{ki} &= \begin{cases} \min\{Q_k, \lfloor (T - t_i)/h + 10^{-9} \rfloor\}, & T + 10^{-9} \geq t_i,\ h > 10^{-12}, \\ Q_k, & T + 10^{-9} \geq t_i,\ h \leq 10^{-12}, \\ 0, & \text{otherwise.} \end{cases} \\ l_i &\leftarrow \max\{l_i, b_i - \max(0, \max_k \min(b_i, Q_k, M_{ki}))\}, \\ u_i &\leftarrow \min\{u_i, b_i + \max(0, \max_k \min(C_i - b_i, Q_k, M_{ki}))\}. \end{aligned} \tag{A20}$$

The outer max with zero transcribes the actual zero-initialized reachability accumulators, including a slightly negative quotient admitted by a floating travel guard. The mathematical version omits the floating guards, uses $h = 0$ for the second case, and clips movement below at zero. Positive published $h = 120$ avoids the small-positive-h threshold ambiguity. If an interval becomes empty, the code flags domain infeasibility, resets that station's storage bounds to $[0, C_i]$, and emits an impossible row. The reset prevents malformed column ranges; it does not reinstate the excluded domain as feasible.

**Proposition A5 (budget and reachability).** Every physical point in $I, F \leq c$ satisfies the ideal versions of (A19)-(A20). Indeed $F = G + \lambda P$, $G \geq \alpha$, and nonnegative weighted deviations imply $\omega_i |Y_i/D_i - 1| \leq (c - \alpha)/\lambda$. Integer $Y_i$ gives the rounded bounds. A visited station requires a depot tour with travel at least $t_i$. Its pickup is at most $Q_k, b_i$, and its drop is at most $Q_k, C_i - b_i$; because $D_k^{\text{drop}} \leq P_k$, either operation is at most cumulative pickup, which has handling cost $hP_k \leq T - t_i$. Maximize over its unique possible vehicle. These are necessary, not sufficient, attainability conditions: a delivery still needs a source. Zero weight or zero lambda supplies no individual penalty bound and is skipped. □

**Actual order and pass count.** The writer first computes budget bounds, intersects reachability, and derives interval extrema; the shared factory recomputes this same sequence from $[0, C_i]$ and replaces the local arrays. It does not feed the first result into a second tightening pass. There is no propagation loop in this enabled path and no iterative fixed point. The preset advertises `iterative` and requested rounds 2, but the legacy completion field is capped at 1. Thus the operative definition is one budget intersection plus one reachability intersection, independently recomputed by the factory. R124's "original two passes" description was incorrect; production is unchanged.

Set $\underline{y}_i = l_i, \bar{y}_i = u_i, \mathcal{Y}_i = \{l_i, \ldots, u_i\}$, and

$$\begin{aligned} \underline{r}_i &= l_i/D_i, \quad \bar{r}_i = u_i/D_i, \\ \underline{S} &= \sum_i \underline{r}_i, \quad \bar{S} = \sum_i \bar{r}_i, \\ \underline{e}_i &= \text{dist}(1, [\underline{r}_i, \bar{r}_i]), \quad \bar{e}_i = \max\{|\underline{r}_i - 1|, |\bar{r}_i - 1|\}, \\ \underline{P} &= \sum_i \omega_i \underline{e}_i, \quad \bar{P} = \sum_i \omega_i \bar{e}_i. \end{aligned} \tag{A21}$$

The last extrema bound the **tight physical completion** of $e$. They do not assert that every nonminimal native epigraph $e_i$ is at most $\bar{e}_i$. There is no stock-weighted knapsack or optimization in the denominator calculation: these are sums over the rectangle, with no extra conservation-driven tightening.

### A.5.2 Physical conservation and activation (P1-P4)

The following rows hold on the full physical domain, independently of (I,c):

$$\sum_i b_i - \sum_k Q_k \le \sum_i Y_i \le \sum_i b_i; \quad -b_i \sum_k z_{ki} \le Y_i - b_i \le (C_i - b_i) \sum_k z_{ki} \quad (i \in N); \tag{A22}$$

$$h \sum_{k,i} p_{ki} \le mT; \qquad z_{ki} \le 0 \quad \text{if } t_i + h > T + 10^{-9}. \tag{A23}$$

**Proof.** Each return load lies in $[0, Q_k]$; insert this into conservation (A10). If no vehicle serves $i$, inventory balance fixes $Y_i = b_i$; otherwise unique assignment makes the visit sum one and the two activation bounds reduce to $0 \le Y_i \le C_i$. Sum (A8) and drop nonnegative travel for global handling. Any nonzero integer service has at least one pickup somewhere on the vehicle: for a drop use empty departure and load nonnegativity. Consequently a tour through $i$ takes at least $t_i + h$. The physical meaning of the historical `transfer_subset_capacity` flag here is this singleton visit exclusion, not a separate general subset-flow formulation. All vehicles and stations are enumerated in increasing index order. These inequalities are implied for integer physical points, but can strengthen the relaxation. The global-handling impossible-row diagnostic based on accumulated required pickup is inactive in the shared-factory path because that legacy accumulator is not populated there. □

### A.5.3 Ordered supply compatibility (P5)

For every vehicle $k$ and receiving station $j$, define

$$\mathcal{C}_{kj} = \{i \in N \setminus \{j\} : \tau_{0i} + \tau_{ij} + \tau_{j0} + h \le T + 10^{-9}\}. \qquad d_{kj} \le \sum_{i \in \mathcal{C}_{kj}} p_{ki}. \tag{A24}$$

All stations passing this test enter the set, even those with zero stock; their pickup column is then fixed at zero. Common $T, h, \tau$ make the sets identical across vehicles, but their pickup bounds still use $Q_k$. No demand assumption or target sign is used.

**Proposition A6.** (A24) is necessary for a physical fleet under metric travel and empty departure. Label the integer units collected by a vehicle. Scan its route in order, assigning each delivered unit to one previously collected, not yet delivered unit. Every prefix load is nonnegative, so this matching always exists. If a unit received at $j$ came from $i$, the route includes the ordered subsequence (0,i,j,0). Repeated triangle inequalities give travel at least $\tau_{0i} + \tau_{ij} + \tau_{j0}$, and its at least one collected unit entails $hP_k \ge h$. Hence $i \in \mathcal{C}_{kj}$. Counting matched units at $j$ proves the row; a pickup unit can be delivered only once. Intermediate stations can add travel or receive other units but cannot create supply. Unmatched collected units may return to depot. An unvisited source has $p_{ki} = 0$; an empty compatibility set forces $d_{kj} = 0$. Preloading, cross-vehicle handoffs, or nonmetric shortcut assumptions would break this proof and are excluded. □

The row is a coarse aggregate necessity, not a claim that all compatible pickups can simultaneously reach $j$. It neither imposes $P_k \le Q_k$ nor substitutes for prefix loads.

### A.5.4 Two- and three-station support (P6)

For $W = \{a, b\}$,

$$\tau(W) = \min\{\tau_{0a} + \tau_{ab} + \tau_{b0}, \tau_{0b} + \tau_{ba} + \tau_{a0}\}.$$

For $W = \{a, b, c\}$, take the minimum of the six expressions

$$\tau_{0\pi_1} + \tau_{\pi_1\pi_2} + \tau_{\pi_2\pi_3} + \tau_{\pi_3 0}, \quad \pi \text{ a permutation of } (a, b, c). \tag{A25}$$

These are explicit finite formulas, not new shortest-route optimization calls. Metric shortcutting of a complete tour yields one such order, so its travel is at least $\tau(W)$.

**Lemma A7 (integer handling).** If all stations in $W$ are served by $k$, their nonzero integer operations give

$$|W| \le \sum_{i \in W} (p_{ki} + d_{ki}) \le P_k + D_k^{\text{drop}} \le 2P_k, \quad P_k \ge \lceil |W|/2 \rceil. \tag{A26}$$

Here $P_k = \sum_{i \in N} p_{ki}$, not $\sum_{i \in W} p_{ki}$: supplies may lie outside $W$. Integer $P_k$ gives the ceiling; the LP relaxation alone cannot justify it. Capacity does not bound cumulative pickup. Therefore the writer emits exactly one of the following rows for each selected support:

$$\begin{cases} \sum_{i \in W} z_{ki} \le |W| - 1, & \tau(W) + h\lceil |W|/2 \rceil > T + 10^{-9}, \\ h \sum_{i \in W} p_{ki} + M_{kW} \sum_{i \in W} z_{ki} \le T - \tau(W) + M_{kW}|W|, & \text{otherwise}, \end{cases}$$
$$M_{kW} = T + \tau(W) + h \max\{1, \sum_{i \in W} \min(b_i, Q_k)\}. \tag{A27}$$

**Proposition A8 (both branches).** The cover branch follows from (A8), (A25), (A26). In the conditional branch, if every station is served, cancel $M_{kW}|W|$. Since full travel is at least $\tau(W)$ and $\sum_W p \le P_k$, the remaining inequality is necessary. If at least one is unserved, put $q = |W| - \sum_W z \ge 1$, $B_W = \sum_W \min(b_i, Q_k)$. The right side after moving the visit terms is $T - \tau(W) + qM_{kW} \ge 2T + h \max(1, B_W) \ge hB_W \ge h \sum_W p$. This proves deactivation for all nonnegative $T, h$, including $B_W = 0$, empty vehicle and $h = 0$. It does not rely on an invalid cumulative-capacity bound. □

**Selection is finite and ordered.** For k increasing, enumerate every increasing pair $a < b$, then every increasing triple $a < b < c$. One shared counter starts at zero outside the vehicle loop. The helper returns without adding a row once the counter is 50000; it increments once before choosing cover versus conditional. Outer loops continue (and can still evaluate cycle formulas), but no more rows are added. This is not "stop after 50000 violated covers", not 50000 per vehicle, and not pairs interleaved with triples. At n=50 each full vehicle contributes 1225+19600=20825 supports; the third vehicle receives only the prefix needed to reach the shared limit. Row validity is independent of this selection policy. The dormant exhaustive-subset and rank3 variants have different coefficients/order and are excluded.

### A.5.5 Dispersion, spread and centering (G1-G4)

For every pair $i < j$, let $d_{ij} = |r_i - r_j|$. For each $u \notin \{i, j\}$, triangle inequality gives $d_{iu} + d_{ju} \ge d_{ij}$. These pairs are distinct in the unordered numerator; the omitted pairs among the other stations are nonnegative. Hence

$$A = d_{ij} + \sum_{u \ne i,j} (d_{iu} + d_{ju}) + \sum_{u<v;\ u,v \ne i,j} d_{uv} \ge [1 + (n-2)]d_{ij} = (n-1)d_{ij}. \tag{A28}$$

For $n = 2$ the other sums are empty and equality holds. For $n = 1$, spread and centering are skipped. For true $G \in [\alpha, \beta]$ and $S > 0$, $n\alpha S \le A \le n\beta S$. At $S = 0$, all ratios and $A$ are zero; only the interval containing true $G = 0$ needs that physical embedding. With **tight** $H_{ij} = d_{ij}$, the emitted rows are

$$n\alpha S \le H_\Sigma \le n\beta S; \qquad (n-1)H_{ij} \le n\beta S \quad (i < j). \tag{A29}$$

This proves existence of an objective-preserving completion. It does **not** prove that an arbitrary inflated native $H_{ij}$, even at integer physical variables, satisfies the new row. For example $r = (1, 1), g = .5, H_{12} = 1$ satisfies the unsliced original epigraph $2gS \ge H$; the true-G-zero cap excludes this completion while preserving $H = 0, g = 0$. Accordingly these static rows are not advertised as callback user cuts preserving all original native integer completions.

Using the retained $H_{ij} \ge \pm(r_i - r_j)$ in the pairwise row gives the two variable-S rows, and $S \le \bar{S}$ then gives the fixed rows:

$$(n-1)(r_i - r_j) \le n\beta S, \quad (n-1)(r_j - r_i) \le n\beta S; \quad \pm(r_i - r_j) \le \frac{n\beta\bar{S}}{n-1}. \tag{A30}$$

The fixed form is the projection of a common min/max ratio band. The allocated extrema columns do not receive that extended formulation in the shared path. All increasing pairs, forward then reverse, are generated. The ideal implication chain is (A29)+absolute-value lower rows → variable-S → fixed-S. Redundancy in ideal arithmetic is not an ablation and authorizes neither row deletion nor an individual speedup claim. The cap/floor, H-spread and signed-difference rows constrain different auxiliary completions even though they preserve the same tight physical embedding.

#### A.5.6 Required movement and penalty closure (C1-C3)

For each station, emit only the applicable bound consequence:

$$\sum_k (d_{ki} - p_{ki}) \ge l_i - b_i \quad (l_i > b_i), \qquad \sum_k (p_{ki} - d_{ki}) \ge b_i - u_i \quad (u_i < b_i). \tag{A31}$$

These follow by substituting inventory balance in $Y_i \ge l_i$ or $Y_i \le u_i$. Their scope inherits the domain: a cutoff-derived $l_i$ is not an unrestricted physical bound. If $b_i \notin \mathcal{Y}_i$, A/B forces service, consistently with required movement. If $\mathcal{Y}_i = \{b_i\}$, A/B forces no service. A singleton zero domain and zero operation bounds require no divisions.

For all active finite positive-cutoff requests write

$$g + \lambda P \le c, \qquad P \ge \underline{P}, \qquad e_i \le \max(0, B/\omega_i) \ (\omega_i > 10^{-12}). \tag{A32}$$

The last row uses the same budget activation tests as A19. The floor follows station by station from the distance to 1 in (A21); the original e upper bound is retained and the budget row intersects it. $\alpha + \lambda\underline{P} > c$ proves ideal cutoff infeasibility. The implementation instead tests $\alpha + \lambda\underline{P} \ge c - 10^{-9}$, writes an impossible row, and uses its numerical closure policy. In particular equality to c is not strict infeasibility of the non-strict cutoff set. A.7 explicitly limits this claim. The factory's "Global" metadata on the incumbent row means the entire current model, not all physical objectives at every cutoff.

Round128 clarification (the frozen ordinary rule is unchanged): if the actual cutoff-domain bound is $L_a = \alpha + \lambda\underline{P}$, its unconditional regional bound is $\min\{L_a, c\}$ by coverage with the cutoff complement. $L_a > c$ excludes the

inclusive cutoff domain; $L_a = c$ permits equality and excludes only strict improvement. For $c - 10^{-9} \leq L_a < c$, the raw supported bound remains $L_a$ and does not rule out every arbitrarily small improvement. The impossible-row shortcut is not a physical embedding inequality for all $F \leq c$. A known witness contradicting its alleged own-domain bound takes priority for investigation. The separately defined MD-R omits only this named shortcut and checks its own Start against the inclusive root; no frozen production numerical policy is upgraded by that control.

### A.5.7 Scalar objective estimate and SP lifting (C4-C6)

For $S > 0$, multiplying the true cutoff inequality by $nS$ gives

$$A + n\lambda SP \leq ncS. \tag{A33}$$

Since $A \geq 0$, $\bar{S} \geq S$ gives $A/(n\bar{S}) \leq A/(nS)$, so $A + n\bar{S}\lambda P \leq n\bar{S}c$ when $\bar{S} > 0$. Equivalently A33 implies $A \leq nS(c - \lambda P) \leq n\bar{S}(c - \lambda P)$, because the original cutoff and nonnegative G imply $c - \lambda P \geq 0$. The emitted scalar row is

$$H_\Sigma + n\bar{S}\lambda \sum_i \omega_i e_i \leq n\bar{S}c \quad (\bar{S} > 10^{-12}). \tag{A34}$$

The bound on S is the rectangular upper estimate (A21), not a new optimum of S. Tight H/e complete every physical point. If $S = 0$, then $A = 0$, $P = \sum_i \omega_i$, and the original condition is $\lambda P \leq c$. Multiplication alone makes A33 tautological and cannot enforce that condition. Thus the original objective cutoff (A32) remains essential. A34 also holds directly at S=0 if that cutoff holds.

Introduce $W = W_{SP}$ and take $W = SP$ on physical completions. The four nonnegative products on the rectangle are $(S - \underline{S})(P - \underline{P})$, $(\bar{S} - S)(\bar{P} - P)$, $(\bar{S} - S)(P - \underline{P})$, and $(S - \underline{S})(\bar{P} - P)$. Expanding them proves respectively

$$\begin{aligned} W &\geq \underline{S}P + \underline{P}S - \underline{SP}, \\ W &\geq \bar{S}P + \bar{P}S - \bar{S}\bar{P}, \\ W &\leq \bar{S}P + \underline{P}S - \bar{S}\underline{P}, \\ W &\leq \underline{S}P + \bar{P}S - \underline{S}\bar{P}. \end{aligned} \tag{A17}$$

These are the standard McCormick envelopes (McCormick 1976), derived here without attributing an unread theorem number. They are relaxations, not an exact continuous-product identity. The writer bounds $W$ by $w_l = \max(0, \underline{SP})$, $w_u = \max(w_l, \bar{S}\bar{P})$, and emits

$$H_\Sigma + n\lambda W - nc \sum_i r_i \leq 0. \tag{A35}$$

Every coefficient is explicit: H coefficients 1; r coefficients $-nc$; W coefficient $n\lambda$; e coefficients in each McCormick row are the corresponding S coefficient times $\omega_i$. The active objective remains $g + \lambda \sum_i \omega_i e_i$ with **ObjCon=0**; W is absent from the objective. Equal S or P bounds cause valid degenerate envelopes. Zero S forces the physical W to zero but does not remove the original penalty. These rows preserve a joint tight completion of g,e,H,W and must not be claimed to preserve all loose auxiliary assignments. No GS or disaggregated-SP lift is enabled.

### A.5.8 Symmetry, skipped rows and dependency order

With the active symmetry option, if **all** capacities are equal and $m > 1$, the writer adds $\sum_i z_{ki} \geq \sum_i z_{k+1,i}$ for adjacent vehicles. Common T, travel and handling make those vehicles interchangeable; relabel routes by nonincreasing visits to obtain an objective-preserving representative. Heterogeneous capacities skip the whole writer family. Start normalization may sort within equal-capacity classes, which is harmless relabelling but does not add mixed-fleet symmetry rows. A physically feasible labelled route need not itself satisfy the symmetry rows before relabelling.

Bit-only interval envelopes are $\alpha b \leq w \leq \beta b$, $g - \beta(1-b) \leq w \leq g - \alpha(1-b)$. They follow by the cases b=0,1; they are **not emitted in active VD-P**, whose q block already embeds its interval bounds. Dormant receiver-source covers, Benders, pricing, D1, exhaustive subset duration, dynamic resource cuts, custom branch priorities, S buckets, GS, AMF/RC, and old timed quanta are not part of this specification. Monitoring/target callbacks do not turn the static strengthening into dynamic separation.

The proof dependency is: physical core → implicit integer service and conservation → reachability/transfer/support; budget plus reachability → rectangle → movement/penalty/S bounds; triangle lemma plus true-G membership → Gini rows; rectangle plus original cutoff → SP/scalar rows. A/B use unique nonzero one-way service independently of the implicit-integer lemma. All can be satisfied by the same physical completion, which avoids circularly assuming the validity of a tightened domain to prove its own construction. Analytic validity and actual partial support selection are separate from computational contribution; no family-level performance attribution is made without an ablation.

## A.6 Cutoff and coverage proof

**Lemma A3 (cutoff complement).** Suppose a returned lower bound $\ell$ is qualified for a model containing objective-preserving embeddings of every physical point with $G_{\text{true}} \in I$ and $F \leq c$. Then $\min(\ell, c)$ is an unconditional regional lower bound.

*Proof.* A point in the cutoff set has $F = f \geq \ell$. A point outside has $F > c$. The minimum covers both. If the cutoff model is infeasible, only that cutoff set is empty, so c, rather than $+\infty$, is the original-region floor. □

**Lemma A4 (complete cover).** Let every possible improving true-G value belong to at least one registered unit $I$, with a qualified unconditional floor $\beta_I$. Include the analytic complement, whose applicable floor is U. Then

$$L = \min\{U, \min_{I \in \mathcal{C}} \beta_I\} \tag{A18}$$

is a lower bound on the original optimum.

*Proof.* Any feasible point is in the complement or at least one covering region. Its corresponding floor bounds its objective. Taking the minimum is safe for the union. Initially $F \geq G_{\text{true}}$ allows $[0, \min((n-1)/n, U)]$ as the improving Gini range. Replacing $[\alpha, \beta]$ by both closed children $[\alpha, \mu]$, $[\mu, \beta]$ preserves coverage, including the shared endpoint. Replacement commits both children atomically with their evidence. An unsplit parent, unresolved leaf, or qualified closed leaf remains represented. Induction over updates proves the invariant. □

Bounds on the same physical domain can be combined by maximum; the union of sibling domains requires minimum. Transfer to a narrower domain requires containing scope and a handled cutoff complement, and does not construct that domain's feasible Start. Proof identities include actual model SHA, input, call, epoch, domain, cutoff and return. A local terminal status cannot certify the whole original problem. Depth/width/scheduling limits can select a request or leave an obligation open; they cannot delete it.

For a compatible own fleet, the mapped complete Start includes unused vehicles and all auxiliaries. Setting $g = G_{\mathrm{true}}$ outside a leaf is not a valid exact-g Start. Submission, API acceptance and an improved native incumbent are different observations. No optimality proof follows from H, repeated physical U or Start acceptance.

## A.7 Numerical implementation boundary

Lemmas A1-A4 concern ideal exact arithmetic and valid mathematical domains. Stored matrices use binary64 arithmetic and Gurobi 13.0.2. Native FeasibilityTol and OptimalityTol are $10^{-6}$, IntFeasTol $10^{-5}$, requested MIPGap/MIPGapAbs zero; retained evidence closure uses $10^{-7}$. A qualified result is a numerical certificate under this policy, not a rational exact proof.

In particular, the implemented guard $\alpha + \lambda \underline{P} \geq c - 10^{-9}$ can serialize $0 \leq -1$ while a mathematical point satisfies $c - 10^{-9} \leq F < c$. Therefore it is numerical closure, not strict infeasibility of every arbitrarily small improver. Its local cutoff discrepancy does not supply a bound on accumulation of all native arithmetic errors; $10^{-9} < 10^{-7}$ alone proves no such global guarantee. This manuscript changes neither guard nor tolerance. No new contradiction with the stated numerical evidence policy was found; claims of error-free rational certification are excluded.

Every numerical original-problem certificate additionally requires a physically verified own U, qualified complete source/return chains and cover, normal lifecycle completion, and the post-seal signed-gap test −1e−7≤U−L≤1e−7. Appendix B.4.1 fully distinguishes that policy, the one-sided native function and the own-zero/source-reduced branches. Tiny signed gaps are reported unchanged. A same-domain counterexample revokes its source and dependent pruning; an independent fleet can remain valid, but floor zero or a good U cannot repair invalid proof dependencies. Source-reduced results are separately labelled and get no dependent nonzero certificate credit.

The single total process deadline guarantees finite execution of this implementation, not certification within every finite window. The finite physical space and ideal complete search do not imply a linear forecast from the remaining gap.

## A.8 Writer and variable map

| Appendix group | Frozen source authority / mapping |
|---|---|
| Data, travel, physical objective | Parser.cpp; Evaluator.cpp; Result.cpp physical objective evaluator around line 109 |
| Variables and VTypes | CplexBaseline.cpp 890-1005; $a \leftrightarrow mName$, $Y \leftrightarrow yName$, $\ell \leftrightarrow lName$, $H \leftrightarrow hName$, $Z \leftrightarrow zprodName$ |
| A1-A4 | Same writer 1600-1695 and 1807-1824; ConnectivityFlow.cpp downstream-count embedding and Round20Current bounds |
| A5-A9 | Same writer 1826-1932; no Round66 arc-load replacement or Round92 handling activation |
| A10-A11 | Same writer 2241-2274 and 2329-2335 |
| A13-A16 | Same writer 2975-3154; stateName and stateGName give $s, q$; bitName and prodName give plain bits/products |
| F0 configuration and domains | PaperK1AmSf.cpp 38-114; IntervalRowFactory.cpp 391-807; writer 1697-1750, 2049-2108, 2203-2238, 2311-2327 |
| Factory global/local metadata | IntervalRowFactory.cpp 402-451; verified-incumbent “Global” remains cutoff-scoped |
| M-B mode isolation | include/Round98StateService.hpp; no projected direction, linked direction, ENS-Q or other opt-in mode |
| AM, targets, cover and lifecycle | PaperExternalGiniTree.cpp; GurobiBaseline.cpp; ProcessPhaseLedger.cpp; MipStartMapping.cpp |

The source is public at the fixed implementation snapshot. The finite production_identity authority records all 216 bindings. Line locations aid review; they do not replace derivations. This appendix supplies constraint groups and validity premises without claiming every engineering identifier as a mathematical contribution.

# Appendix B. Executable method specification

**Current-version map (Round130).** S2/MS2 uses the unchanged H and controller specified below with uniform pair-only support. Historical full-M-B traces and development variants retain their named identity. S2R is a diagnostic deletion control, not a second proposed method.

This supplement defines the controller and finite feasible startup H shared by the adopted S2/MS2 method and its full-M-B predecessor. The current S2 row policy is uniformly pair-only; the underlying H and controller definitions are unchanged. Algorithm 1 and its two complete subprocedures are in manuscript Section 5; this appendix supplies their data, numerical decisions, decoding and return semantics. Mathematical symbols refer to Appendix A. Historical names are source-map aliases, not additional algorithm variants. The definitions and proofs are retained from R125. Round126 executes only the fixed complete M-B and cold P-GRB entries on the declared new panel; it changes no algorithm.

## B.1 Active entry and parameters

The inherited settings below originate in the saved R123 full-M-B command. The current S2 command additionally selects `--round127-mode ms2`, so every three-station support predicate is false. Both versions select `gcap-frontier`, preset `research-round83-vds-equal-net-exchange`, `round98-state-service=m-binary`, `round90-lp-g-split=false`, candidate mode off, witness audit on and H verified-zero stop on. The preset recursively selects joint-seeded interroute decoded descent, VD-P and verified Start; the canonical paper configuration supplies the static row package and first-class interval controller. R122 uses this same M-B method with the panel's declared limits/Seeds. Input, lambda, capacities and physical times remain data, not tuned choices in this round.

| Item | Frozen active value / interpretation |
|---|---|
| Initial geometry | One true-G interval $[0, \min(U, (n-1)/n)]$, n>1; n=1 has G=0 |
| Split | Midpoint; two closed children; depth <8 and width $>10^{-4}+10^{-12}$ |
| Score | Balanced normalized closure, threshold .08; exact formula B.3 |
| Infeasible child policy | Two-child split, then mark proved children empty; contraction_ enabled=false |
| Native work | Gurobi 13.0.2; one thread; Seed from arm; PresolveAuto; requested MIP gaps 0 |
| Numerical policy | Scheduler epsilon $10^{-7}$; FeasibilityTol/OptimalityTol $10^{-6}$, IntFeasTol $10^{-5}$ |
| Cutoff | Current own U; writer epsilon=0; ObjCon=0 |
| Resource | One process deadline; 30 s shutdown reserve in the saved runs; preparation and H are charged |
| Products | VD-P; continuous p/d; binary direction retained; A/B on; integer loads retained |
| Row generation | Uniform two-station support, no three-station support; static physical plus shared interval factory; no dynamic user/lazy cut separation; native Gurobi cuts remain available |
| Domain option vs execution | Requested iterative/2; operative one budget intersection plus reachability intersection, recomputed by shared factory; no feedback iteration |
| H orders | 24 shuffled chromosomes from mt19937 seed 20260626; appended constructive order; no genetic generations |
| H decoder | Full compaction mode 1; cache limit 200000 complete decoded orders; no generation/stagnation quota in this active branch |

The "interval-mip-core-no-exhaustive-subset-duration" policy is overridden by the VD-P preset to `round55-vd-p`, still excluding exhaustive subset duration. Compiler identity is the frozen MSYS2 UCRT64 g++ binary

SHA256 `8cd0aa578687e4182532222c8734439b3f34ac78b6ce23cba1071c5838025fc1`. Library shuffle and equivalent-key sort order are implementation-version details; the paper does not invent a canonical cross-library tie rule absent in the source.

The unique paper entry is fully resolved in paper_method_config.json; argv_freeze.json contains the exact six current S2 argument vectors, six independent cold controls and twelve diagnostic control vectors. PE and DLL SHA256 bind every preset/default value that has no separate CLI option. H seed 20260626 and all native settings are explicit configuration values, not instance-dependent choices. The template below is one token vector (line breaks are presentation only); INPUT, OUT, physical T, producer B and B_MINUS_6 and SEED take exactly their fixed group-table values. The existing supervisor starts the complete production clock before admission, invokes this vector, and seals normal output; direct native Runtime alone is not T_cert.

```
PE --input INPUT --lambda 0.15 --T T
   --pickup-time 60 --drop-time 60 --time-limit B_MINUS_6
   --process-wall-time-limit B --process-shutdown-margin 30
   --threads 1 --mip-threads 1 --gurobi-seed SEED --gurobi-presolve -1
   --method gcap-frontier
   --algorithm-preset research-round83-vds-equal-net-exchange
   --round90-lp-g-split false --round61-candidate-mode off
   --round65-witness-audit true --round65-hga-zero-stop true
   --out OUT/result.json --log OUT/native.log
   --process-phase-ledger OUT/phases.csv
   --external-gini-artifact-dir OUT/external
   --primal-heuristic-generation-log OUT/hga.csv
   --progress-log OUT/progress.csv --native-evidence-dir OUT/journal
   --round98-state-service m-binary --round127-mode ms2
```

## B.2 State, identity and order

A coverage unit has id, interval endpoints, parent id, child index, depth, base/inherited bound, current bound and sources, cutoff, status, and replacement flag. Active statuses are Open (Invalid may remain selectable as an unresolved obligation); Closed, Empty and Fathomed retain their proof role. Replaced parents are represented by their committed children. TerminalReady/Coalesced are dormant sibling-coalescing states, not normal M-B actions. Native status is not interchangeable with coverage status.

Runtime state separately contains canonical artifact path/SHA/row signature, generation and incumbent epoch; LP result (terminal-valid, optimal/infeasible, bound available, raw bound), LP epoch/SHA/endpoints; cached speculative child objects; terminal-started state; native phase count; and the NEXT milestone bit. Own witness state contains complete routes, integer operations, inventory, recomputed U and true G. The model's epigraph G never replaces the latter. Epoch starts at zero and increments on an independently verified incumbent improvement exceeding $10^{-9}$. Callback observations and physical witnesses retain separate identities.

Selection uses the **current own U** as cutoff, not an old source cutoff. It considers unreplaced Open/Invalid units with α<U−epsilon and b<U−epsilon. Find their minimum bound and form the epsilon plateau $b_I \le b_{min}+\epsilon$. Sort the plateau by decreasing width, increasing lower endpoint, increasing upper endpoint, then lexicographic id (the first three comparisons use $10^{-12}$). Cycle through this order. Reset the cursor when the plateau membership/order changes or its minimum changes by more than epsilon; after a full cycle increment the tie round. Effort, elapsed time and requested_quantum fields do not affect this `selectNextByBoundOnly` call.

On a new epoch, before reusing a selected artifact verify its old saved bytes, discard its retained backend model, clear artifact/LP state, terminal flags and child cache, and build a new generation at the current cutoff. Qualified old-domain lower bounds remain on their covering units: a smaller cutoff does not invalidate a bound already completed by its old cutoff complement. The NEXT milestone bit is initialized false in each newly created runtime and set true only by a returned NEXT target. **It is not reset in the active epoch invalidation block.**

Child-target returns do not set it. No stronger child bound is sent upward to the parent merely because the child has a higher value.

Parent LP reuse requires complete evidence and the same incumbent epoch. A retained artifact must exist and match its SHA. The ordinary cached-child reuse gate requires the parent's ready flag, exactly two cached children and complete child runtime LPs; epoch invalidation clears that flag before reuse. The more elaborate endpoint/path/SHA `validRound90LpGCachedChild` check is conditional on round90 and is off here; its existence is not claimed as an active gate. When reusing children, their stored bounds inherit max(old child bound,current parent bound) and their cutoff is refreshed. AM still reads the **raw LP bound fields**, not those inherited maxima. Missing/incomplete cache falls back to constructing and solving both children; it is never interpreted as infeasible.

### B.3 NEXT, child target and terminal decisions

Call NEXT after the selected parent has completed its LP and merged any qualified bound; re-read the current bound b. The function first rejects nonfinite bounds, and first requeues if any other relevant bound satisfies $b_j + \epsilon < b$, even if the milestone was already reached. Otherwise, if the milestone is false and any $b_j > b + \epsilon$, return NEXT with $t = \min\{b_j : b_j > b + \epsilon\}$. If there is no such bound, or the milestone is true, allow lookahead. The caller supplies only other **Open/Invalid schedulable units** whose current bound is below current U−epsilon (and whose interval remains relevant). Closed, Empty, Fathomed and replaced units are excluded from NEXT competitors even though their floors remain in proof coverage. Thus a lone closed unit at .40 does not supply a NEXT target to an open unit at .20. Source-cutoff metadata governs bound scope, never this current scheduling filter. The function does not use a timed quantum.

If geometry is ineligible, proceed to terminal MIP after NEXT permits lookahead. If eligible, form closed midpoint children in index order 0,1. Each is speculative until the complete replacement transaction. Solve both complete LP relaxations, or use the current admissible cache. A terminal valid child means either proved infeasible, or optimal with an available finite bound; TimeLimit/UNKNOWN is not terminal valid. Deadline exits retain the parent; an unexpected nonterminal/invalid LP is an engineering failure with the parent retained, not permission to prune it. The optional UNKNOWN→core-due research branch is off.

If either qualified child LP is infeasible, AM immediately requests a two-child atomic split. With contraction=false this applies also when both are infeasible; only after installing both does the controller mark the infeasible child or children Empty and discard those backend models. The inactive one-child contraction/both-child parent-close code is not part of M-B.

For two finite optimal LPs with raw bounds $\ell 0$,$\ell 1$, use

$$\Delta = \max(U - b, \max(\epsilon, 10^{-12})), \quad v_j = \min(1, \max(0, (\ell_j - b)/\Delta)), \quad \eta = \min(v_0, v_1), \quad \mu = (v_0 + v_1)/2. \tag{B1}$$

$$\begin{aligned} s &= \max(1, |b|, |\ell_0|, |\ell_1|, |U|), \\ \delta_b &= \max(\epsilon, 32\epsilon_{mach} s), \\ \delta_{score} &= \min(1, \delta_b/\Delta + 32\epsilon_{mach}), \quad \epsilon_{mach} = 2^{-52}. \end{aligned} \tag{B2}$$

Put $b_+ = \min(\ell_0, \ell_1)$. If $b_+ \le b + \epsilon$, return terminal; else if $\eta\mu + \delta_{score} \ge .08$, split; else request a **parent-domain MIP** with bound target $b_+$. A child target is an intended stopping threshold, not a bound to merge into the parent before native evidence returns. A returned target requeues the parent and re-evaluates the current rule; it does not force a delayed split. This distinguishes NEXT (competitors), child target (lookahead), and terminal (complete parent MIP without a bound-target stop). The recorded adaptive rho is $\min(1, .08/\max(\mu, \max(\epsilon, 10^{-12})))$; it is diagnostic, not another active score rule.

### B.4 Native requests, Starts and returned actions

An LP request relaxes the actual original VTypes, keeps the complete row set, and retains the model when permitted. A MIP request restores the exact saved VTypes before optimization. Targets and terminal requests receive the remaining common deadline after preparation, with native TimeLimit set/read back; no new per-leaf budget is created. Native bound callbacks can request termination when the qualified target is reached. Such a request is not the native return itself.

For enabled verified Start, normalize the current own fleet only among equal-capacity vehicle classes, stable sorting by decreasing number of operations and assigning smaller vehicle ids first. Clear explicit previous Starts even if the replacement is ineligible. Reverify routes, true G and F. Check interval membership with $10^{-8}$ and non-strict cutoff with $10^{-8}\max(1,|F|,|c|)$. An outside-interval/cutoff fleet is simply ineligible, and the MIP can run without it. Other mapping failures fail the request's engineering gate; they do not establish infeasibility.

For each used route set selected arcs to one, visit z=1, direction to pickup indicator, quantities to its integers, load to each prefix and rank to position. Flow along a q-station route is q,q−1,…,1,0. All absent route columns, including unused vehicles, are zero. Set (Y,r,e,H,G,Z) to physical values, state selector to one at Y, state_g to true G there and zero elsewhere, W_SP=SP, and the unused extrema to actual min/max ratios. Check every actual column bound/type, every actual linear row at $10^{-7}$, and actual objective including ObjCon at $10^{-7}$; require minimization and no unexpected nonlinear structure. Then submit a complete Start and read it back. Submission, native-log acceptance, integer-vector observation and a later improved incumbent are different observations.

The outcome object is consumed only after the optimize call returns and request finalization detaches callbacks/settles native data. Retaining the same in-memory model can preserve useful solver state, but this paper does not promise that the entire native branch-and-bound tree survives type changes or later calls. Once returned, merge finite qualified bounds only on the actual interval/cutoff; replay a proposed fleet before accepting an improved own U. This order precedes the next controller transaction.

| Returned condition after required identity, finalization, zero-gap-setting and consistency gates | Coverage action |
|---|---|
| LP optimal with finite qualified bound | Merge into that parent, or save on speculative child; child alone changes no active cover |
| LP infeasible | Parent Empty, or speculative infeasibility retained until atomic split |
| Target MIP OPTIMAL with bound / INFEASIBLE | Close selected cutoff obligation; retain cutoff-complement floor; discard its model and speculative child models |
| Target reached flag, termination-request flag, available bound ≥target−epsilon | Merge returned bound; set milestone only for NEXT; requeue; preserve parent and siblings |
| MIP interrupted/TimeLimit before target | Keep qualified partial bound if engineering gates passed; leave region open and end the common lifecycle |
| Terminal MIP OPTIMAL / INFEASIBLE | Close its own cutoff obligation; never certify siblings by this status |
| Terminal MIP interrupted | Leave region open and stop |
| Unsupported status, missing optimal bound, failed identity, incomplete LP or native error | Stop with explicit failure; preserve unresolved coverage; dependent unqualified evidence is not admitted |
| No launch because deadline expired | Leave region and previous evidence unchanged; finalize open result |

The implementation merges same-unit bounds with max(old bound,new bound), without clipping to U. Empty/Fathomed stores max(previous bound,cutoff); Closed keeps its qualified returned floor. The raw returned L is the scheduler's complete-cover minimum, also **without clipping to U**. Tiny negative signed gaps can therefore occur and remain recorded. On the empty relevant-set path the scheduler helper alone returns 0; the enclosing lifecycle/closure logic determines the outcome.

For the exact mathematical proof, each cutoff-bound source has unconditional regional floor min(bound,source

cutoff), and the analytic complement contributes its cutoff floor (A.6). This is proof accounting, not an extra min-assignment performed by the frozen scheduler. Raw numerical evidence and exact proof-accounting expressions must not be interchanged. A.7's numerical closure exception still applies.

### B.4.1 Certificate admission: native function and sealed evidence

The native certificate function requires valid complete coverage, all relevant leaves closed, all used bounds certified, monotone bound history, a verified own finite U, completed lifecycle, and source consistency. It then accepts the one-sided test L+max(0,tolerance)≥U. It does **not** enforce a lower bound on the signed gap U−L. In particular, a negative gap of −2e−7 is not rejected by that predicate alone.

The published post-seal reader applies the stronger admission policy. Each native source must have a finished request with matching interval, epoch, model bytes/row signature and parentage; the saved parameters must match the frozen original model (including requested zero relative/absolute MIP gap); every used bound must be finite, available and sourced to the corresponding returned native request, without a counterexample or consistency failure. It reconstructs the represented children and their full interval union, preserves unresolved obligations, requires every relevant unit closed for certification, replays the own physical fleet and objective, and requires normal process completion (no killed, timed-out producer, missing seal, or failed engineering gate). These gates are conjunctions, not substitutes for one another. A clean nonzero certificate additionally requires −1e−7≤U−L≤1e−7; gaps are never clipped. Thus the above −2e−7 case is rejected after sealing.

The clean independent own-zero branch requires a physically replayed own U≤1e−12, nonnegative lambda and weights, and **no started native request**; it returns L=0 and a synthetic complete zero cover. The startup early-stop test is 0≤verified F≤1e−12; the neutral physical loop itself stops only on exact F=0. If native sources are damaged, the source-reduced fallback is stricter: it needs every Y_i=D_i and recomputed F=G=P=0 exactly. Its independent L=0 does not restore any invalid native source or dependent nonzero certificate. A retained positive feasible fleet alone supplies U, never certificate credit for damaged proof dependencies.

These are distinct functions in the frozen evidence chain: ExternalGiniTree.cpp:460-489 for native admission; read_round122/123 → read_round118.item/finalize → read_round116.reconstruct_arm → round108_reader.endpoint (328-391), with CLOSURE_TOL=1e−7 and ZERO_TOL=1e−12; round114_claims/round114_audit supplies damaged-source fallback. The rules are stated here so those links are audit references, not missing algorithm definitions.

## B.5 Teaching states and a separately sourced event

These first numbers are **synthetic teaching states**, not measurements. Take U=1, epsilon=$10^{-7}$. With selected b=.20 and competitor bounds (.20,.35,.60), the plateau contains the tied .20 units and NEXT=.35. After a qualified NEXT return b=.35 the milestone is true; if all other relevant bounds are at least .35, lookahead is permitted. (If the tied .20 unit still exists, the requeue check wins first.)

For a fresh controlling parent b=.20 with no strict competitor, children (.21,.30) give Δ=.8, v=(.0125,.125), score=.000859375, b+=.21: run parent MIP to .21. Children (.20,.90) instead give b+=b: terminal, regardless of the strong sibling. Children (.50,.60) give v=(.375,.5), score=.1640625: split, install both children, with union floor .50. If the other covering unit has floor .25, global L remains .25. A target TimeLimit at .205 keeps the parent open at its qualified .205 bound; a missing Start changes none of these implications.

The following is an **actual R123 saved event** from M-B's `adaptive_mass_decision_ledger.csv` and `paper_tree_events.csv`, selected as a finite trace excerpt. On L0.0.0=[0,.09308544387476185], depth 2, b=.33371261877766062, raw child LPs (.34112380854511071,.35882784229299913), U=.3723417754990474

give eta=.19185481632186216, mu=.42100858578645295, score=.08077252489598688 and score tolerance=.0000025887181797865216. The decision is split; the event at telemetry 127.2749697 records atomic replacement and global floor .34112380854511071. Earlier recorded splits at 38.3552841 and 72.829994 used the infeasible-child two-child branch. These times are native-run telemetry, not newly measured clocks or a reconstructed unknown checkpoint. Compact source excerpts and hashes accompany the finite checks.

### B.6 H encoding and top-level procedure

Physical routes are labelled lists $0, i_1, \ldots, i_q, 0$ with one signed integer operation $a_i = p_i - d_i \neq 0$ per station; missing vehicle means empty route. H's **order encoding** is different: M ordered lists partition all n stations, including a tail that the decoder may not execute. A chromosome is a permutation of station ids 1..n and M−1 identical separator tokens; splitting at separators preserves empty vehicle lists. Reordering an encoded tail does not itself describe a feasible service plan. Decoding returns signed operations; materialization deletes zero-operation stations, closes used routes at depot and verifies the entire fleet.

```
H(instance, common deadline):
    C := empty physical fleet, verify Y=b
    repeat choose and admit best joint insertion (B.7) until none/deadline
    keep verified C as an own candidate
    generate all 24 shuffled order chromosomes with mt19937(20260626)
    append C's order after deterministic tail completion (below)
    initialize a shared complete-decode cache, empty
    decode all 25 orders in order, publishing verified improvements
    for each seed in order:
        repeat generate guided candidates (B.10), order by proxy fitness
        fully decode candidates in that order
        accept first fitness improvement >1e-12; restart this seed's pass
        if every candidate checked without improvement: finish this seed
        honor verified-zero/deadline between calls and candidates
    take the best independently verified candidate, including C
    run strict insertion/quantity closure followed by best neutral block move;
        repeat these two phases (B.11) until zero/no neutral/deadline/failure
    return best admitted complete fleet and stop reason
```

Constructive tail completion starts with C's visited order and each vehicle's complete physical duration as a proxy. For each unvisited station i in increasing id, append it to the vehicle minimizing proxy[k]+tau(last,i)+tau(i,0)−tau(last,0), choosing the first vehicle on an exact tie; update that proxy. The remaining operations are decided by the same full decoder. Twenty-four shuffles are generated before appending this extra seed, preserving random draws. `constructive_mode=0` means there are no additional biased population members. The snapshot's C++ shuffle is authoritative for exact seed orders; the algorithm specifies the generator/seed/order and permits cross-library sequence differences. HGA crossover, mutation, education generations, 2000-generation stagnation and obsolete "10 iterations" fields do not run in this decoded-descent branch.

The outer physical candidate store keeps a candidate only after full verification and an improvement greater than $10^{-10}$, unless no incumbent exists. Verified-zero can terminate startup early. This publication threshold differs from inner $10^{-12}$ move thresholds; an internally accepted tiny improvement need not replace the outer witness.

### B.7 Joint insertion

At a verified fleet let $L_k(a)$ be load on leg a, numbering the original legs from 0 through the number of visited stations. For each ordered (pickup station i or 0,drop station j or 0), excluding (0,0), i=j, and already visited nonzero stations, enumerate vehicles and positions. Zero denotes an absent side, not depot stock. Both sides

use the same integer quantity q. For a pair choose original legs a≤b, inserting i then j; on one leg the sequence is predecessor,i,j,successor. Travel increment is the sum of replaced arc differences (one combined expression when a=b).

The load upper amount is $Q_k - \max_{v=a..b} L_k(v)$ for a pair; $Q_k - \max_{v=a..end} L_k(v)$ for pickup only; $\min_{v=a..end} L_k(v)$ for drop only. Intersect with available inventory at i, room at j and max vehicle capacity. Complete duration increment is travel increment plus hq when a pickup is inserted, and just travel increment for drop only (that drop replaces return unloading). The time test uses T+$10^{-7}$. For each amount, retain the cheapest feasible travel placement, ties (vehicle,a,b), by a suffix scan of placement amount limits. Evaluate all amounts 1..limit using F on the changed one/two inventories; accept only gain>$10^{-12}$.

Choose globally: nonpositive added duration before positive; then greatest gain per added duration (compare cross products, or gain for both nonpositive), greater gain, smaller added duration, and lexicographically (i,j,q,k,a,b). Apply and fully replay; require objective agreement within $10^{-10}$ and verified gain>$10^{-12}$. Every insertion adds at least one previously unvisited station, so construction has at most n accepted insertions. This finite placement/amount scan is a heuristic neighborhood, not an optimization backend.

### B.8 Single and paired quantity modification

Only currently served stations participate. Enumerate a increasing, then b=0 (single) and each served b>a (pair). For every nonzero integer t set $a'_a = a_a + t$, and $a'_b = a_b - t$ if b exists. Station bounds give

$$b_a - C_a - a_a \le t \le b_a - a_a; \qquad a_b - b_b \le t \le a_b - b_b + C_b \quad (b \neq 0). \tag{B3}$$

Intersect these with every affected prefix load $0 \le L + \chi t \le Q_k$, where chi is +1 after a, −1 after b, zero before/beyond the cancelling pair. This includes cross-vehicle modifications; each route has its own capacity. Enumerate all integers in the resulting interval increasing, skip zero. Convert signed operations to pickup=max(a',0), drop=max(−a',0), remove zero stops and shortcut travel. Sign reversal is allowed. Check complete duration, including depot unloading, at $10^{-7}$; evaluate F; require gain>$10^{-12}$. Choose lowest proposed objective, tie (a,b,t). Full replay and objective agreement within $10^{-10}$ precede acceptance. Missing stations cannot be added by this operator; insertion handles them.

Strict closure computes both the best insertion and best quantity move on the same current fleet, chooses smaller proposed F, insertion on an exact tie, admits the verified strict improvement, and repeats. If neither exists, that union neighborhood is exhausted. Verification failure retains the previous fleet and records failure. This is exactly the Round76 subprocedure called by active exchange descent; the standalone Round75/76 preset entry branches are not separately run by main.

### B.9 Complete greedy decoder and its compensating moves

For each encoded route take its maximal prefix whose travel with direct return stays within T+$10^{-12}$, stopping at the first exceeding station. Fix this travel during that decoder pass. Set the cumulative pickup budget $B_k = \max(0, \lfloor (T - travel + 10^{-12})/h \rfloor)$ (an integer sentinel max-int/4 if $h \le 10^{-12}$); this is a handling budget, not Q. Initialize all operations zero and Y=b. The single-route proxy version instead keeps other-route operations fixed. The active mode is 1: visit vehicles and stations forward, greedily match each current surplus above clipped target to later deficits, as much as remaining supply, demand, station bounds, all intervening load bounds and pickup budget allow; leftover surplus may be collected for depot return. This constructive initialization does not require each step to decrease F.

Define an interval move by vehicle k, positions a<b, sign $\sigma \in \{+1, -1\}$, amount $q \geq 1$. Increase a's signed operation by sigma q and decrease b's by sigma q. The sink may instead be the depot after the last position, in which case only a changes. Loads on [a,b) change by sigma q. Each final inventory must stay in [0,C], loads in [0,Q], and total positive operations in [0,B_k]. Thus an earlier delivery can be reduced/reversed and a pickup compensated elsewhere; there is still only one net one-way operation per physical station. Feasible amount bounds come from station room and the minimum/maximum load on the changed interval; the positive-part pickup change is piecewise linear with breakpoints at operation sign changes. The implementation finds the largest feasible q by these segments, then evaluates every positive integer up to it. The feasible amount sublevel contains q=0 and is an interval, so no integer hole is skipped. Objective arithmetic updates the changed ratios, pairwise absolute sum and penalties; it is not an LP/DP solve.

The active decoder loop is:

```
repeat:
    repeatedly apply the best strict atomic interval move, until none
    if a strict same-route composite exists: apply best; restart
    if a strict two-route composite exists: apply best; restart
    if a strict unit interval polish exists: apply best; restart
    if a strict flow-pair redistribution exists: apply best; restart
    if no atomic move occurred in this pass: return current operations
```

Atomic enumeration is vehicle, source position, later sink positions then depot, sign + then −, amount increasing. Keep the best move and the eight best atoms per route. To update the best, initialize value F−EPS and replace only when candidate+EPS<best, EPS=$10^{-12}$; this gives an effective initial gain test >2EPS. For each bounded list, append candidates until size K ($K = 8$ atoms or $K = 6$ local candidates), maintaining a max heap with comparator a.objective<b.objective. Once full, replace its front only if candidate.objective+EPS<front.objective, then restore the heap. Final sorting compares only ascending objective, with no secondary key. Equivalent-key heap positions consequently depend on the frozen standard library. Local candidates comprise each retained atom and each pair of distinct atoms on one route, adding their operation deltas simultaneously. Recheck all combined loads, positive-operation budget and station bounds, then retain at most six best candidates per route using the same bounded-heap objective logic. Candidates can individually worsen F: their combination may compensate that worsening. Same-route best selection uses F−EPS and candidate+EPS<best. Cross-route candidates combine one retained local candidate from each pair of vehicles, recompute the full common Gini/penalty objective for their jointly changed (up to eight) stations, and use the same strict test. Distinct routes have disjoint stations, so combined physical feasibility follows from their checked local feasibility; joint objective evaluation remains necessary.

Unit polish checks amount 1 for every legal atomic interval, with initial best F−$10^{-14}$ and candidate+$10^{-14}$<best. Flow-pair polish ignores zero operations, partitions the route into consecutive same-sign runs, and pairs each pickup run with its immediately following drop run. For each ordered pair of distinct such run pairs, choose one pickup and drop position in each: subtract one unit from the first pickup/drop flow and add one to the second. Check both interval load changes together, station bounds and original signs; pickup total is unchanged. Its candidate helper still requires gain>$10^{-12}$, followed by the $10^{-14}$ best-comparison guard. Residual pair-unit, route-exact and pair-exact neighborhoods return false in this active implementation; they are not hidden optimization oracles.

After a first full decode, if its executed prefix contains zero-operation stations, delete them, keep only its nonzero active order, and run this same full decoder from zero operations **once more** on those compacted routes. Return whichever has smaller F, first decode on exact tie. This does not repeat compaction to a fixed point and does not restore the removed tail. The cached object contains complete signed operations and fitness=−F. Materializing that object removes remaining zero operations and independently checks the original physical route duration.

Decoder cache key is compaction mode plus the ordered lists with separators; it includes tail positions, even if they decode identically. On a hit return the entire stored decode. On a miss at size≥200000 clear the whole map

before inserting; this is neither LRU nor a permanent visited-state ban. Internal atomic/local caches are tied to an incremented operation-state epoch and cleared after accepted updates. The static ratio/penalty preprocessing cache keys capacities, targets and weights (including source slice offsets); it affects arithmetic reuse, not candidate admission. No general accepted-state de-duplication is claimed.

### B.10 Guided order descent and ties

For each route compute the travel-only executable prefix as above. Within it record first negative-operation position and last positive-operation position. A station's role is pickup if its decoded operation>0, drop if <0; if zero, use surplus/shortage of decoded final inventory relative to target, otherwise neutral. Supply anchor is first drop, or position 0; demand anchor is immediately after last pickup, otherwise prefix end.

Generate, in vehicle order: move the last zero-operation prefix station to that route's end; collect every nonneutral prefix position plus the first supply and first demand in its tail. Move a supply lying after first drop, or in tail, to the supply anchor; move a demand before last pickup, or in tail with a last pickup present, to the demand anchor. Duplicate generated candidates are not globally removed. Same-route erase/insert decrements destination position if it originally lay after the erased position; skip unchanged order. For interroute proposals take only the first supply/demand in each source tail, each different destination in order: insert supply at destination's supply anchor if that route contains any demand-role station; insert demand at its demand anchor if it has an executed pickup.

The proxy fully runs the single-route incremental decoder with other-route operations fixed. For a two-route change try both route decoding orders and take the larger fitness (smaller F). Proxies only order proposals: sort decreasing proxy fitness, then fully decode in that order and accept the first fitness increase>$10^{-12}$. An unsuccessful pass evaluates every generated proposal, including a poor proxy. There is no top-k rejection here.

The guided comparator has no secondary key: equal proxy values are equivalent to C++ `std::sort`, which is not stable. Likewise internal bounded-heap/objective-only sorts have implementation-dependent equivalent-key order. For the frozen compiler/library they supply the recorded run's repeatable behavior; a cross-language implementation may choose a different equivalent-key order and consequently a different heuristic path. Specifying lexicographic ties here would change the frozen algorithm. By contrast B.7/B.8 and the physical block operators do have explicit deterministic ties. This limitation affects path-level reproduction, not feasibility, exact model scope or native lower-bound validity.

### B.11 Neutral block moves and exit argument

Strict closure is followed by one best neutral move from the union below. The potential is the **complete** vector of all vehicle durations, sorted decreasingly, including zeros for unused vehicles; compare it with exact C++ lexicographic <, never by an aggregate sum or a tolerance that ignores an earlier increase.

Balanced relocation enumerates source vehicle, all nonempty contiguous half-open position blocks [a,b), other target vehicle and every insertion leg. Require net signed operation zero. If relative prefix load extrema inside the block are lo,hi, require target load+lo≥0 and target load+hi≤Q_target. Removing a net-zero block leaves source suffix loads unchanged; inserting it leaves target suffix loads unchanged. Keep operation quantities and order inside the block; recompute both full durations and accept only if each≤T+$10^{-7}$ and the sorted vector strictly improves. Choose smallest potential, tie (source,a,b,target,leg).

Equal-net exchange enumerates every contiguous nonempty block per route, bucketed by integer net operation in increasing net order; pair vehicles s<t and blocks from equal-net buckets. At each insertion prefix require its old load plus the incoming block's relative extrema in that vehicle's [0,Q]. Equal net preserves all suffix loads, even when the common net is nonzero. Replace the two blocks in their original internal order,

preserve all station operations, recompute complete durations and compare the same potential. Choose smallest potential, tie $kind, source, a, b, target, target_a, target_b$, with kind 0 relocation before kind 1 exchange. `bestRound83Neutral` evaluates the whole relocation neighborhood first and then exchanges.

Before accepting a neutral candidate, independent replay requires identical final-inventory vector, **exactly equal recomputed objective**, and exact equality between recomputed duration potential and the proposed potential, which must strictly improve the prior one. Failed admission retains the prior fleet. After an accepted neutral move, run strict closure again. Stop on exact verified objective zero, common deadline, failed verification, or closure plus no improving neutral move. A selected no-move event is an exhaustion claim only for these enumerated neighborhoods.

In ideal arithmetic, after the finite construction the accepted physical phases strictly decrease (F,sorted durations) lexicographically, and the finite space of labelled routes/integer quantities excludes cycles. The order-descent phase strictly increases its deterministic decoded fitness on a finite set of orders. The implementation uses incremental floating arithmetic and tolerances in the decoder; these comparisons are not a proof of exact rational monotonicity or an accumulated error bound. The physical block phases have exact inventory/objective equality for neutral steps, unlike a merely tolerance-neutral rule. There is no fabricated de-duplication argument. The common deadline and external lifecycle cap are the operational stopping boundary; expensive decoder calls are finite routines and do not have a separate native optimization timeout. H neighborhood closure never certifies the original optimum.

### B.12 Separately identified inclusive-root research control (Round128)

MD-R version 1 is a research ablation, not another proposed algorithm. Section 6.8 and the root contract define its same-H inclusive domain, semantic shortcut omission, mandatory actual own-Start pre-LP check, one LP plus one integer MIP, fixed cutoff and complete native optimal-root return obligation. It retains the physical and signed numerical evidence policy. TIME_LIMIT/INTERRUPTED may yield a qualified open endpoint with null T_cert; a near-equal bound alone cannot satisfy its missing optimal-return obligation. Its extra checks and complete costs are paid. These control-specific guards leave the frozen ordinary procedure and the historical evidence in this appendix unchanged.

### B.13 Uniform support candidate confirmation (Round129)

Round129 changes no production source, H, model generator, numerical policy or controller. It reuses original E2 modes mb/ms2/p. The sole candidate MS2 suppresses triple support for every input while retaining pairs, ordered supply and the shared enumeration order/cap. The five fixed candidate rules and thirty complete paid observations are in Section 6.9. At the R129 selection stage, MS2 was the proposed candidate and full M-B was the then-current frozen method. That historical distinction and all tables remain; R130 now adopts S2/MS2 as the sole paper method. The independent finite review and public recovery are evidence checks, not new native solves or rational proofs of all matrices.

### B.14 Current pair-support version and fixed attribution controls

The sole paper method is original E2 `ms2`, called S2. E3 retains its source behavior and adds the off-by-default unique `s2-root` identity plus PS parameter evidence readback. E2 is retained unchanged. PS reuses the full original-compact mapper/validator; S2R reuses R128 inclusive-root semantics with triple emission disabled. Section 6.10 states the paid workflows, exact cutoff complement, nonzero LP/MIP obligations, source/return/coverage requirements, actual exposure and attribution limits. No individual cut, H, threshold or native parameter is optimized here. The passed R129 candidate selection is not rerun.

## Complete historical-result display supplement

This display is generated one way from the six unchanged R125 table authorities. Numeric display uses twelve significant figures; the adjacent original CSV/Markdown files retain their exact published bytes and full precision. N is a nonzero numerical certificate; O is a qualified open endpoint. Historical WIN labels retain their old endpoint rule and are not new certification wins.

## Table S1a. All 32 R122 physical/bound endpoints

| Role | Seed | Method | U | L | Signed gap | Tier |
|---|---|---|---|---|---|---|
| PB-C20-B | 0 | P-GRB | 0.224680311823 | 0.176578375157 | 0.0481019366668 | O |
| PB-C20-B | 0 | ENS-C | 0.224680311823 | 0.224680311823 | -1.11022302463e-16 | N |
| PB-C20-B | 0 | M-B | 0.224680311823 | 0.224680311823 | 1.05471187339e-15 | N |
| PB-C20-B | 0 | MB-D1 | 0.224680311823 | 0.224680311823 | 2.49800180541e-16 | N |
| PB-B20-S | 0 | ENS-C | 0.863152320565 | 0.863152320565 | -2.22044604925e-16 | N |
| PB-B20-S | 0 | M-B | 0.863152320565 | 0.863152320565 | -2.22044604925e-16 | N |
| PB-B20-S | 0 | MB-D1 | 0.863152320565 | 0.863152320565 | -2.22044604925e-16 | N |
| PB-B20-S | 0 | P-GRB | 0.863152320565 | 0.863152320565 | 1.11022302463e-16 | N |
| PB-C35-U | 0 | M-B | 0.169636402195 | 0.0839518542929 | 0.0856845479019 | O |
| PB-C35-U | 0 | MB-D1 | 0.179300436585 | 0.08353399353 | 0.0957664430545 | O |
| PB-C35-U | 0 | P-GRB | 0.174771887556 | 0.0597029031719 | 0.115068984384 | O |
| PB-C35-U | 0 | ENS-C | 0.164237487795 | 0.0815753937702 | 0.0826620940247 | O |
| PB-B35-B | 0 | MB-D1 | 0.234581409857 | 0.0920079836183 | 0.142573426238 | O |
| PB-B35-B | 0 | P-GRB | 0.305555778217 | 0 | 0.305555778217 | O |
| PB-B35-B | 0 | ENS-C | 0.246504470413 | 0.13228796803 | 0.114216502383 | O |
| PB-B35-B | 0 | M-B | 0.246504470413 | 0.127634301846 | 0.118870168567 | O |
| PB-C50-S | 0 | MB-D1 | 0.372341775499 | 0.356874739941 | 0.0154670355577 | O |
| PB-C50-S | 0 | M-B | 0.372341775499 | 0.356874739941 | 0.0154670355577 | O |
| PB-C50-S | 0 | ENS-C | 0.372341775499 | 0.357980799883 | 0.014360975616 | O |
| PB-C50-S | 0 | P-GRB | 0.425950075294 | 0.349220695801 | 0.0767293794932 | O |
| PB-B50-U | 0 | P-GRB | 1.17695575854 | 0.0683793824136 | 1.10857637613 | O |
| PB-B50-U | 0 | MB-D1 | 0.677298419001 | 0.471234869552 | 0.206063549448 | O |
| PB-B50-U | 0 | M-B | 0.697473509598 | 0.45534539996 | 0.242128109638 | O |
| PB-B50-U | 0 | ENS-C | 0.832548015217 | 0.452360367383 | 0.380187647834 | O |
| PB-B35-B | 1 | ENS-C | 0.234581409857 | 0.114641011444 | 0.119940398412 | O |
| PB-B35-B | 1 | P-GRB | 0.279798434442 | 2.02718966417e-18 | 0.279798434442 | O |
| PB-B35-B | 1 | MB-D1 | 0.234581409857 | 0.138322042028 | 0.0962593678281 | O |
| PB-B35-B | 1 | M-B | 0.246504470413 | 0.0982245517985 | 0.148279918614 | O |
| PB-C50-S | 1 | M-B | 0.372341775499 | 0.357008333973 | 0.0153334415264 | O |
| PB-C50-S | 1 | ENS-C | 0.372341775499 | 0.358161762014 | 0.0141800134847 | O |
| PB-C50-S | 1 | P-GRB | 0.391780558579 | 0.349450859646 | 0.0423296989334 | O |
| PB-C50-S | 1 | MB-D1 | 0.372341775499 | 0.357008333973 | 0.0153334415264 | O |

## Table S1b. All 32 R122 complete clocks

| Role | Seed | Method | Cap | T_produce | T_verify | T_total |
|---|---|---|---|---|---|---|
| PB-C20-B | 0 | P-GRB | 900 | 872.589019559 | 1.23015520003 | 873.819174759 |
| PB-C20-B | 0 | ENS-C | 900 | 430.5575901 | 3.21318039999 | 433.7707705 |
| PB-C20-B | 0 | M-B | 900 | 395.8595413 | 3.24970759999 | 399.1092489 |
| PB-C20-B | 0 | MB-D1 | 900 | 665.0687819 | 4.85072250001 | 669.9195044 |
| PB-B20-S | 0 | ENS-C | 900 | 4.62243311576 | 1.44725659999 | 6.06968971575 |
| PB-B20-S | 0 | M-B | 900 | 2.6319677 | 1.57636520002 | 4.20833290002 |
| PB-B20-S | 0 | MB-D1 | 900 | 2.6983887 | 2.1314255 | 4.8298142 |
| PB-B20-S | 0 | P-GRB | 900 | 116.0327756 | 1.02186559999 | 117.0546412 |
| PB-C35-U | 0 | M-B | 2400 | 2372.52643333 | 8.951676 | 2381.47810933 |
| PB-C35-U | 0 | MB-D1 | 2400 | 2370.6203998 | 11.0386199 | 2381.6590197 |
| PB-C35-U | 0 | P-GRB | 2400 | 2370.6161682 | 1.09987460001 | 2371.7160428 |
| PB-C35-U | 0 | ENS-C | 2400 | 2370.5989279 | 8.8968437 | 2379.4957716 |
| PB-B35-B | 0 | MB-D1 | 2400 | 2372.44035804 | 10.1910831 | 2382.63144114 |
| PB-B35-B | 0 | P-GRB | 2400 | 2370.5561489 | 0.518086399999 | 2371.0742353 |
| PB-B35-B | 0 | ENS-C | 2400 | 2370.552666 | 9.1507964 | 2379.7034624 |
| PB-B35-B | 0 | M-B | 2400 | 2371.7861588 | 8.1583865 | 2379.9445453 |
| PB-C50-S | 0 | MB-D1 | 3600 | 3572.88123229 | 40.0557249 | 3612.93695719 |
| PB-C50-S | 0 | M-B | 3600 | 3570.5953431 | 40.3517955 | 3610.9471386 |
| PB-C50-S | 0 | ENS-C | 3600 | 3571.8953901 | 40.0828359 | 3611.978226 |
| PB-C50-S | 0 | P-GRB | 3600 | 3583.0201592 | 1.57247429999 | 3584.5926335 |
| PB-B50-U | 0 | P-GRB | 3600 | 3572.99104926 | 1.30260140001 | 3574.29365066 |
| PB-B50-U | 0 | MB-D1 | 3600 | 3570.5630142 | 25.6511639 | 3596.2141781 |
| PB-B50-U | 0 | M-B | 3600 | 3571.8623909 | 19.9168553 | 3591.7792462 |
| PB-B50-U | 0 | ENS-C | 3600 | 3570.5730882 | 19.7308828 | 3590.303971 |
| PB-B35-B | 1 | ENS-C | 2400 | 2372.97340551 | 18.6435541 | 2391.61695961 |
| PB-B35-B | 1 | P-GRB | 2400 | 2370.5116857 | 1.97982149999 | 2372.4915072 |
| PB-B35-B | 1 | MB-D1 | 2400 | 2370.6168373 | 22.4221823 | 2393.0390196 |
| PB-B35-B | 1 | M-B | 2400 | 2370.4895614 | 18.640514 | 2389.1300754 |
| PB-C50-S | 1 | M-B | 3600 | 3572.60278401 | 85.3058211 | 3657.90860511 |
| PB-C50-S | 1 | ENS-C | 3600 | 3570.6597372 | 84.4460346 | 3655.1057718 |
| PB-C50-S | 1 | P-GRB | 3600 | 3570.672291 | 930.1450991 | 4500.8173901 |
| PB-C50-S | 1 | MB-D1 | 3600 | 3572.56418137 | 85.1722841 | 3657.73646547 |

## Table S2. All 48 R122 directed comparisons

| Role | Seed | Candidate | Control | Class | Severe |
|---|---|---|---|---|---|
| PB-C20-B | 0 | MB-D1 | P-GRB | WIN | False |
| PB-C20-B | 0 | M-B | P-GRB | WIN | False |
| PB-C20-B | 0 | ENS-C | P-GRB | WIN | False |
| PB-C20-B | 0 | MB-D1 | M-B | LOSS | False |
| PB-C20-B | 0 | MB-D1 | ENS-C | LOSS | False |
| PB-C20-B | 0 | M-B | ENS-C | TIE | False |
| PB-B20-S | 0 | MB-D1 | P-GRB | WIN | False |
| PB-B20-S | 0 | M-B | P-GRB | WIN | False |
| PB-B20-S | 0 | ENS-C | P-GRB | WIN | False |
| PB-B20-S | 0 | MB-D1 | M-B | TIE | False |
| PB-B20-S | 0 | MB-D1 | ENS-C | TIE | False |
| PB-B20-S | 0 | M-B | ENS-C | TIE | False |
| PB-C35-U | 0 | MB-D1 | P-GRB | MIXED | False |
| PB-C35-U | 0 | M-B | P-GRB | WIN | False |
| PB-C35-U | 0 | ENS-C | P-GRB | WIN | False |
| PB-C35-U | 0 | MB-D1 | M-B | LOSS | False |
| PB-C35-U | 0 | MB-D1 | ENS-C | LOSS | False |
| PB-C35-U | 0 | M-B | ENS-C | LOSS | False |
| PB-B35-B | 0 | MB-D1 | P-GRB | WIN | False |
| PB-B35-B | 0 | M-B | P-GRB | WIN | False |
| PB-B35-B | 0 | ENS-C | P-GRB | WIN | False |
| PB-B35-B | 0 | MB-D1 | M-B | MIXED | False |
| PB-B35-B | 0 | MB-D1 | ENS-C | MIXED | False |
| PB-B35-B | 0 | M-B | ENS-C | TIE | False |
| PB-C50-S | 0 | MB-D1 | P-GRB | WIN | False |
| PB-C50-S | 0 | M-B | P-GRB | WIN | False |
| PB-C50-S | 0 | ENS-C | P-GRB | WIN | False |
| PB-C50-S | 0 | MB-D1 | M-B | TIE | False |
| PB-C50-S | 0 | MB-D1 | ENS-C | TIE | False |
| PB-C50-S | 0 | M-B | ENS-C | TIE | False |
| PB-B50-U | 0 | MB-D1 | P-GRB | WIN | False |
| PB-B50-U | 0 | M-B | P-GRB | WIN | False |
| PB-B50-U | 0 | ENS-C | P-GRB | WIN | False |
| PB-B50-U | 0 | MB-D1 | M-B | WIN | False |
| PB-B50-U | 0 | MB-D1 | ENS-C | WIN | False |
| PB-B50-U | 0 | M-B | ENS-C | WIN | False |
| PB-B35-B | 1 | MB-D1 | P-GRB | WIN | False |
| PB-B35-B | 1 | M-B | P-GRB | WIN | False |
| PB-B35-B | 1 | ENS-C | P-GRB | WIN | False |
| PB-B35-B | 1 | MB-D1 | M-B | WIN | False |

| Role | Seed | Candidate | Control | Class | Severe |
|---|---|---|---|---|---|
| PB-B35-B | 1 | MB-D1 | ENS-C | WIN | False |
| PB-B35-B | 1 | M-B | ENS-C | LOSS | False |
| PB-C50-S | 1 | MB-D1 | P-GRB | WIN | False |
| PB-C50-S | 1 | M-B | P-GRB | WIN | False |
| PB-C50-S | 1 | ENS-C | P-GRB | WIN | False |
| PB-C50-S | 1 | MB-D1 | M-B | TIE | False |
| PB-C50-S | 1 | MB-D1 | ENS-C | TIE | False |
| PB-C50-S | 1 | M-B | ENS-C | TIE | False |

## Table S3a. All three known R123 endpoints

| Method | U | L | Signed gap | Tier |
|---|---|---|---|---|
| P-GRB | 0.37764170855 | 0.350377933041 | 0.0272637755094 | O |
| M-B | 0.372341775499 | 0.358011485552 | 0.0143302899472 | O |
| ENS-C | 0.372341775499 | 0.358943666168 | 0.0133981093313 | O |

## Table S3b. All three R123 complete clocks

| Method | Cap | T_produce | T_verify | T_total |
|---|---|---|---|---|
| P-GRB | 18000 | 17971.7112989 | 1.96611939999 | 17973.6774183 |
| M-B | 18000 | 17970.5725766 | 40.964555 | 18011.5371316 |
| ENS-C | 18000 | 17970.8751446 | 58.2420274 | 18029.117172 |

## Table S4. All three R123 directed comparisons

| Candidate | Control | Class | Severe |
|---|---|---|---|
| M-B | P-GRB | WIN | False |
| ENS-C | P-GRB | WIN | False |
| M-B | ENS-C | TIE | False |

## Table S5: existing checkpoints

| Method | Time (s) | U/last witness | L | Source |
|---|---|---|---|---|
| P-GRB | 3600 | 0.425950075294 | 0.349220695801 | last_prior_existing_committed_records |
| P-GRB | 7200 | 0.42083291094 | 0.349949886971 | last_prior_existing_committed_records |
| P-GRB | 10800 | 0.389701007874 | 0.350050746873 | last_prior_existing_committed_records |
| P-GRB | 14400 | 0.378043868561 | 0.350180030064 | last_prior_existing_committed_records |
| P-GRB | end 17971.711298929702 | 0.37764170855 | 0.350377933041 | actual_sealed_endpoint_before_checkpoint |
| M-B | 3600 | 0.372341775499 | 0.356874739941 | last_prior_existing_committed_records |
| M-B | 7200 | 0.372341775499 | 0.357080008664 | last_prior_existing_committed_records |
| M-B | 10800 | 0.372341775499 | 0.357681585834 | last_prior_existing_committed_records |
| M-B | 14400 | 0.372341775499 | 0.357815221734 | last_prior_existing_committed_records |
| M-B | end 17970.572576599981 | 0.372341775499 | 0.358011485552 | actual_sealed_endpoint_before_checkpoint |
| ENS-C | 3600 | unknown | unknown | last_prior_existing_committed_records |
| ENS-C | 7200 | unknown | unknown | last_prior_existing_committed_records |

| Method | Time (s) | U/last witness | L | Source |
|---|---|---|---|---|
| ENS-C | 10800 | unknown | unknown | last_prior_existing_committed_records |
| ENS-C | 14400 | unknown | unknown | last_prior_existing_committed_records |
| ENS-C | end 17970.875144599995 | 0.372341775499 | 0.358943666168 | actual_sealed_endpoint_before_checkpoint |

## S.131 Final-version full endpoint and clock authority

**Table R131.S. All 52 formal arms and three clocks.** Layer, Seed and repeat preserve denominators. Open certification time is null. Display: clocks to 0.001 seconds, U/L to six significant digits and signed gap to three; CSV/JSON retain full precision.

| Layer / problem | Seed / repeat | Method | Certified | T_cert | T_produce | T_verify | T_total | U | L | signed U-L |
|---|---|---|---|---|---|---|---|---|---|---|
| A / SC-C15-S | 0 / 1 | P | True | 460.809 | 460.809 | 6.174 | 466.983 | 0.462806 | 0.462806 | 1.11e-16 |
| A / SC-C15-S | 0 / 1 | S2 | True | 33.167 | 33.167 | 1.035 | 34.202 | 0.462806 | 0.462806 | 7.42e-10 |
| A / SC-C15-S | 1 / 1 | S2 | True | 35.538 | 35.538 | 2.544 | 38.082 | 0.462806 | 0.462806 | 1.11e-16 |
| A / SC-C15-S | 1 / 1 | P | True | 469.536 | 469.536 | 16.436 | 485.973 | 0.462806 | 0.462806 | -1.67e-16 |
| A / SC-B15-S | 0 / 1 | S2 | True | 2.851 | 2.851 | 0.788 | 3.639 | 0.947005 | 0.947005 | 1.22e-15 |
| A / SC-B15-S | 0 / 1 | P | True | 12.875 | 12.875 | 0.352 | 13.227 | 0.947005 | 0.947005 | 8.88e-16 |
| A / SC-B15-S | 1 / 1 | P | True | 16.181 | 16.181 | 1.285 | 17.466 | 0.947005 | 0.947005 | 2.22e-16 |
| A / SC-B15-S | 1 / 1 | S2 | True | 2.245 | 2.245 | 2.067 | 4.311 | 0.947005 | 0.947005 | -3.33e-16 |
| A / SC-C15-B | 0 / 1 | P | True | 242.256 | 242.256 | 0.288 | 242.544 | 0.00676078 | 0.00676078 | -1.73e-17 |
| A / SC-C15-B | 0 / 1 | S2 | True | 57.928 | 57.928 | 0.898 | 58.826 | 0.00676078 | 0.00676077 | 2.42e-09 |
| A / SC-C15-B | 1 / 1 | S2 | True | 33.444 | 33.444 | 2.311 | 35.755 | 0.00676078 | 0.00676078 | 1.39e-17 |
| A / SC-C15-B | 1 / 1 | P | False | null | 570.429 | 1.225 | 571.654 | 0.0399198 | 0.0005342 | 0.0394 |
| A / SC-B20-S | 0 / 1 | S2 | True | 982.279 | 982.279 | 5.453 | 987.731 | 0.315389 | 0.315389 | -1.67e-16 |
| A / SC-B20-S | 0 / 1 | P | False | null | 1170.605 | 1.589 | 1172.194 | 0.34025 | 0.209048 | 0.131 |
| A / SC-B20-S | 1 / 1 | P | False | null | 1170.903 | 3.589 | 1174.493 | 0.375037 | 0.214415 | 0.161 |
| A / SC-B20-S | 1 / 1 | S2 | True | 676.371 | 676.371 | 10.964 | 687.335 | 0.315389 | 0.315389 | 2.47e-10 |
| A / SC-C20-S | 0 / 1 | P | False | null | 1171.043 | 1.406 | 1172.449 | 0.28058 | 0.161053 | 0.12 |
| A / SC-C20-S | 0 / 1 | S2 | False | null | 1170.676 | 4.953 | 1175.628 | 0.21643 | 0.208683 | 0.00775 |
| A / SC-C20-S | 1 / 1 | S2 | False | null | 1171.035 | 9.836 | 1180.871 | 0.216734 | 0.208248 | 0.00849 |

| Layer / problem | Seed / repeat | Method | Certified | T_cert | T_produce | T_verify | T_total | U | L | signed U-L |
|---|---|---|---|---|---|---|---|---|---|---|
| A / SC-C20-S | 1 / 1 | P | False | null | 1170.577 | 3.273 | 1173.850 | 0.234721 | 0.162552 | 0.0722 |
| A / SC-B20-U | 0 / 1 | S2 | True | 26.893 | 26.893 | 1.485 | 28.377 | 0.173762 | 0.173762 | 1.67e-16 |
| A / SC-B20-U | 0 / 1 | P | False | null | 1170.574 | 1.720 | 1172.294 | 0.221769 | 0.013354 | 0.208 |
| A / SC-B20-U | 1 / 1 | P | False | null | 1171.024 | 7.360 | 1178.384 | 0.20523 | 0.019548 | 0.186 |
| A / SC-B20-U | 1 / 1 | S2 | True | 15.998 | 15.998 | 3.447 | 19.444 | 0.173762 | 0.173762 | 4.08e-10 |
| B / SM-C8-S | 0 / 1 | P | True | 5.568 | 5.568 | 0.371 | 5.939 | 0.516077 | 0.516077 | 1.11e-16 |
| B / SM-C8-S | 0 / 1 | S2 | True | 2.645 | 2.645 | 0.405 | 3.050 | 0.516077 | 0.516077 | 1.11e-16 |
| B / SM-C8-S | 0 / 2 | S2 | True | 2.865 | 2.865 | 0.407 | 3.272 | 0.516077 | 0.516077 | 1.11e-16 |
| B / SM-C8-S | 0 / 2 | P | True | 5.143 | 5.143 | 0.372 | 5.515 | 0.516077 | 0.516077 | 1.11e-16 |
| B / SM-C8-S | 1 / 1 | S2 | True | 3.227 | 3.227 | 1.199 | 4.426 | 0.516077 | 0.516077 | -4.44e-16 |
| B / SM-C8-S | 1 / 1 | P | True | 3.904 | 3.904 | 0.907 | 4.811 | 0.516077 | 0.516077 | 1.11e-16 |
| B / SM-C8-S | 1 / 2 | P | True | 4.197 | 4.197 | 0.911 | 5.108 | 0.516077 | 0.516077 | 1.11e-16 |
| B / SM-C8-S | 1 / 2 | S2 | True | 2.845 | 2.845 | 1.194 | 4.039 | 0.516077 | 0.516077 | -4.44e-16 |
| B / SM-C8-S | 2 / 1 | P | True | 4.611 | 4.611 | 0.942 | 5.553 | 0.516077 | 0.516077 | 1.11e-16 |
| B / SM-C8-S | 2 / 1 | S2 | True | 3.781 | 3.781 | 1.090 | 4.871 | 0.516077 | 0.516077 | 1.33e-15 |
| B / SM-C8-S | 2 / 2 | S2 | True | 4.126 | 4.126 | 1.136 | 5.262 | 0.516077 | 0.516077 | 1.33e-15 |
| B / SM-C8-S | 2 / 2 | P | True | 4.307 | 4.307 | 0.948 | 5.254 | 0.516077 | 0.516077 | 1.11e-16 |
| B / SM-B10-B | 0 / 1 | S2 | True | 0.792 | 0.792 | 0.159 | 0.951 | 0.535891 | 0.535891 | 0 |
| B / SM-B10-B | 0 / 1 | P | True | 0.778 | 0.778 | 0.136 | 0.914 | 0.535891 | 0.535891 | 1.11e-16 |
| B / SM-B10-B | 0 / 2 | P | True | 1.122 | 1.122 | 0.140 | 1.262 | 0.535891 | 0.535891 | 1.11e-16 |
| B / SM-B10-B | 0 / 2 | S2 | True | 0.461 | 0.461 | 0.153 | 0.614 | 0.535891 | 0.535891 | 0 |
| B / SM-B10-B | 1 / 1 | P | True | 1.067 | 1.067 | 0.653 | 1.720 | 0.535891 | 0.535891 | 0 |
| B / SM-B10-B | 1 / 1 | S2 | True | 0.511 | 0.511 | 0.632 | 1.143 | 0.535891 | 0.535891 | 0 |
| B / SM-B10-B | 1 / 2 | S2 | True | 0.801 | 0.801 | 0.633 | 1.435 | 0.535891 | 0.535891 | 0 |
| B / SM-B10-B | 1 / 2 | P | True | 0.708 | 0.708 | 0.641 | 1.349 | 0.535891 | 0.535891 | 0 |
| B / SM-B10-B | 2 / 1 | S2 | True | 0.800 | 0.800 | 0.626 | 1.426 | 0.535891 | 0.535891 | 0 |
| B / SM-B10-B | 2 / 1 | P | True | 0.794 | 0.794 | 0.654 | 1.448 | 0.535891 | 0.535891 | 0 |

| Layer / problem | Seed / repeat | Method | Certified | T_cert | T_produce | T_verify | T_total | U | L | signed U-L |
|---|---|---|---|---|---|---|---|---|---|---|
| B / SM-B10-B | 2 / 2 | P | True | 1.129 | 1.129 | 0.669 | 1.797 | 0.535891 | 0.535891 | 0 |
| B / SM-B10-B | 2 / 2 | S2 | True | 0.461 | 0.461 | 0.628 | 1.089 | 0.535891 | 0.535891 | 0 |
| C / PB-C35-U | 0 / 1 | P | False | null | 2370.823 | 1.107 | 2371.930 | 0.174772 | 0.0597029 | 0.115 |
| C / PB-C35-U | 0 / 1 | S2 | False | null | 2370.592 | 5.475 | 2376.067 | 0.183702 | 0.0847283 | 0.099 |
| C / PB-B35-B | 0 / 1 | S2 | False | null | 2370.807 | 5.246 | 2376.053 | 0.246504 | 0.121942 | 0.125 |
| C / PB-B35-B | 0 / 1 | P | False | null | 2370.555 | 0.546 | 2371.101 | 0.305556 | 0 | 0.306 |